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Have a Cow, Man (2010 Version)

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Dissertation

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Abstract

Field-programmable gate arrays (FPGAs) are increasingly attractive for nuclear instrumentation and control (I&C) because they offer deterministic hardware execution, reconfigurability, and high computational efficiency in compact, low-power platforms. However, their use in reactor-relevant environments is limited by radiation-induced effects such as single-event upsets and related functional disturbances. Conventional qualification campaigns can also be expensive and inefficient because they depend on scarce beam time, custom setups, and substantial external instrumentation. This work proposes a low-overhead irradiation-test methodology for commercial off-the-shelf SRAM-based FPGAs that emphasizes on-chip benchmarking, high-rate telemetry, and application-relevant fault observability.

The methodology combines two complementary workloads executed on the FPGA under identical beam conditions. The first is a flip-flop scan-farm benchmark that stresses a large population of explicitly instantiated registers and produces time-tagged mismatch events suitable for upset-rate and effective cross-section extraction. The second is a closed-loop digital control benchmark based on an I-PD controller coupled to a digital two-node thermal plant. Together, these workloads provide both fabric-level sensitivity information and application-level performance metrics, enabling comparison between raw upset behavior and functional degradation during

33 irradiation. Time-tagged telemetry from both workloads is streamed over UART for
34 post-processing and comparison across test conditions.

35 Planned irradiations will be performed at The Ohio State University Research
36 Reactor Fast Beam Facility, which provides a collimated fast-neutron beam with se-
37 lectable gamma and thermal-neutron suppression. The intended configuration uses
38 a bismuth gamma filter, a lead shutter, and an external cadmium filter to establish
39 a reactor-relevant fast-neutron environment while suppressing unwanted beam com-
40 ponents. Under the selected configuration, the facility characterization indicates a
41 representative fast-neutron flux of approximately $2.6 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}$ with an associ-
42 ated gamma dose rate of about 50 mR/h.

43 The digital control workload is implemented entirely on-chip using signed fixed-
44 point arithmetic and includes a discretized two-node thermal plant, measurement lag,
45 actuator dynamics, filtered derivative action, and anti-windup. This structure allows
46 radiation-induced perturbations to be observed directly in control-relevant variables
47 such as tracking error, control effort, saturation behavior, and output response. In
48 parallel, the scan-farm benchmark repeatedly executes LOAD, HOLD, and READ
49 phases across 55 scan chains of 512 flip-flops each, corresponding to 28,160 registers,
50 with a PRBS31 stimulus source and embedded signature-based checking.

51 The goal of this work is to establish a repeatable and transferable test approach
52 that reduces setup overhead, improves statistical confidence during limited beam al-
53 locations, and better reflects realistic FPGA workloads in nuclear environments. If
54 successful, this combined fabric-stress and application-benchmark methodology could
55 provide a scalable path for lower-cost comparative radiation testing of candidate
56 FPGA platforms for reactor-relevant service.

This is dedicated to the one I love ... la la la ...

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64

Vita

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69

Publications

70 Research Publications

71

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Chapter 1: Introduction

411 Nuclear power occupies a singular position among low-carbon energy sources: it is
412 the only technology that delivers firm, dispatchable, high-density electricity without
413 direct greenhouse-gas emissions at a scale already proven across decades of opera-
414 tion. Yet the environments that make nuclear power possible, intense neutron and
415 gamma fields, high temperatures, and chemically aggressive primary coolants (e.g.,
416 liquid sodium, molten fluoride salts, and high-pressure water), are precisely the en-
417 vironments that destroy the instruments needed to monitor and control it. Reliable
418 radiation monitoring is not incidental to safe reactor operation; it is the primary
419 mechanism by which operators verify that every defense-in-depth barrier to fission-
420 product release — fuel cladding, reactor pressure boundary, and containment — re-
421 mains intact. As the existing fleet ages and a generation of advanced and small
422 modular reactors (SMRs) moves from concept to construction, the instrumentation
423 technologies that have served the industry for half a century are confronting hard
424 limits in both technical performance and economic viability. This dissertation inves-
425 tigates two complementary approaches to resolving that tension: the development
426 of perovskite semiconductors as radiation-responsive detector materials, and the im-
427 plementation of radiation-tolerant digital monitoring systems built on commercially
428 available field-programmable gate arrays (FPGAs). Taken together, these efforts span

the full stack from detector material to deployable system and aim to demonstrate that next-generation nuclear instrumentation can be both technically capable and represent a meaningful improvement in economic viability over current approaches.

1.1 Nuclear Power and the Global Decarbonization Context

The scale of the decarbonization challenge facing the global electricity system makes any realistic pathway to net-zero heavily dependent on nuclear energy. At the COP28 climate summit in December 2023, twenty-two nations signed a declaration to triple global nuclear generating capacity from approximately 375 GWe to more than 1,125 GWe by 2050; that coalition has since grown to thirty-three countries [95]. The International Energy Agency’s net-zero pathway requires nuclear capacity to reach approximately 1,079 GWe by 2050, demanding that annual investment in the sector roughly double from its current level of \$65–75 billion to \$120–150 billion per year by 2030 [41]. The International Atomic Energy Agency (IAEA) has revised its projections upward for five consecutive years, with its 2025 high-case scenario projecting 992 GWe by 2050 [40].

The existing global fleet comprises approximately 408 operating reactors in thirty-one countries, with a combined net capacity that reached a record 369 GWe in 2024 [84]. Nuclear electricity generation also set an all-time high of 2,667 TWh in 2024, narrowly surpassing the 2006 record, though nuclear’s share of global electricity has fallen to approximately 9% as variable renewables have expanded at a far faster pace [84]. The fleet’s average age of 32.4 years — with 141 units exceeding 41 years of operation — makes life extension, instrumentation modernization, and new construction simultaneously urgent priorities.

Development of advanced reactors is more active and better funded than at any point in the last several decades. In the United States, TerraPower’s 345 MW(e) Natrium sodium-cooled fast reactor received its NRC construction permit in March 2026, the first commercial non-light-water reactor approval in over forty years [79]. Kairos Power’s Hermes fluoride-salt-cooled test reactor began nuclear safety-related construction in 2025 [44], and X-energy’s Xe-100 pebble-bed high-temperature gas-cooled reactor is under development for an industrial host site with over \$1.2 billion in DOE cost-share [108]. Among light-water SMRs, GE-Hitachi’s BWRX-300 is under construction at Ontario’s Darlington site and has received a \$400 million DOE award for deployment at TVA’s Clinch River site [103]. Each of these programs introduces novel coolant chemistries, fuel geometries, and multi-module control architectures that existing monitoring technologies were never designed to address.

1.2 The Regulatory Foundation for Radiation Monitoring

Continuous radiation monitoring across neutron flux, gamma dose, primary coolant activity, and effluent pathways is mandated by the regulatory foundations of nuclear operation, not merely encouraged by best practice. The NRC’s General Design Criteria (GDC), codified in 10 CFR 50 Appendix A, establish this requirement at the highest level of the licensing framework. GDC 13 requires instrumentation capable of monitoring all variables that affect the reactor core, coolant pressure boundary, and containment integrity; GDC 64 mandates monitoring of the containment atmosphere, recirculation spaces, and effluent pathways for radioactivity; and GDC 19 requires that control room operators maintain safe-shutdown capability with adequate instrumentation even under accident conditions [98]. The IAEA mirrors these

475 requirements through the Safety Requirements for Nuclear Power Plants (SSR-2/1)
476 and provides detailed I&C system guidance in SSG-39 [39, 38].

477 The technical scope of these monitoring requirements is exceptionally broad. Neu-
478 tron flux measurement alone spans approximately twelve decades — from a few counts
479 per second during refueling to approximately $10^{15} \text{ n cm}^{-2} \text{ s}^{-1}$ at full power — requiring
480 three overlapping detector subsystems. Source-range monitors based on proportional
481 counters or fission chambers operate in pulse-counting mode across the lowest six
482 to seven decades during startup and shutdown. Intermediate-range monitors em-
483 ploy compensated ionization chambers in Campbell mode across approximately eight
484 to ten decades. Power-range monitors using uncompensated boron-lined ionization
485 chambers cover the upper two decades and are calibrated against secondary-side heat
486 balance. Each of these subsystems must meet independent safety-channel redundancy
487 requirements, typically two to four independent channels per range [69].

488 Beyond neutron flux, plants deploy area gamma monitors throughout contain-
489 ment, auxiliary buildings, fuel handling areas, and spent fuel pools to enforce occu-
490 pational dose limits under 10 CFR 20. On-line gamma spectroscopy of the primary
491 coolant monitors fission-product signatures that indicate fuel-cladding breaches. Con-
492 tainment atmosphere monitors classified as Category 1 under Regulatory Guide 1.97,
493 Revision 5, must survive combined temperature, pressure, radiation, and chemical-
494 spray environments for the full duration of a design-basis accident [97]. Stack and
495 liquid effluent monitors verify compliance with 10 CFR 50 Appendix I dose objec-
496 tives, while the Radiological Environmental Monitoring Program tracks off-site air,
497 water, and vegetation samples under RG 4.1.

498 The Fukushima Daiichi accident of March 2011 provided a severe demonstration
499 of the consequences when monitoring fails. The tsunami rendered 23 of 24 on-site
500 radiation monitoring stations nonfunctional. The subsequent loss of all AC power and
501 battery depletion eliminated reactor vessel pressure, temperature, and water-level
502 indication, forcing operators to manage three simultaneous core-melt progressions
503 with essentially no real-time process data [60, 61]. Spent fuel pool instrumentation
504 was unavailable for nearly two weeks. The NRC’s post-Fukushima response, including
505 Order EA-12-051 requiring backup-powered spent fuel pool level instrumentation, the
506 FLEX diverse-coping strategy described in NEI 12-06, and the addition of Type F
507 severe-accident management variables to RG 1.97 Rev. 5, underscored that reliable
508 monitoring is not a compliance formality but the operational foundation of accident
509 management.

510 **1.3 Limitations of Current Radiation Detector Technologies**

511 The detector materials deployed across the nuclear industry each carry critical
512 limitations that constrain their use in existing plants and threaten to preclude their
513 adoption in advanced reactor environments. High-purity germanium (HPGe) remains
514 the gold standard for gamma spectroscopy, achieving better than 0.2% full-width at
515 half-maximum energy resolution at 662 keV. However, its narrow bandgap of 0.67 eV
516 mandates cryogenic operation below 120 K. Liquid-nitrogen dewars are impractical
517 for unattended or remote monitoring; mechanical coolers add substantial cost, com-
518 plexity, and microphonic noise that degrades resolution at low energies. HPGe is
519 therefore confined almost exclusively to laboratory and assay applications rather than
520 continuous in-plant monitoring [19].

521 Silicon detectors dominate charged-particle detection and, in research settings, are
522 used for fast-neutron spectrometry via proton-recoil interactions [64], but suffer irre-
523 versible displacement damage above approximately 10^{14} – 10^{15} n_{eq} cm^{−2} fluence, with
524 charge-collection efficiency falling toward 50% — a threshold easily surpassed in the
525 lifetime of a power-reactor instrument [64]. Silicon’s low atomic number ($Z = 14$)
526 also provides poor photoelectric stopping power for gamma-ray energies above a few
527 hundred keV, making it poorly suited to the mixed neutron-gamma fields present
528 throughout a reactor plant.

529 Scintillators constitute the most widely deployed class of gamma-detection instru-
530 ments and are valued for their ability to produce measurable light from a wide range of
531 ionizing radiations. Sodium iodide thallium-doped [NaI(Tl)] crystals, the workhorse
532 material, achieve approximately 6–7% energy resolution at 662 keV but are highly
533 hygroscopic, mechanically fragile, and must be coupled to photomultiplier tubes that
534 are sensitive to magnetic fields, fragile under shock and vibration, and require high-
535 voltage supplies of 800–1200 V [54]. Advanced scintillators such as LaBr₃(Ce) achieve
536 approximately 3% resolution but contain intrinsic radioactivity from ¹³⁸La that cre-
537 ates a flat background continuum complicating low-level measurements. Organic
538 scintillators, including plastic scintillators and stilbene crystals, are well established
539 for fast neutron detection through proton-recoil interactions, but their low effective
540 atomic number renders them essentially transparent to gamma rays, requiring either
541 active discrimination electronics or companion gamma-shielding that adds mass, cost,
542 and complexity.

543 Cadmium zinc telluride (CZT) represents the most mature room-temperature
544 semiconductor alternative, offering energy resolutions approaching 0.7% at 662 keV

545 in pixelated geometries with full single-carrier-transport correction. However, CZT
 546 crystal growth requires extremely precise control of stoichiometry and vapor-phase
 547 conditions during directional solidification: high cadmium vapor pressure promotes
 548 vacancy defects, tellurium inclusions limit boule diameter, and critical shear stresses
 549 of only 0.3 MPa make the material mechanically fragile during processing [104]. Pro-
 550 duction costs remain high, supply is constrained, and the situation deteriorated sig-
 551 nificantly in 2025 when the People’s Republic of China imposed export controls on
 552 germanium, gallium, and related critical materials including cadmium telluride com-
 553 pounds.

554 For thermal neutron detection, the ^3He tube has been the standard technology
 555 since the 1950s by virtue of its high neutron absorption cross-section (5330 barns at
 556 0.025 eV), excellent gamma discrimination, and compatibility with conventional pulse-
 557 counting electronics. The supply of ^3He — produced exclusively as a decay product of
 558 tritium from the nuclear-weapons stockpile — was recognized as critically constrained
 559 following the post-September 11 deployment of approximately 1,300 radiation portal
 560 monitors for border-security applications. Annual demand reached roughly 70,000 L
 561 against a production rate of only approximately 8,000 L yr $^{-1}$, driving prices from
 562 approximately \$100 L $^{-1}$ to over \$2,000 L $^{-1}$ and prompting DOE to begin rationing
 563 allocation in 2009 [16]. The episode exposed the fragility of depending on a single non-
 564 renewable detection medium and has accelerated development of alternative neutron
 565 detectors using ^{10}B -lined tubes and $^6\text{LiF}:\text{ZnS}$ scintillator screens.

566 Halide perovskites have emerged as a compelling alternative class of radiation-
 567 responsive materials. Their high effective atomic number — CsPbBr_3 has $Z_{\text{eff}} =$
 568 65.9, compared to 50.2 for CZT — provides excellent photoelectric stopping power

for gamma radiation [25]. Single-crystal MAPbBr₃ and CsPbBr₃ devices, operating at room temperature, have demonstrated mobility-lifetime products of up to $8 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$ and energy resolutions approaching 1.4% at 662 keV [33], approaching CZT performance without cryogenic cooling or Bridgman growth. A distinctive self-healing mechanism — in which the soft perovskite lattice allows radiation-displaced ions to migrate back toward equilibrium positions at room temperature — has been confirmed under proton irradiation to $10^{13} \text{ p cm}^{-2}$ and extended gamma exposures [24]. The methylhydrazinium lead chloride (MHyPbCl₃) composition investigated in this work introduces a high-hydrogen-density 3D perovskite framework with a hydrogen density of $3.79 \times 10^{22} \text{ cm}^{-3}$ comparable to organic scintillators, enabling direct fast-neutron conversion through proton recoil without requiring a separate moderator or conversion layer [73].

Significant challenges remain. Ion migration under applied electric fields and elevated humidity drives hysteresis in current-voltage characteristics and accelerates device failure at the biases required for thick-crystal direct-conversion operation. Long-term stability data at power-reactor fluence levels (10^{14} – $10^{15} \text{ n cm}^{-2}$) remain absent from the literature. Lead toxicity presents regulatory and environmental considerations for deployed instruments. These challenges motivate the stabilization strategies — fluorinated surface passivation, hexagonal boron nitride composites, and vacuum operation — investigated in the present work.

1.4 The Economics of Nuclear Instrumentation

Cost overruns in nuclear construction have become the defining obstacle to expansion in liberalized electricity markets. The two AP1000 units completed at Plant

592 Vogtle in Georgia cost approximately \$35 billion against an original estimate of \$14
 593 billion [6]. France’s Flamanville 3 EPR reached €23.7 billion — seven times its orig-
 594 inal budget — after seventeen years of construction. The UK’s Hinkley Point C has
 595 escalated to an estimated £35 billion in 2015 prices against an initial projection of
 596 approximately £18 billion [18]. New nuclear levelized cost of electricity in the United
 597 States ranges from approximately \$110 MWh⁻¹ (EIA) to \$180 MWh⁻¹ (Lazard), com-
 598 pared to approximately \$55 MWh⁻¹ for utility-scale solar and \$40 MWh⁻¹ for onshore
 599 wind. The existing U.S. fleet operates far more competitively — NEI reports a 2023
 600 average total generating cost of \$31.76 MWh⁻¹, the lowest on record [68] — but twelve
 601 commercial reactors have closed since 2013 due to competitive pressure, and emer-
 602 gency support programs including a \$6 billion Civil Nuclear Credit allocation have
 603 been required to prevent further premature retirements [65].

604 Instrumentation and control systems, while representing an estimated 5–10% of
 605 total plant capital cost, exert a disproportionate influence on both construction time-
 606 lines and long-term operating expenses. A typical large light-water reactor contains
 607 approximately 10,000 sensors, 5,000 km of I&C cabling, and hundreds of electronic
 608 cabinets. Nuclear-grade quality assurance requirements under 10 CFR 50 Appendix B
 609 impose a cost premium that pervades the entire supply chain: components manufac-
 610 tured to identical functional specifications have been documented at price ratios of
 611 4:1 or greater between nuclear-qualified and commercial-grade procurement, driven by
 612 QA documentation requirements, environmental qualification testing, and the severe
 613 scarcity of nuclear-qualified suppliers [4]. Dominion Energy’s comprehensive digital
 614 I&C modernization across four units at Surry and North Anna is budgeted at \$5.45
 615 billion as part of subsequent license renewal [78]. The DOE Light Water Reactor

616 Sustainability (LWRS) Program has explicitly cautioned that “wide-scale moderniza-
617 tion of legacy I&C systems could prove to be cost-prohibitive unless implemented to
618 enable significant business innovation.”

619 The cost of radiation-hardened digital electronics amplifies this problem at the
620 component level. Purpose-built radiation-hardened FPGAs — the Microchip RTG4,
621 AMD/Xilinx XQRKU060, and their equivalents — carry price tags of tens of thou-
622 sands to hundreds of thousands of dollars per device, a premium of $10\times$ to $100\times$
623 over commercial equivalents that cost \$10–500 [11]. The radiation-hardened FPGA
624 market, valued at \$465.5 million in 2024, serves primarily space and defense appli-
625 cations where unit volumes are small and alternatives do not exist. These devices
626 trail commercial FPGAs by two to three process generations: the Microchip RTG4
627 uses a 65 nm flash-based process while commercial devices operate at 3–5 nm, a gap
628 representing multiple orders of magnitude in logic density, performance, and power
629 efficiency [57, 3]. Long lead times of 6–18 months, ITAR export restrictions, and the
630 absence of advanced intellectual property cores compound the challenge. For SMRs
631 targeting a levelized cost of electricity of \$60–90 MWh⁻¹, this cost structure is unten-
632 able: I&C requirements do not scale down proportionally with reactor output, making
633 instrumentation a higher fraction of total cost in smaller plants. The DOE’s GEM-
634 INA program has set an aspirational target of reducing advanced reactor operations
635 and maintenance cost from \$23 MWh⁻¹ to \$2 MWh⁻¹ through autonomous operations
636 and advanced diagnostics — a goal that cannot be achieved without fundamentally
637 cheaper and more capable sensing and processing hardware [96].

638 The alternative to purpose-built rad-hard electronics is software-defined radiation
639 tolerance using commercial off-the-shelf FPGAs with designer-inserted mitigation.

640 Triple modular redundancy (TMR) triplicates critical logic with majority voting to
 641 correct single-event upsets in real time; configuration memory scrubbing periodically
 642 corrects bit-flip errors in SRAM-based FPGAs before they propagate; and partial
 643 dynamic reconfiguration enables real-time repair of corrupted fabric regions. CERN's
 644 two-year study of Kintex-7 FPGAs under mixed-field radiation found no permanent
 645 operational failures to 300 krad total ionizing dose with TMR and scrubbing [23].
 646 Montana State University's RadSat program demonstrated COTS FPGA-based ra-
 647 diation tolerance at "a cost 100 $\times$ lower than existing radiation-hardened computers."
 648 A TMR-with-spares architecture for nuclear reactor protection system logic has been
 649 validated under cyclotron irradiation [26]. These results suggest that COTS FP-
 650 GAs with mature mitigation strategies can meet nuclear monitoring requirements
 651 at a fraction of the cost of purpose-built rad-hard devices, provided the mitigation
 652 architecture is co-designed with the application from the outset.

653 **1.5 Research Objectives and Dissertation Organization**

654 The research presented in this dissertation is organized around three complemen-
 655 tary objectives aligned with the materials-to-systems hierarchy described above.

656 The first objective is to characterize the radiation response of halide perovskite
 657 materials across a range of compositions, device architectures, and radiation types
 658 relevant to nuclear monitoring. This encompasses gamma-photon response from Cs-
 659 137 and Co-60 sources, X-ray sensitivity as a function of applied bias, fast-neutron re-
 660 sponse, and the material instability mechanisms — ion migration, humidity-dependent
 661 breakdown, and charge trapping — that currently limit perovskite detector lifetimes.

662 The second objective is to develop and demonstrate perovskite-based radiation
663 detection devices optimized for specific applications in the nuclear fuel cycle: high-
664 energy photon monitoring for post-detonation forensics, direct-conversion X-ray de-
665 tection, and direct fast-neutron conversion using the hydrogen-rich MHyPbCl_3 com-
666 position. Each device is characterized through electrical transport measurements,
667 Monte Carlo simulation, and experimental irradiation at a research reactor facility.

668 The third objective is to implement and validate a radiation-tolerant FPGA-based
669 digital monitoring system suitable for nuclear plant instrumentation applications.
670 This includes design and testing of a fault-tolerant flip-flop scrubbing register (FFSR)
671 architecture, integration of TMR voter logic, and validation of the combined system
672 against a digital plant simulation environment representing nuclear I&C functions.

673 The remainder of this dissertation is organized as follows. Chapter 2 provides
674 the perovskite-focused background necessary for the materials and device investiga-
675 tions, encompassing perovskite crystal structure, radiation interaction physics, ion
676 migration mechanisms, and the current state of perovskite-based radiation detectors.
677 Chapter 3 provides the electronics background, covering radiation effects on digital de-
678 vices, single-event effects in SRAM-based FPGAs, and the taxonomy of fault-tolerant
679 design approaches including triple modular redundancy and configuration scrubbing.
680 Chapter 4 presents the characterization of $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ perovskite solar cells un-
681 der extended gamma irradiation. Chapter 5 addresses the voltage-induced instability
682 problem in two-dimensional perovskite detectors and the 5F-PEAI surface passivation
683 strategy developed to overcome it. Chapter 6 describes the MAPbI_3 -hBN compos-
684 ite detector and its characterization for X-ray detection. Chapter 7 presents the

685 MHyPbCl₃ fast neutron detector, from crystal synthesis through beam-facility test-
686 ing and gamma discrimination analysis. Chapter 8 describes the radiation-tolerant
687 FPGA monitoring system, including the FFSR design, TMR integration, and digital
688 plant simulation validation. Chapter 9 provides overall conclusions and recommen-
689 dations for future work.

Chapter 2: Background: Halide Perovskite Semiconductors for Radiation Detection

The research reported in Chapters 4–?? rests on three intersecting bodies of knowledge: the crystal chemistry and electronic structure of halide perovskites, the physics governing how ionizing radiation deposits energy in matter, and the device physics that determines how efficiently that deposited energy is converted into a measurable electrical signal. This chapter develops each body of knowledge in sufficient depth to motivate the experimental choices made in later chapters and to place the results in the context of the broader literature on semiconductor radiation detectors. The FPGA-based systems work is treated separately in Chapter 3.

2.1 The Perovskite Crystal Structure and Material Families

2.1.1 The ABX_3 Archetype

The term “perovskite” derives from the mineral calcium titanate ($CaTiO_3$), named for the Russian mineralogist Lev von Perovski and first synthesized in the nineteenth century [102]. The defining structural motif is a three-dimensional network of corner-sharing BX_6 octahedra in which a larger A-site cation occupies the 12-coordinate cuboctahedral void at the unit-cell body-center (in the ideal cubic setting, space group $Pm\bar{3}m$). The versatility of this archetype arises from the wide range of chemistries

that can be accommodated without destroying the framework: in the oxide family, B-sites range from Ti^{4+} to Pb^{4+} , while in the halide family addressed here, B is almost exclusively Pb^{2+} (or Sn^{2+}) and X is a monovalent halide (I^- , Br^- , or Cl^-).

The stability of a given ABX_3 composition is conveniently predicted by the Goldschmidt tolerance factor,

$$t = \frac{r_A + r_X}{\sqrt{2}(r_B + r_X)}, \quad (2.1)$$

where r_i denotes the ionic radius of species i [30]. Perovskite formation is generally favored for $0.80 \leq t \leq 1.05$; values close to unity favor the ideal cubic structure, while smaller values drive cooperative octahedral tilting that lowers the symmetry to tetragonal or orthorhombic phases. Kieslich et al. extended this framework to hybrid organic-inorganic systems by defining effective radii for polyatomic organic cations from the root-mean-square displacement of their hydrogen envelope [46], enabling quantitative phase stability predictions for the methylammonium and formamidinium compounds central to this work.

A complementary criterion is the octahedral factor $\mu = r_B/r_X$, which must satisfy $0.414 \leq \mu \leq 0.732$ for the B cation to fit stably inside the halide octahedron. Pb^{2+} ($r = 1.19 \text{ \AA}$) paired with I^- ($r = 2.20 \text{ \AA}$) gives $\mu = 0.54$, comfortably within the stable range and consistent with the broad tunability of the Pb-halide framework.

2.1.2 Three-Dimensional Structures

Methylammonium lead iodide (MAPbI_3). The most extensively studied hybrid halide perovskite, MAPbI_3 (methylammonium = CH_3NH_3^+ , MA^+), adopts a cubic structure above 327 K, a tetragonal structure between 162 K and 327 K, and an orthorhombic structure below 162 K [7]. At room temperature, the tetragonal phase

(space group $I4/mcm$) is therefore the operative structure for detector applications. MAPbI₃ has a direct bandgap of approximately 1.55 eV and a high optical absorption coefficient ($> 10^4 \text{ cm}^{-1}$ above the bandgap), along with the excellent charge transport described in Section 2.4.2. Large single crystals with centimeter-scale dimensions can be grown from solution by inverse temperature crystallization [83], making bulk detector fabrication feasible. The principal weakness of MAPbI₃ for detector applications is the light A-site cation: the low atomic masses of C, N, and H contribute minimally to gamma-ray stopping power, and the modest hydrogen density in the cubic CH₃NH₃⁺ cation limits fast-neutron response. The MAPbI₃-hBN composite detector described in Chapter ?? addresses the neutron sensitivity limitation by incorporating high- n boron-containing nanoflakes within a MAPbI₃ matrix.

Cesium lead bromide (CsPbBr₃). Replacing the organic MA⁺ cation with the monoatomic Cs⁺ ($r = 1.67 \text{ \AA}$) yields an all-inorganic perovskite with significantly improved thermal and photochemical stability [91]. CsPbBr₃ undergoes orthorhombic-to-tetragonal ($\sim 88 \text{ }^\circ\text{C}$) and tetragonal-to-cubic ($\sim 130 \text{ }^\circ\text{C}$) transitions, remaining in the orthorhombic phase at room temperature. Its bandgap of approximately 2.3 eV is wide enough to suppress intrinsic carrier excitation at 300 K, yielding resistivities exceeding $10^9 \text{ } \Omega \text{ cm}$ in single crystals [33] — a prerequisite for the low dark-current operation required in radiation counting. Most critically for gamma detection, the presence of both Cs ($Z = 55$) and Pb ($Z = 82$) gives CsPbBr₃ a calculated effective atomic number of approximately 65.9, far exceeding that of conventional detector materials such as silicon ($Z_{\text{eff}} = 14$) and approaching that of cadmium zinc telluride ($Z_{\text{eff}} \approx 49.1$) [25]. This high Z_{eff} makes CsPbBr₃ among the most promising semiconductor candidates for room-temperature gamma spectroscopy.

754 **Mixed cation $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$.** Pure formamidinium lead iodide (FAPbI_3 , FA^+
755 $= \text{CH}(\text{NH}_2)_2^+$) has a narrower bandgap of approximately 1.48 eV and superior thermal
756 stability relative to MAPbI_3 , but the desired black α -phase is metastable at room tem-
757 perature, spontaneously converting to the photoinactive yellow δ -phase [101]. Partial
758 substitution of FA^+ with Cs^+ (typically 5–15 mol%) suppresses this phase transition
759 by reducing the tolerance factor toward unity, entropy-stabilizing the cubic α -phase at
760 ambient conditions [111]. The composition $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ investigated in Chapter 4
761 exploits this stabilization while retaining the narrow bandgap and large absorption
762 cross-section of FAPbI_3 , enabling its use as a photon-responsive active layer in a
763 photovoltaic architecture repurposed as a radiation monitor.

764 2.1.3 Two-Dimensional Ruddlesden-Popper Structures

765 The three-dimensional perovskite framework can be sliced along specific crystal-
766 lographic planes and the resulting slabs separated by layers of large organic spacer
767 cations to form two-dimensional (2D) or quasi-2D structures. In the Ruddlesden-
768 Popper series the general formula is $(R)_2(\text{MA})_{n-1}\text{Pb}_n\text{X}_{3n+1}$, where R is a large or-
769 ganic cation that cannot fit the A-site of the 3D structure, n is the number of corner-
770 sharing PbX_6 octahedral layers between adjacent organic spacer bilayers, and the
771 ratio of inorganic to organic content increases with n [94]. In the limit $n \rightarrow \infty$ the
772 three-dimensional structure is recovered.

773 For the present work, the relevant composition is the $n = 3$ member $\text{PEA}_2\text{MA}_2\text{Pb}_3\text{I}_{10}$,
774 where PEA^+ denotes the phenylethylammonium cation ($\text{C}_6\text{H}_5(\text{CH}_2)_2\text{NH}_3^+$). The al-
775 ternating organic-inorganic layering creates a natural quantum-well electrostatic land-
776 scape in which charge is preferentially confined within the higher-dielectric inorganic

777 slabs. From a detector perspective, this layered architecture confers two important
 778 advantages over its 3D counterpart: the bulky PEA^+ layers act as physical barriers
 779 to halide vacancy migration, substantially suppressing the ion-transport instabilities
 780 detailed in Section 2.6; and the organic layers provide intrinsic passivation of the inor-
 781 ganic slab surfaces, reducing trap-state densities at the heterojunction interfaces [15].
 782 The price paid is a modest reduction in charge-carrier mobility relative to three-
 783 dimensional analogs, since carriers must traverse the organic-inorganic boundaries.
 784 Chapter ?? investigates whether surface-passivation strategies can recover sufficient
 785 transport quality to enable practical detector operation while retaining the stability
 786 advantages of the 2D architecture.

787 **2.1.4 Methylhydrazinium Lead Chloride (MHyPbCl_3)**

788 Methylhydrazinium lead chloride, MHyPbCl_3 ($\text{MHy}^+ = \text{CH}_3\text{NH}_2\text{NH}_2^+$), is a rela-
 789 tively recently reported member of the halide perovskite family in which the conven-
 790 tional monovalent cation is replaced by the methylhydrazinium ion [67]. This compo-
 791 sition is of particular interest for fast-neutron detection because of its exceptionally
 792 high hydrogen content: the $\text{CH}_3\text{NH}_2\text{NH}_2^+$ cation carries five hydrogen atoms, and the
 793 resulting crystallographic hydrogen density is approximately $3.79 \times 10^{22} \text{ H cm}^{-3}$ [73].
 794 This exceeds the hydrogen density of many purpose-built neutron moderator and
 795 detector materials. Fast neutrons (energies above $\sim 1 \text{ MeV}$) interact primarily with
 796 hydrogen nuclei via elastic scattering, transferring up to 100% of their kinetic energy
 797 to the recoil proton in a head-on collision; those recoil protons then ionize the de-
 798 tector medium in the same manner as a directly ionizing particle. The combination
 799 of high hydrogen density (for neutron moderation and proton production), moderate

bandgap (~ 3.0 eV, appropriate for room-temperature operation), and the semiconducting perovskite framework (for direct electronic readout) makes MHyPbCl₃ a compelling candidate for a compact direct-conversion fast-neutron detector. Chapter ?? presents the synthesis, characterization, and fast-neutron response measurements for this material.

2.2 Radiation Detection Fundamentals

2.2.1 Mechanisms of Ionizing Radiation Detection

A radiation detector converts the energy deposited by an ionizing particle or photon into a measurable quantity — in semiconductor detectors, this quantity is an electrical charge pulse whose amplitude is proportional to the deposited energy. The process proceeds in three stages: (1) the primary ionizing event, in which the incident radiation transfers energy to the detector medium, producing a spatial distribution of electron-hole pairs along the particle track or at the absorption site; (2) charge transport, in which the liberated carriers drift under the applied bias field toward the collection electrodes; and (3) signal induction and readout, in which the moving charges induce a current pulse on the electrodes that is amplified and digitized [48].

The mean energy required to create one electron-hole pair in a semiconductor is given by Klein’s empirical rule,

$$W \approx 2.67 E_g + 0.87 \text{ eV}, \quad (2.2)$$

where E_g is the bandgap energy [47]. For MAPbI₃ ($E_g \approx 1.55$ eV), this gives $W \approx 5.0$ eV per pair; for CsPbBr₃ ($E_g \approx 2.3$ eV), $W \approx 7.0$ eV per pair. These values compare favorably with conventional semiconductor detectors: silicon ($W = 3.6$ eV) and germanium ($W = 2.9$ eV) produce more pairs per unit energy, but those materials

require cryogenic cooling or high-purity growth technologies incompatible with cost-effective deployment. The higher W of perovskites is partially offset by their far greater stopping power per unit thickness (Section 2.4), so the total charge produced by a given photon energy can be comparable to or exceed that in a thinner silicon detector despite the higher pair creation energy.

2.2.2 Energy Resolution and Pulse Height Analysis

The ability to resolve closely spaced photon energies — and thereby identify specific radionuclides — is quantified by the energy resolution R , conventionally expressed as the fractional full-width at half-maximum (FWHM) of a spectral peak:

$$R = \frac{\Delta E_{\text{FWHM}}}{E_0}, \quad (2.3)$$

where E_0 is the incident photon energy. In an ideal semiconductor detector in which Poisson statistics govern pair production, the minimum achievable resolution is set by the Fano limit:

$$\Delta E_{\text{FWHM}}^{\text{Fano}} = 2.35 \sqrt{F \cdot W \cdot E_0}, \quad (2.4)$$

where F is the Fano factor ($F \approx 0.1$ – 0.2 for most semiconductors) [48]. In practice, the measured resolution is degraded by incomplete charge collection due to carrier trapping, electronic noise from the amplifier chain, and dielectric polarization from ion migration (the last factor being specific to halide perovskites and discussed in Section 2.6). The benchmark figure commonly cited in the perovskite detector literature is the 1.4% resolution at 662 keV (^{137}Cs) achieved by He et al. for a CsPbBr_3 single crystal coupled to a custom application-specific integrated circuit (ASIC) [33], which at the time of publication was the best room-temperature resolution demonstrated by any non-CZT semiconductor detector.

Energy-resolved radiation measurements are performed by connecting the detector preamplifier output to a pulse-height analyzer (PHA), which bins each charge pulse by its amplitude and accumulates a histogram — the pulse-height spectrum — whose peak positions encode incident photon energies and whose areas are proportional to the detected activity. The shaping time of the amplifier is a critical parameter: it must be long enough to integrate the slow hole component of the charge pulse (hole mobilities in halide perovskites are typically $5\text{--}10\times$ lower than electron mobilities) but short enough to avoid excessive pileup at practical count rates.

2.2.3 X-Ray Detector Sensitivity

For applications in medical and industrial X-ray imaging, the relevant figure of merit is not energy resolution but sensitivity S , defined as the charge collected per unit area per unit X-ray exposure:

$$S = \frac{\Delta Q}{A \cdot \Delta X} \quad [\mu\text{C Gy}^{-1} \text{ cm}^{-2}], \quad (2.5)$$

where ΔX is the absorbed dose rate. Sensitivity is controlled by the X-ray absorption efficiency (determined by the stopping power and detector thickness), the charge collection efficiency (determined by $\mu\tau$ products and applied field), and the dark current level (which sets the minimum detectable signal). Perovskite thick-film detectors fabricated from polycrystalline MAPbI_3 or MAPbBr_3 have demonstrated sensitivities exceeding $4,500 \mu\text{C Gy}^{-1} \text{ cm}^{-2}$ under 8 keV X-rays, more than 10^4 times higher than conventional amorphous selenium panels [100]. The high atomic number of the Pb sublattice is the primary driver of this extraordinary sensitivity at soft X-ray energies, where the photoelectric cross-section scales roughly as $Z^{4.5}$.

2.2.4 Neutron Detection and Discrimination

Neutron detection in semiconductor materials is indirect: neutrons carry no charge and interact negligibly with electrons, so they must first undergo a nuclear reaction or elastic scattering event that produces a charged secondary particle capable of ionizing the detector. Thermal neutrons (energies below ~ 0.5 eV) are most efficiently captured via exothermic reactions such as $^{10}\text{B}(\text{n},\alpha)^7\text{Li}$ and $^6\text{Li}(\text{n},\alpha)^3\text{H}$. Fast neutrons above ~ 1 MeV interact principally through elastic scattering from hydrogen nuclei, producing recoil protons with a continuous energy spectrum up to the full neutron energy. The detection efficiency for fast neutrons is proportional to the hydrogen areal density in the active medium, making hydrogen-rich materials intrinsically advantageous. The measured ^1H density in MHyPbCl_3 of $3.79 \times 10^{22} \text{ cm}^{-3}$ exceeds that of common polymer moderator materials and places this perovskite in a unique position as a combined moderating and detecting medium for fast neutrons [73].

Discrimination between neutron and gamma signals is accomplished by exploiting differences in the ionization density (linear energy transfer, LET) of the respective secondary particles. Recoil protons from fast-neutron elastic scattering and alpha particles from thermal capture reactions deposit energy far more densely than the Compton electrons produced by gamma photons of similar incident energy. In systems with sufficient time resolution, pulse-shape discrimination (PSD) can separate the fast and slow scintillation components that arise from these different ionization densities; in solid-state detectors, differences in carrier extraction dynamics under the high local ionization density of heavy charged particles can be exploited in a similar manner [48].

886 2.3 Wide Bandgap Semiconductors for Room-Temperature De- 887 tection

888 2.3.1 The Room-Temperature Constraint

889 The operating requirement most often overlooked in academic detector compar-
890 isons — yet most consequential for deployment in nuclear plants, field instruments,
891 and space systems — is that the detector must function reliably at room temperature
892 without active cooling. This single constraint eliminates two of the highest-performing
893 semiconductor detector technologies: germanium (HPGe) must be operated below
894 120 K to suppress thermally generated leakage current, requiring liquid-nitrogen de-
895 wares or mechanical cryocoolers that add cost, mass, and failure modes incompatible
896 with unattended monitoring [48]. Silicon drift detectors, while operable at modest
897 Peltier cooling (-20 to -40 °C), still suffer from displacement damage under high
898 neutron fluences that degrade their spectroscopic performance irreversibly.

899 The root cause of the temperature sensitivity in narrow-gap semiconductors is the
900 intrinsic carrier concentration n_i , which increases exponentially with temperature as:

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2kT}\right), \quad (2.6)$$

901 where N_c and N_v are the effective density of states in the conduction and valence
902 bands, respectively, and k is Boltzmann’s constant [92]. At room temperature ($T =$
903 300 K), germanium ($E_g = 0.67$ eV) has $n_i \approx 2.4 \times 10^{13}$ cm $^{-3}$, producing a leak-
904 age current density many orders of magnitude larger than the signal currents from
905 typical radiation fields. Silicon ($E_g = 1.12$ eV) fares considerably better with $n_i \approx$

906 $1.5 \times 10^{10} \text{ cm}^{-3}$, but displacement damage from neutron irradiation introduces com-
 907 pensating defects that effectively raise the intrinsic carrier density and degrade the
 908 depletion voltage over time.

909 For a semiconductor to operate as a room-temperature detector without active
 910 cooling or impractically high reverse bias, n_i must be sufficiently low that the thermal
 911 generation current is smaller than the smallest radiation-induced signal of interest. A
 912 widely cited rule of thumb requires $n_i \lesssim 10^7 \text{ cm}^{-3}$, which for silicon-like band struc-
 913 ture translates to a minimum bandgap of approximately 1.5 eV [71]. This threshold
 914 defines the wide bandgap semiconductor regime relevant to room-temperature radia-
 915 tion detection.

916 **2.3.2 Figures of Merit for Wide Bandgap Detector Materials**

917 The suitability of a wide bandgap semiconductor for radiation detection is char-
 918 acterized by several interrelated figures of merit beyond the bandgap itself:

919 **Resistivity** (ρ , $\Omega \text{ cm}$). High bulk resistivity minimizes the dark current $I_{\text{dark}} =$
 920 $V/R = VA/(L\rho)$ that flows under the applied bias field and sets the noise floor.
 921 For spectroscopic applications, resistivities above $10^9 \Omega \text{ cm}$ are typically required; for
 922 imaging applications where high flux dominates shot noise, values above $10^7 \Omega \text{ cm}$ may
 923 be acceptable. Wide bandgap materials achieve high resistivity by a combination of
 924 large E_g (reducing n_i) and the absence of shallow donor or acceptor impurities in the
 925 bandgap. CsPbBr₃ single crystals have been measured at $\rho > 10^{10} \Omega \text{ cm}$ at room
 926 temperature [33], exceeding the resistivity of commercial CdZnTe by approximately
 927 one order of magnitude.

928 **Mobility-lifetime product** ($\mu\tau$, $\text{cm}^2 \text{V}^{-1}$). As derived in Section 2.4.2, $\mu\tau$ di-
929 rectly governs the mean free path of carriers between generation and trapping events.
930 Wide bandgap semiconductors generally have lower carrier mobilities than narrow-
931 gap materials (because heavier effective masses accompany larger bandgaps in many
932 crystal systems), but the lifetime can be extraordinarily long in materials with low
933 defect densities, partially compensating. The target for spectroscopic detectors is
934 $\mu_e\tau_e \gtrsim 10^{-3} \text{ cm}^2 \text{V}^{-1}$; for hole collection, the equivalent target is often relaxed to
935 $10^{-4} \text{ cm}^2 \text{V}^{-1}$ when single-polarity collection schemes are employed [53].

936 **Effective atomic number** (Z_{eff}). For gamma and X-ray detection, Z_{eff} deter-
937 mines the photoelectric stopping power, which scales as $Z^{4.5}$. Wide bandgap semicon-
938 ductors based on heavy elements (Pb, Bi, Hg, Cd, Te) combine the thermal benefits
939 of a large bandgap with high stopping power in a way that narrow-gap elemental
940 semiconductors cannot achieve. Silicon ($Z = 14$) requires centimeters of active thick-
941 ness to stop 662 keV photons, while a few millimeters of CsPbBr_3 ($Z_{\text{eff}} = 65.9$) or
942 CdZnTe ($Z_{\text{eff}} = 49.1$) provide equivalent absorption [59].

943 **Pair creation energy** (W , eV). As noted in Eq. 2.2, W is approximately pro-
944 portional to E_g . Wide bandgap materials therefore produce fewer electron-hole pairs
945 per unit of deposited energy than narrow-gap ones, increasing the statistical (Fano)
946 contribution to the energy resolution. This is the principal penalty paid for room-
947 temperature operability: it must be offset by large $\mu\tau$ products that ensure complete
948 charge collection, since any additional variance from trapping further degrades the
949 resolution beyond the Fano limit.

2.3.3 The Wide Bandgap Material Landscape

Table 2.1 contextualizes the halide perovskites investigated in this dissertation within the wider landscape of wide bandgap semiconductor detector materials.

Table 2.1: Selected properties of wide bandgap semiconductor detector materials at 300 K. CZT = cadmium zinc telluride $\text{Cd}_{0.9}\text{Zn}_{0.1}\text{Te}$. Data compiled from [71, 33, 25, 48].

Material	E_g (eV)	Z_{eff}	ρ ($\Omega\text{ cm}$)	$\mu_e\tau_e$ ($\text{cm}^2\text{ V}^{-1}$)	W (eV)	Best R_{662}
Si	1.12	14	$> 10^4$	> 1	3.6	$< 0.1\%$ (cryo)
CdTe	1.50	49.5	10^9	3×10^{-3}	4.4	1.7%
CZT	1.57	49.1	10^{10}	10^{-2}	4.6	0.5%
GaAs	1.42	31.6	10^7	10^{-5}	4.2	$\sim 2\%$
4H-SiC	3.26	14	$> 10^{12}$	10^{-6}	8.0	$\sim 1\%^*$
HgI ₂	2.13	62.2	10^{13}	10^{-4}	4.2	0.5%
CsPbBr ₃	2.30	65.9	$> 10^{10}$	1.3×10^{-2}	7.0	1.4%
MAPbI ₃	1.55	51.4	10^8	8×10^{-3}	5.0	$\sim 6\%$

*For alpha particles; gamma spectroscopy performance is limited by the low Z_{eff} .

Several observations emerge from this comparison. CdZnTe has long occupied the upper tier of room-temperature gamma spectrometers, combining a near-ideal bandgap for thermal leakage suppression, high Z_{eff} , and a $\mu_e\tau_e$ product sufficient for millimeter-to-centimeter-scale crystals — but its dominance rests entirely on the extraordinary effort invested in Bridgman crystal growth and post-growth processing that has accumulated over three decades. HgI₂ offers the highest Z_{eff} and resistivity of the conventional materials but suffers from mechanical fragility and a phase transition at 127 °C that has limited commercial uptake. Silicon carbide (4H-SiC)

is exceptional in its radiation hardness and thermal tolerance but its low Z_{eff} confines it to charged-particle and neutron applications where stopping power from the heavy-element sublattice is not required [71].

CsPbBr_3 occupies a remarkably favorable position in this landscape. Its Z_{eff} of 65.9 is the highest of any material in common laboratory study, its $\mu_e\tau_e$ is competitive with CdZnTe despite far simpler solution-based crystal growth, and its resistivity exceeds that of CdZnTe by an order of magnitude. The 1.4% energy resolution at 662 keV achieved by He et al. is already within a factor of three of the best CdZnTe single-element detectors, and it was achieved after only a few years of focused development [33]. The remaining gap is attributable primarily to the ion-migration polarization effect (Section 2.6), which introduces a time-dependent component to charge collection efficiency not present in purely electronic semiconductors.

2.3.4 Why Bandgap Engineering Matters: The Perovskite Advantage

The halide perovskite family is unique among wide bandgap detector candidates in that the bandgap is continuously tunable over the range 1.2–3.5 eV by compositional substitution at all three crystallographic sites, without changing the crystal structure or growth method. Replacing iodide with bromide shifts the bandgap from 1.55 eV (MAPbI_3) to 2.3 eV (MAPbBr_3); substituting Pb^{2+} with Sn^{2+} narrows the gap to ~ 1.2 eV; replacing MA^+ with the larger FA^+ cation narrows the gap marginally to ~ 1.48 eV [91]. This tunability enables the optimization of three competing requirements simultaneously: (1) E_g large enough to suppress thermal leakage at the maximum operating temperature; (2) E_g small enough to keep W — and therefore

the Fano noise — acceptably low; and (3) Z_{eff} high enough for efficient photon stopping in the energy range of interest. No other material family offers all three degrees of freedom within a single, solution-processable crystal system, which is the central motivation for investigating multiple perovskite compositions across the chapters that follow.

2.4 Radiation Interaction with High- Z Semiconductors

2.4.1 Photon Interactions

Three mechanisms govern the interaction of gamma and X-ray photons with matter: the photoelectric effect, Compton scattering, and pair production. Each dominates over a different energy range and scales differently with the atomic number of the absorber, making the choice of detector material central to the detection efficiency for a given source energy.

The *photoelectric effect* is dominant below approximately 100 keV in high- Z materials. An incident photon is completely absorbed by an inner-shell electron, which is ejected with kinetic energy $E_e = h\nu - E_b$, where E_b is the binding energy of that shell. The photoelectric cross-section scales strongly with atomic number and inversely with photon energy as $\sigma_{\text{pe}} \propto Z^{4.5}/E^{3.5}$, so high- Z semiconductors provide orders-of-magnitude higher photoelectric absorption than silicon or germanium for the same detector volume. The complete energy deposition of the photoelectric process produces the full-energy peak in the pulse-height spectrum, making it the principal mechanism for photon spectroscopy.

Compton scattering dominates for photon energies between approximately 100 keV and several MeV in heavy materials. An incident photon transfers a fraction of its

energy to a loosely bound or free electron, with the scattered photon carrying the remainder. Because only partial energy is deposited, Compton events populate a continuum below the full-energy peak rather than contributing to it. The Compton cross-section is proportional to Z , so high- Z absorbers still offer a substantial geometric advantage, but the spectroscopic information is contained only in the photopeak fraction.

Pair production becomes significant above the threshold energy of 1.022 MeV and dominates at multi-MeV photon energies. The photon is converted into an electron-positron pair in the Coulomb field of the nucleus; the 511-keV annihilation photons from the subsequent positron annihilation appear as characteristic escape peaks in the spectrum. The pair-production cross-section scales as Z^2 .

The combined effect of these mechanisms is captured by the mass attenuation coefficient μ/ρ , tabulated as a function of energy for all elements by the National Institute of Standards and Technology (NIST XCOM database). For a compound material, the effective atomic number is calculated as:

$$Z_{\text{eff}} = \left(\sum_i w_i Z_i^n \right)^{1/n}, \quad (2.7)$$

where w_i is the weight fraction of element i and the exponent $n \approx 3.5$ is chosen to match the photoelectric cross-section scaling [59]. Applying this formula to CsPbBr₃ (Cs: $Z = 55$, Pb: $Z = 82$, Br: $Z = 35$; stoichiometric weight fractions of 0.229, 0.449, and 0.322, respectively) yields $Z_{\text{eff}} \approx 65.9$, substantially higher than silicon ($Z_{\text{eff}} = 14$), germanium (32), or even CdZnTe ($Z_{\text{eff}} \approx 49.1$) [25].

1027 2.4.2 Charge Transport and the Hecht Equation

1028 Even if radiation deposits charge in the bulk of a detector, that charge must
 1029 be transported to the collection electrodes without recombination or trapping if a
 1030 proportional signal is to be measured. The quality of charge transport is characterized
 1031 by the mobility-lifetime product $\mu\tau$ for electrons and holes separately. Carrier *mobility*
 1032 μ ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$) quantifies the drift velocity per unit electric field; carrier *lifetime* τ
 1033 (s) is the mean time before a carrier is irreversibly trapped at a deep defect and
 1034 removed from the circuit. The product $\mu\tau$ has units of $\text{cm}^2 \text{V}^{-1}$ and represents the
 1035 mean distance a carrier drifts per unit of applied field before being trapped; a large
 1036 $\mu\tau$ indicates that carriers can traverse a thick crystal without significant charge loss.

1037 The fractional charge collection for a detector of thickness L under applied field E
 1038 can be derived rigorously for a planar single-carrier case. For a general double-carrier
 1039 detector with the interaction site at depth x from the cathode, the Hecht equation
 1040 gives:

$$Q(x) = Q_0 \left[\frac{\mu_e \tau_e E}{L} (1 - e^{-(L-x)/\mu_e \tau_e E}) + \frac{\mu_h \tau_h E}{L} (1 - e^{-x/\mu_h \tau_h E}) \right], \quad (2.8)$$

1041 where $Q_0 = eN_0$ is the charge generated by the initial ionizing event [34]. For uni-
 1042 form irradiation, averaging Eq. 2.8 over x from 0 to L gives the mean charge col-
 1043 lection efficiency (CCE), which approaches unity only when both $\mu_e \tau_e E/L \gg 1$ and
 1044 $\mu_h \tau_h E/L \gg 1$. In practice this requires either very high applied fields, very thin
 1045 detectors, or very large $\mu\tau$ products.

1046 Halide perovskites have demonstrated $\mu\tau$ products competitive with those of ma-
 1047 ture compound semiconductors. Single-crystal MAPbI₃ exhibits $\mu_e \tau_e \approx 8 \times 10^{-3} \text{ cm}^2 \text{V}^{-1}$
 1048 and $\mu_h \tau_h \approx 3 \times 10^{-3} \text{ cm}^2 \text{V}^{-1}$ [83], values that in a 1-mm-thick crystal at a modest field

1049 of 100 V cm^{-1} yield a mean electron drift length of $\sim 8 \text{ mm}$ — far exceeding the crys-
1050 tal thickness. CsPbBr₃ single crystals have achieved $\mu_e \tau_e$ up to $1.34 \times 10^{-2} \text{ cm}^2 \text{ V}^{-1}$,
1051 supporting the 1.4% energy resolution reported by He et al. [33]. These values are
1052 within a factor of two to three of CdZnTe ($\mu_e \tau_e \sim 10^{-2} \text{ cm}^2 \text{ V}^{-1}$), a material that
1053 has required decades of crystal growth refinement to achieve its current performance
1054 levels.

1055 The hole transport is typically the limiting factor in halide perovskites, as in most
1056 compound semiconductors. The hole $\mu\tau$ in both MAPbI₃ and CsPbBr₃ is typically
1057 $5\text{--}10\times$ lower than the electron value, leading to a characteristic single-polarity col-
1058 lection signature in which events occurring near the cathode (from which electrons
1059 must traverse the full crystal thickness) are preferentially detected. Electronic tech-
1060 niques such as single-carrier collection (using a Frisch collar or virtual Frisch ring)
1061 and waveform analysis can mitigate poor hole collection [53], and perovskite detectors
1062 are amenable to these approaches.

1063 2.5 Schottky Diode Contacts in Semiconductor Detectors

1064 2.5.1 Metal-Semiconductor Junction Physics

1065 When a metal is brought into contact with a semiconductor, charge redistributes
1066 at the interface until the Fermi levels align, creating a built-in potential difference
1067 and a depletion region in the semiconductor. For a metal in contact with an n-type
1068 semiconductor, the Schottky barrier height is:

$$\phi_B = \phi_m - \chi_s, \quad (2.9)$$

1069 where ϕ_m is the metal work function and χ_s is the electron affinity of the semicon-
 1070 ductor [92]. The depletion region extends a distance W into the semiconductor:

$$W = \sqrt{\frac{2\varepsilon\varepsilon_0(V_{\text{bi}} - V)}{eN_D}}, \quad (2.10)$$

1071 where ε is the static dielectric constant, V_{bi} is the built-in voltage, V is the applied
 1072 reverse bias, and N_D is the donor concentration. The rectifying current-voltage rela-
 1073 tionship of the Schottky diode follows the thermionic emission model:

$$J = J_0 (e^{qV/nkT} - 1), \quad J_0 = A^{**}T^2 e^{-q\phi_B/kT}, \quad (2.11)$$

1074 where A^{**} is the effective Richardson constant and n is the ideality factor [92].

1075 In contrast, an *ohmic* contact is formed when the metal work function is matched
 1076 to the Fermi level of the semiconductor such that no barrier to majority-carrier flow
 1077 exists. Ohmic contacts allow free exchange of majority carriers with the external
 1078 circuit and are necessary on the collection side of the detector to avoid field distortion
 1079 near the electrode.

1080 2.5.2 Application to Halide Perovskite Detectors

1081 The asymmetric contact architecture — one ohmic and one Schottky electrode
 1082 — has become the standard configuration for halide perovskite radiation detectors
 1083 because it simultaneously achieves low dark current and efficient charge collection [99].
 1084 Under reverse bias, the Schottky junction is depleted, suppressing majority-carrier
 1085 injection from the blocking contact; minority carriers generated by ionizing radiation
 1086 within the depletion region are swept by the electric field toward the appropriate
 1087 electrode. Because the dark current is thermally limited by the Schottky barrier
 1088 rather than set by the bulk resistivity, even moderate barrier heights of 0.5–0.8 eV

1089 can reduce the dark current density by several orders of magnitude relative to the
1090 ohmic-contact geometry.

1091 For MAPbI₃-based detectors, gold (Au, $\phi_m = 5.1$ eV) forms a near-ohmic contact
1092 on the p-type surface, while chromium (Cr, $\phi_m = 4.5$ eV), gallium, or indium con-
1093 tacts on the opposite face exhibit Schottky character [99]. For CsPbBr₃, the typical
1094 asymmetric pair is a carbon paste or Au ohmic contact on one face and a Ga or
1095 indium-gallium eutectic Schottky contact on the other. The net result is a device
1096 structure functionally analogous to a *p-i-n* junction in which the intrinsic region (the
1097 semi-insulating bulk crystal) contains the active detector volume, and the contact
1098 asymmetry enforces unipolar drift under bias [33].

1099 For the 2D perovskite devices studied in Chapter ??, the layered crystal architec-
1100 ture introduces an additional consideration: charge transport perpendicular to the
1101 organic spacer layers (the out-of-plane direction, which is the relevant direction for
1102 planar detector geometry) is more resistive than in-plane transport, and the contact
1103 interface with the topmost organic spacer layer must be carefully engineered to avoid a
1104 secondary barrier that would impede collection. Surface passivation using Lewis-base
1105 molecules that coordinate to surface Pb²⁺ sites can simultaneously reduce trap-state
1106 density and tune the contact work function alignment.

1107 2.6 Ion Migration in Halide Perovskites

1108 2.6.1 Origins and Mechanisms

1109 The same structural softness that enables solution-processable crystal growth from
1110 precursor salts also makes halide perovskites susceptible to ionic transport under ap-
1111 plied fields — a property that distinguishes them fundamentally from conventional

covalent or ionic semiconductors such as silicon or CdZnTe. Density functional theory calculations and activation energy measurements converge on halide vacancy migration as the dominant ionic conduction mechanism, with activation energies of approximately 0.1–0.6 eV depending on the halide and A-site cation [21]. For comparison, the thermal energy at room temperature ($kT \approx 26$ meV) is sufficient to drive significant vacancy hopping across these barriers on detector-relevant timescales, particularly under the electric fields of 10–1,000 V cm⁻¹ routinely applied to semiconductor detectors.

The migration pathway for iodide vacancies in MAPbI₃ proceeds via nearest-neighbor hopping through the faces of the PbI₆ octahedra, with a computed activation barrier of approximately 0.58 eV [21]. Bromide vacancies in CsPbBr₃ have a somewhat higher barrier of ~ 0.74 eV, consistent with the greater thermal stability of that compound. Chloride vacancies, which are the operative mobile species in MHPbCl₃, have not been as extensively studied computationally, but experimental activation energies inferred from temperature-dependent dark current measurements suggest similar or slightly higher barriers than for iodide.

2.6.2 Consequences for Detector Performance

Ion migration manifests in detector operation through three interrelated phenomena. First, under sustained bias, mobile ions drift to the electrodes and accumulate, establishing an internal dipole that *screens* the applied field and reduces the effective field within the crystal bulk. This polarization effect depresses charge collection efficiency over time and produces the characteristic dark current spike followed by slow decay observed in the current transient of a freshly biased perovskite crystal [33].

1135 Second, the accumulation of ions at the electrode interfaces modifies the contact bar-
 1136 rier height, causing the rectification ratio and dark current to evolve continuously
 1137 during operation — a direct threat to the baseline stability required in nuclear in-
 1138 strumentation. Third, if the bias polarity is reversed (as during AC-coupled readout
 1139 or impedance spectroscopy measurements), the accumulated ion profiles transiently
 1140 drive the internal field in the opposite direction to the intended collection field, pro-
 1141 ducing severe measurement artifacts.

1142 These effects are not merely theoretical concerns: in a systematic study of CsPbBr₃
 1143 single-crystal detectors, Pan et al. demonstrated that the charge collection efficiency
 1144 under a fixed field of 200 V cm⁻¹ dropped by approximately 30% over the first 100 sec-
 1145 onds of bias application before reaching a new quasi-equilibrium dominated by ionic
 1146 rather than electronic currents [72]. The voltage-induced degradation was partially
 1147 reversible upon removal of bias, with recovery occurring on timescales of minutes to
 1148 hours — consistent with thermal relaxation of the accumulated ion profiles [112].

1149 2.6.3 Mitigation Strategies

1150 Several strategies have been demonstrated to suppress or circumvent ion-migration
 1151 instability in perovskite detectors. *2D spacer layers* (the Ruddlesden-Popper architec-
 1152 ture) interpose wide-gap organic bilayers at regular intervals through the crystal thick-
 1153 ness, acting as electrostatic barriers to halide vacancy drift while preserving charge
 1154 transport via tunneling or thermally activated hopping [94]. *Surface passivation* with
 1155 Lewis-base ligands such as phenethylammonium iodide (PEAI), methylammonium
 1156 chloride, or thiophene-2-carboxylic acid coordinates unsaturated surface Pb²⁺ sites
 1157 that would otherwise serve as ion-migration nucleation points and electronic trap

states [15]. *All-inorganic compositions* such as CsPbBr₃ exhibit significantly slower ion drift than their hybrid organic-inorganic analogs because the rigid Cs⁺ sublattice provides fewer low-energy migration pathways. Finally, *low-temperature operation* reduces the ionic conductivity exponentially (Arrhenius behavior) and has been used to virtually eliminate polarization artifacts in experimental detector characterizations, though at the cost of the room-temperature operability that is one of perovskite’s primary advantages [33]. Chapter ?? investigates the combination of 2D architecture with chemical surface passivation as a room-temperature solution to the voltage-induced instability problem.

2.7 Prior Art in Perovskite Radiation Detectors

2.7.1 Early Single-Crystal Gamma Detectors

The first demonstration of halide perovskite single crystals as room-temperature radiation detectors appeared in a pair of landmark papers in 2016. Wei et al. grew large MAPbI₃ single crystals by inverse temperature crystallization and contacted them in an asymmetric Au/perovskite/Cr geometry; at 662 keV they observed a resolved ¹³⁷Cs photopeak with an energy resolution of approximately 6.8% [99]. While this was inferior to the best CdZnTe devices, the demonstration that a solution-processed hybrid perovskite could rival polycrystalline CdZnTe — at a fraction of the material cost and with far simpler growth conditions — generated immediate and sustained interest. Concurrently, Yakunin et al. reported sensitivity values exceeding 10³ μC Gy⁻¹ cm⁻² for MAPbI₃ thick-film X-ray detectors, establishing the perovskite platform for medical and industrial X-ray imaging [110].

1180 **2.7.2 CsPbBr₃ and the Resolution Record**

1181 The shift from hybrid to all-inorganic CsPbBr₃ was motivated by the thermal
1182 and photo-instability of MAPbI₃, which loses iodide at temperatures above $\sim 85^\circ\text{C}$
1183 and degrades rapidly in ambient humidity. Pan et al. demonstrated in 2019 that
1184 centimeter-scale CsPbBr₃ single crystals grown by the Bridgman method exhibit a
1185 resistivity exceeding $10^9\ \Omega\text{cm}$, $\mu_e\tau_e$ products above $10^{-2}\ \text{cm}^2\text{V}^{-1}$, and a 662 keV en-
1186 ergy resolution of approximately 3.9% [72]. The subsequent integration of CsPbBr₃
1187 detectors with custom-designed ASICs — rather than laboratory charge-sensitive am-
1188 plifiers — addressed the electronic noise limitation and enabled the record 1.4% res-
1189 olution at 662 keV reported by He et al. in 2021 [33], subsequently approached and
1190 extended by other groups employing refined crystal growth and improved contact
1191 engineering [27].

1192 **2.7.3 X-Ray Detection and Imaging**

1193 Polycrystalline thick films of MAPbI₃, MAPbBr₃, and CsPbI₂Br have been investi-
1194 gated extensively as direct-conversion X-ray detector arrays for applications in digital
1195 radiography and computed tomography. The high atomic number of the Pb sublattice
1196 confers exceptional sensitivity at soft X-ray energies (below $\sim 30\ \text{keV}$), where the
1197 photoelectric cross-section makes lead-bearing compounds intrinsically more absorb-
1198 ing than the amorphous selenium currently deployed in flat-panel detectors. Wei et
1199 al. reported a sensitivity of $4,500\ \mu\text{C Gy}^{-1}\text{cm}^{-2}$ for a 2-mm-thick solution-processed
1200 MAPbI₃ film under an 8 keV X-ray beam [100], and subsequent optimization of film
1201 deposition and contact passivation has pushed sensitivities above $10^4\ \mu\text{C Gy}^{-1}\text{cm}^{-2}$

1202 in several laboratories. The principal challenges remaining for commercial deploy-
1203 ment are long-term operational stability (particularly under the high photon fluences
1204 of medical imaging), the toxicity of lead-bearing materials in a clinical environment,
1205 and the uniformity of thick-film deposition over large detector areas.

1206 **2.7.4 Neutron Detection**

1207 Neutron detection by perovskite materials has been demonstrated via two com-
1208plementary approaches: neutron-capture via the incorporation of ^{10}B or ^6LiI into the
1209 perovskite matrix, and direct fast-neutron detection via hydrogen-recoil in hydrogen-
1210 rich perovskites. The composite approach was demonstrated in this group’s own prior
1211 work: MAPbI_3 crystals incorporating hexagonal boron nitride (hBN) nanoflakes were
1212 shown to exhibit a measurable response to thermal neutrons, with the neutron signal
1213 distinguished from the ambient gamma background by pulse-shape analysis [73]. The
1214 MHyPbCl_3 results reported in Chapter ?? represent the first systematic investigation
1215 of a perovskite designed from the outset for high hydrogen content, aiming to max-
1216 imize the fast-neutron elastic-scatter cross-section in a single-phase semiconducting
1217 medium. The availability of a structurally well-characterized perovskite with both
1218 high H density and a bandgap suitable for room-temperature electronic detection
1219 opens a path toward compact direct-conversion fast-neutron detectors that do not
1220 require external moderators or ^3He -filled proportional tubes.

1221 2.7.5 Mixed-Cation and 2D Detector Architectures

1222 The instability of single-phase 3D perovskites under sustained bias has driven
1223 interest in compositionally engineered architectures designed to suppress ion migra-
1224 tion while retaining the favorable electronic properties of the perovskite host. Two-
1225 dimensional Ruddlesden-Popper compounds were first applied to radiation detection
1226 by Xu et al., who demonstrated that $\text{PEA}_2\text{MA}_3\text{Pb}_4\text{I}_{13}$ ($n = 4$) single crystals exhib-
1227 ited suppressed polarization current compared to 3D MAPbI_3 under identical bias
1228 conditions, attributed to the PEA^+ barriers impeding halide vacancy drift [109]. The
1229 compositions studied in Chapter ?? extend this concept to the $n = 3$ compound
1230 $\text{PEA}_2\text{MA}_2\text{Pb}_3\text{I}_{10}$, where the additional organic layer density provides even stronger
1231 suppression of ion migration, and investigates the interaction between the 2D ar-
1232 chitecture and chemical surface passivation as a dual strategy for achieving stable,
1233 low-dark-current detector operation.

1234 2.8 Summary

1235 This chapter has established the conceptual framework necessary to interpret the
1236 experimental results reported in Chapters 4 through ??. The ABX_3 halide perovskite
1237 framework accommodates a diverse family of compositions spanning 3D, quasi-2D,
1238 and layered architectures, with A-site chemistry tunable from inorganic Cs^+ through
1239 hybrid organic-inorganic MA^+ and FA^+ to the hydrogen-rich methylhydrazinium
1240 cation of MHyPbCl_3 . The physics of semiconductor radiation detectors connects
1241 material properties — bandgap, effective atomic number, $\mu\tau$ product — to measur-
1242 able figures of merit such as energy resolution, sensitivity, and neutron detection
1243 efficiency via the Hecht equation and Klein’s pair-creation rule. The Schottky diode

1244 contact architecture suppresses dark current to levels compatible with radiation count-
1245 ing without sacrificing charge collection, provided that ion migration is controlled by
1246 compositional engineering or surface passivation. The problem of voltage-induced
1247 instability driven by halide vacancy migration is identified as the central unsolved
1248 challenge for room-temperature perovskite detector operation, motivating the 2D and
1249 passivation strategies pursued in the experimental chapters that follow.

Chapter 3: Background: Radiation Effects on Digital Electronics and FPGA-Based Instrumentation

The deployment of field-programmable gate arrays (FPGAs) in nuclear instrumentation and control (I&C) systems represents a convergence of two historically separate engineering disciplines: the radiation effects community, which has characterized and mitigated the response of microelectronics to ionizing radiation primarily in the context of space and defense applications, and the nuclear industry, which is confronting the simultaneous pressures of fleet aging, digital modernization, and cost reduction described in Chapter ???. This chapter provides the technical background required to understand the FPGA-based monitoring system developed in Chapter ??. It begins with the radiation environments of concern, then develops the physical mechanisms by which ionizing particles damage or disturb microelectronic devices, surveys the taxonomy of single-event effects (SEEs) with emphasis on those most relevant to SRAM-based FPGAs, describes the fault-tolerance architectures used to mitigate those effects, and concludes with the testing methodologies used to characterize and qualify devices for radiation environments.

1266 3.1 Radiation Environments for Digital Electronics

1267 Digital electronics encounter ionizing radiation in three principal deployment con-
1268 texts relevant to this work: the near-Earth space environment, the nuclear reactor
1269 plant environment, and the terrestrial cosmic-ray environment at ground level. Each
1270 environment differs in particle species, energy spectra, and dose rates, and each im-
1271 poses distinct qualification requirements.

1272 3.1.1 The Space Radiation Environment

1273 The space environment exposes electronics to three primary sources of energetic
1274 particles. *Trapped protons and electrons* in the Van Allen belts (primarily 1–10 MeV
1275 protons in the inner belt, up to several hundred MeV in the South Atlantic Anomaly)
1276 produce continuous total ionizing dose accumulation and displacement damage that
1277 degrades device performance over mission lifetime [74]. *Galactic cosmic rays* (GCRs)
1278 are fully ionized nuclei spanning the entire periodic table, with energies from tens of
1279 MeV/nucleon to above 10^{20} eV; they are isotropic, difficult to shield, and responsible
1280 for the bulk of single-event effects at high altitude [113]. *Solar energetic particles*
1281 (SEPs), produced during coronal mass ejections and solar flares, deliver intense but
1282 episodic fluences of protons (up to hundreds of MeV) and heavy ions that can cause
1283 acute upsets far exceeding the GCR background [22].

1284 The cumulative radiation dose received by a spacecraft depends strongly on or-
1285 bit, shielding, and mission duration. Low-Earth orbit (LEO) satellites at Interna-
1286 tional Space Station altitude (~ 400 km, 51.6° inclination) accumulate roughly 5–
1287 20 krad(Si)/year behind 100 mil aluminum shielding, while geostationary (GEO) and
1288 medium-Earth orbit (MEO) satellites may see 30–100 krad(Si)/year due to prolonged

1289 residence in the Van Allen belts [85]. Qualification standards such as MIL-STD-883
1290 and ESA ESCC 22900 require demonstration of margin over these mission doses be-
1291 fore flight hardware can be approved.

1292 **3.1.2 The Nuclear Plant Radiation Environment**

1293 Within an operating nuclear power plant, electronics installed in or near the re-
1294 actor building are exposed to a mixed neutron-gamma field whose intensity depends
1295 strongly on location. In-core instrumentation must withstand integrated fast-neutron
1296 fluences of 10^{17} – 10^{19} n/cm² and total gamma doses of 10^6 – 10^8 rad over the reactor
1297 lifetime. Ex-core instrumentation in the primary containment sees lower but still sig-
1298 nificant fluences of 10^{13} – 10^{16} n/cm² and gamma doses of 10^4 – 10^6 rad [37]. Safety I&C
1299 systems in the auxiliary building and control room operate at much lower dose rates
1300 (typically well below 1 rad/year) but must maintain functionality under design-basis
1301 accident conditions that could spike the local dose rate by several orders of magnitude
1302 for hours to days [63].

1303 The neutron fluence in reactor environments produces displacement damage in
1304 semiconductor lattices that is qualitatively distinct from ionizing dose effects: ener-
1305 getic neutrons displace silicon atoms from lattice sites, creating vacancy-interstitial
1306 pairs and vacancy clusters that act as generation-recombination centers, increase
1307 leakage current, reduce minority carrier lifetime, and degrade transistor gain [56].
1308 SRAM-based FPGAs, whose configuration memory cells rely on stable stored charge,
1309 are particularly sensitive to neutron-induced displacement damage in the pass tran-
1310 sistors that gate cell access.

1311 **3.1.3 The Terrestrial Cosmic-Ray Environment**

1312 At sea level, the secondary cosmic-ray flux consists primarily of muons, with a
1313 smaller but device-relevant component of high-energy neutrons (“spallation” neutrons)
1314 produced by hadronic showers in the upper atmosphere. The terrestrial fast-neutron
1315 flux at sea level is approximately 1.3×10^{-2} n/cm²/s for neutrons above 10 MeV,
1316 increasing by roughly a factor of three per kilometer of altitude [66, 113]. Although
1317 this flux is orders of magnitude lower than in accelerator or reactor environments, it
1318 is ubiquitous and non-negligible for high-density memories operating in safety-critical
1319 ground applications. The JEDEC standard JESD89A quantifies the terrestrial soft-
1320 error rate environment and defines the reference neutron spectrum at sea level against
1321 which device qualification measurements are normalized [42].

1322 **3.2 Total Ionizing Dose Effects**

1323 Total ionizing dose (TID) refers to the cumulative energy deposited in a material
1324 by ionizing radiation — principally energetic electrons, protons, and gamma photons
1325 — integrated over the operating lifetime of the device. In metal-oxide-semiconductor
1326 (MOS) structures, which form the transistor basis of all modern digital FPGAs, the
1327 primary site of TID damage is the gate oxide and the field oxide used to isolate
1328 adjacent transistors.

1329 **3.2.1 Charge Trapping in Gate Oxides**

1330 When ionizing radiation passes through a SiO₂ gate oxide, it generates electron-
1331 hole pairs with a yield proportional to the local electric field and the oxide thickness.
1332 Electrons, which have high mobility in SiO₂ ($\mu_e \approx 20$ cm²/V·s), sweep out of the

oxide rapidly under the applied gate field on nanosecond timescales. Holes, with a mobility orders of magnitude lower, undergo a dispersive hopping transport toward the Si/SiO₂ interface, where a fraction become trapped at pre-existing or radiation-induced oxide trap sites [70]. The resulting positive oxide trapped charge shifts the transistor threshold voltage V_T in the negative direction (for n-channel MOSFETs), eventually causing the device to conduct when it should be off — a failure mode known as threshold-voltage rolloff [29].

In parallel, radiation generates interface trap states at the Si/SiO₂ boundary by breaking Si-H bonds. These interface traps contribute both a fixed charge and a frequency-dependent recombination current that degrades transistor transconductance, increases $1/f$ noise, and reduces drive current. Unlike oxide trapped charge, which anneals partially at room temperature, interface traps are thermally stable on device-lifetime timescales [70].

3.2.2 TID Effects in Modern FPGAs

The susceptibility of a CMOS process to TID is strongly dependent on gate oxide thickness: thinner oxides, which dominate in sub-micron nodes, trap fewer holes because the thin-oxide regime favors rapid recombination before hole transport can occur. A 65-nm process FPGA may tolerate 300–500 krad(Si) before functional failure, compared to 30–100 krad(Si) for older 1- μ m process devices [85]. However, the isolation structures (shallow trench isolation, STI) in modern processes are relatively thick and remain a TID-sensitive element, causing leakage current between adjacent transistors at doses that do not yet affect the gate oxides.

For the nuclear instrumentation context, the TID tolerance of commercial SRAM FPGAs in the control room or auxiliary building is generally adequate without special hardening, as cumulative doses over a plant lifetime at those locations are well below the few-krad level [4]. The primary concern for FPGAs in nuclear plants is therefore not TID but the stochastic single-event effects described in the next section, which occur at arbitrarily low dose rates and cannot be mitigated by simply limiting exposure time.

3.3 Single-Event Effects: Physical Mechanisms

3.3.1 Ion Track Formation and Charge Collection

A single-event effect (SEE) is a functional or physical disturbance in a microelectronic device caused by the passage of a single energetic particle. The mechanism begins with the energy deposition of the incident particle along its track through the semiconductor. Charged particles — heavy ions, protons, and the recoil nuclei from neutron elastic scattering or nuclear reactions — lose energy primarily through Coulomb interactions with the semiconductor electrons, creating a dense column of electron-hole plasma along the track.

The key parameter governing SEE sensitivity is the linear energy transfer (LET), defined as the energy deposited per unit path length:

$$\text{LET} = -\frac{1}{\rho} \frac{dE}{dx} \quad [\text{MeV} \cdot \text{cm}^2/\text{mg}], \quad (3.1)$$

where ρ is the material density. For a silicon target, an LET of 1 MeV·cm²/mg deposits approximately 1.6×10^{-13} J per micron of path length, producing roughly 10^4 electron-hole pairs per micron. The LET of a given ion depends on its atomic number Z , mass A , and kinetic energy; heavy ions near the middle of the periodic

1377 table (e.g., Fe, Kr, Xe, Bi) with energies of tens of MeV/nucleon achieve the highest
 1378 LETs, up to $\sim 100 \text{ MeV}\cdot\text{cm}^2/\text{mg}$ in silicon [74].

1379 Following the primary ionization event, the excess carriers are collected at the
 1380 nearest reverse-biased junction by two mechanisms illustrated in Figs. 3.1 and 3.2.
 1381 *Drift collection* occurs because the high carrier density temporarily collapses the
 1382 depletion region and extends the electric field deep into the substrate — a phe-
 1383 nomenon known as *funneling* — sweeping charge toward the junction on picosecond
 1384 timescales [20]. *Diffusion collection* then operates on slower nanosecond timescales
 1385 as carriers generated outside the funnel diffuse through the neutral substrate and are
 1386 eventually collected. The total collected charge Q_{coll} is determined by the integral of
 1387 the current transient at the junction and must exceed the critical charge Q_{crit} of the
 1388 struck node to cause a SEE.

1389 3.3.2 The SEE Sensitivity Figure of Merit: Cross-Section

1390 The probability that a particle causes a SEE is characterized by the device SEE
 1391 cross-section σ , defined as the number of events observed per unit incident particle
 1392 fluence:

$$\sigma(L) = \frac{N_{\text{events}}}{\Phi} \quad [\text{cm}^2/\text{device or cm}^2/\text{bit}] , \quad (3.2)$$

1393 where Φ is the fluence in particles/cm². The cross-section is measured as a function
 1394 of LET or proton energy and typically exhibits a threshold below which no events
 1395 are observed and a plateau at high LET where every ion strike within the sensitive
 1396 volume causes an event. The threshold LET (LET_{th}) and the saturation cross-section
 1397 (σ_{sat}) together characterize the inherent hardness of the device [76].

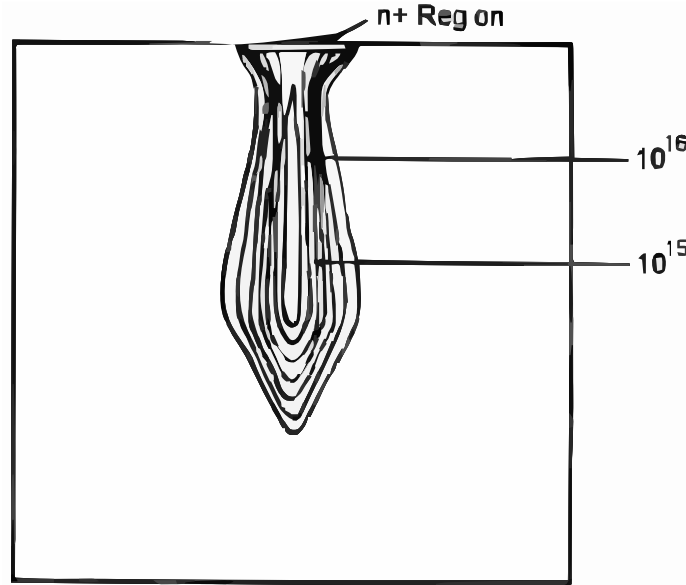


Figure 3.1: Simulated electron concentration during the funneling phase following a heavy-ion strike in a reverse-biased n^+/p junction. The dense electron-hole plasma temporarily extends the high-field depletion region deep into the substrate, collecting charge from a volume far larger than the geometric depletion region.

1398 The on-orbit SEE rate is computed by convolving the measured cross-section
 1399 curve with the environment's differential LET spectrum, a calculation codified in
 1400 tools such as CREME96 and OMERE [75]. For a device with $\sigma_{\text{sat}} = 10^{-6} \text{ cm}^2/\text{bit}$
 1401 and $\text{LET}_{\text{th}} = 3 \text{ MeV}\cdot\text{cm}^2/\text{mg}$ in a LEO orbit, the predicted upset rate is typically
 1402 10^{-5} – 10^{-3} upsets/bit/day, meaning a 16-Mbit configuration memory will experience
 1403 roughly 10–1,000 upsets per day without mitigation.

1404 3.4 Single-Event Effects Taxonomy

1405 SEEs span a range of severity from fully recoverable bit flips to permanent device
 1406 destruction. Fig. 3.3 presents the standard SEE taxonomy tree, which organizes
 1407 effects by their recoverability and the circuit element affected.

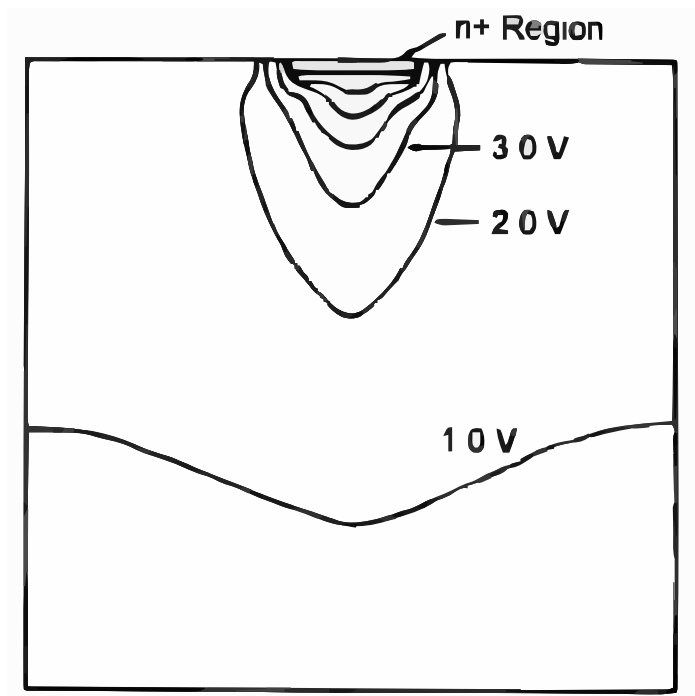


Figure 3.2: Electrostatic potential distribution during funneling, showing the field distortion caused by the ion track. The funnel depth scales with LET and substrate doping, reaching several micrometers for high-LET heavy ions in lightly doped substrates [20].

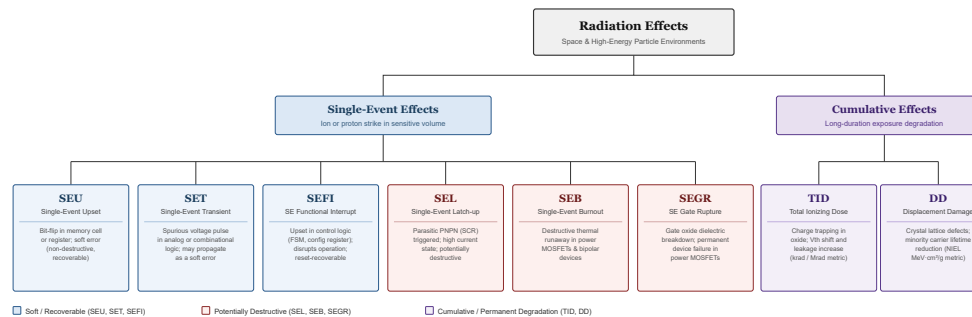


Figure 3.3: Taxonomy of single-event effects in microelectronic devices, organized by recoverability (soft vs. hard) and affected circuit element. Effects in bold are those most critical for SRAM-based FPGA operation in nuclear instrumentation applications.

1408 **3.4.1 Single-Event Upset (SEU)**

1409 A single-event upset (SEU) is a non-destructive, recoverable change in the logic
1410 state of a bistable storage element — a flip-flop, SRAM cell, or register — caused by
1411 charge collection from a single particle strike. The struck node’s voltage is transiently
1412 driven past the switching threshold of the latch, causing it to adopt the opposite
1413 state [20]. In an SRAM-based FPGA, the configuration memory that encodes the
1414 user’s design is implemented as an array of SRAM cells; an SEU in this array changes
1415 the programmed function of a logic element, routing multiplexer, or interconnect
1416 switch, potentially altering the circuit’s behavior in ways that are not immediately
1417 apparent without continuous monitoring [81].

1418 The critical charge for SEU in a modern 28-nm SRAM cell is on the order of
1419 1–10 fC, achievable by the recoil nucleus from a proton nuclear interaction or by
1420 a moderate-LET heavy ion [8]. As process nodes have scaled, Q_{crit} has decreased
1421 roughly proportionally with feature size, partially offset by reductions in node capac-
1422 itance and improvements in cell drive strength.

1423 **3.4.2 Multiple Cell Upset (MCU)**

1424 A multiple cell upset (MCU) occurs when a single particle deposits sufficient
1425 charge to simultaneously flip two or more adjacent memory cells. MCUs are of par-
1426 ticular concern because they can defeat single-bit error correction codes (SECDED)
1427 that are effective against isolated SEUs. The probability of MCU scales with LET
1428 and is strongly geometry-dependent: cells laid out in adjacent rows and columns of
1429 the SRAM array are co-vulnerable, while cells interleaved across different physical
1430 locations are not [12]. In modern planar CMOS processes, MCU events can involve

1431 tens of cells from a single high-LET heavy-ion strike, while proton-induced MCUs
 1432 typically involve two to four cells from the recoil of a proton nuclear reaction [8].

1433 3.4.3 Single-Event Transient (SET)

1434 A single-event transient (SET) is a voltage glitch on a combinational logic node
 1435 caused by charge collection at a circuit junction. Unlike an SEU, which corrupts a
 1436 storage element, an SET propagates through combinational logic and may or may not
 1437 be captured by a downstream flip-flop, depending on the timing relationship between
 1438 the transient and the clock edge. Fig. 3.4 illustrates the generation of an SET at an
 1439 inverter output, and Fig. 3.5 shows the conditions under which an SET propagating
 1440 through a logic cone is captured by a flip-flop.

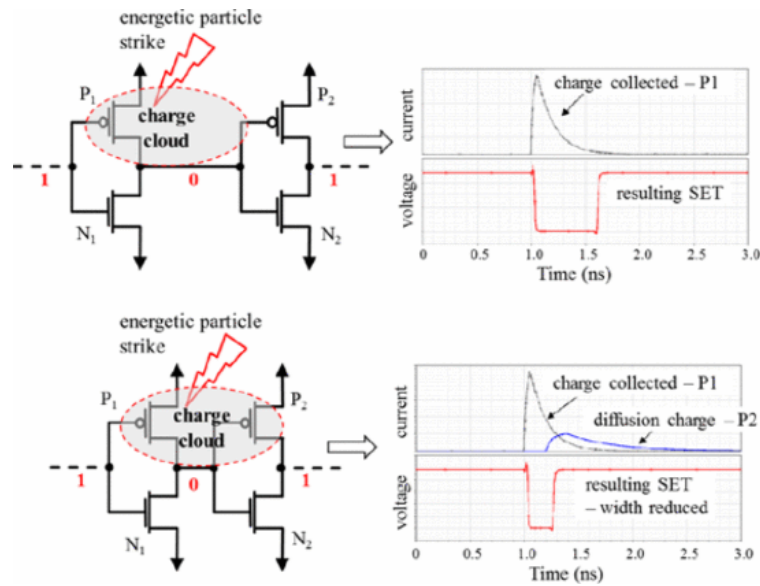


Figure 3.4: Generation of a single-event transient (SET) at a CMOS inverter output. The ion track deposits charge at a reverse-biased drain junction, driving the output voltage transiently toward the supply rail. Pulse width is determined by the collected charge and the drive strength of adjacent transistors [28].

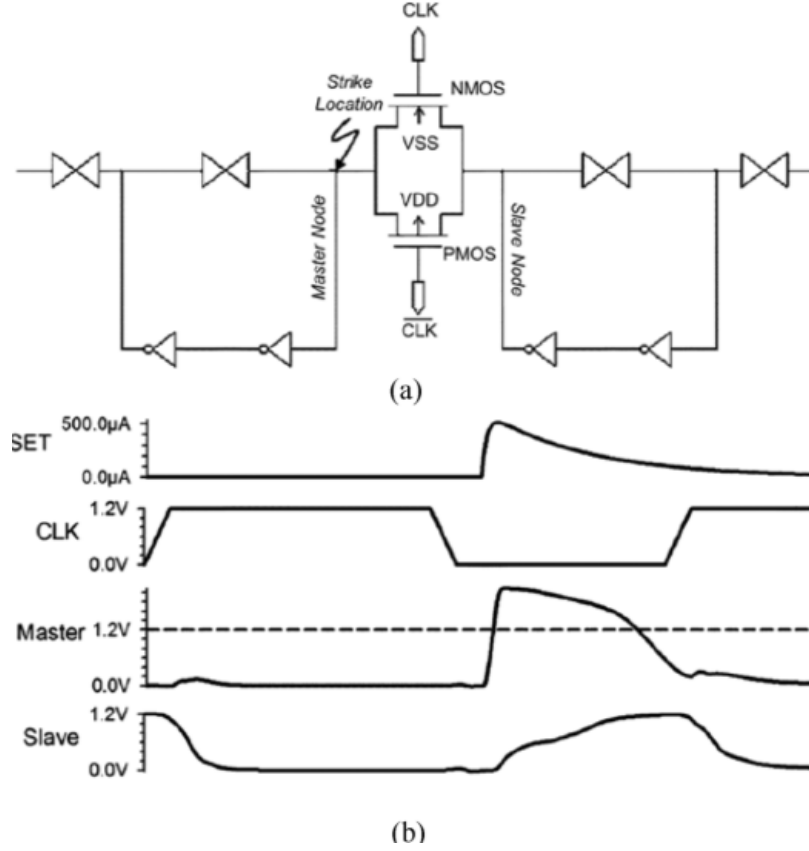


Figure 3.5: SET propagation and capture by a flip-flop. The SET (shaded pulse) traverses the combinational logic cone and arrives at the flip-flop's data input. It is captured and causes an SEU only if it overlaps with the flip-flop's setup-and-hold window around the clock edge [28].

1441 In high-speed digital logic, SET pulse widths of tens to hundreds of picoseconds are
 1442 comparable to clock periods, making SET-induced SEUs a dominant error mechanism
 1443 in modern FPGAs operating above 100 MHz [28]. At the lower clock rates typical
 1444 of nuclear I&C applications, SET capture probability is reduced, but long transients
 1445 from high-LET events remain a concern.

1446 **3.4.4 Single-Event Functional Interrupt (SEFI)**

1447 A single-event functional interrupt (SEFI) is an SEU in a control register, finite
1448 state machine, or configuration memory that causes a device to enter an erroneous
1449 operating mode — resetting itself, locking up, or outputting invalid data — that
1450 requires a system-level reset to clear. In FPGAs, SEFIs in the configuration con-
1451 troller or global control registers can cause a partial or complete loss of the user’s
1452 programmed logic, a far more serious consequence than a single flipped bit in the
1453 data path [81]. SEFI rates are typically much lower than SEU rates per bit (be-
1454 cause control registers represent a small fraction of total bits), but their operational
1455 consequence is disproportionately severe.

1456 **3.4.5 Single-Event Latchup (SEL)**

1457 Single-event latchup (SEL) is a potentially destructive condition in which a par-
1458 ticle strike triggers the parasitic p-n-p-n thyristor inherent in bulk CMOS processes,
1459 causing a low-impedance path to form between the supply rails. If not cleared by
1460 power cycling within milliseconds, the resulting current can permanently destroy the
1461 device through thermal runaway [86]. SEL susceptibility is strongly suppressed in
1462 silicon-on-insulator (SOI) processes (which lack the bulk substrate needed for the
1463 parasitic thyristor) and by epitaxial substrates with heavily doped buried layers.
1464 Modern commercial SRAM FPGAs fabricated in bulk CMOS are SEL-susceptible
1465 and require current-limiting power supply protection and power-cycling circuitry in
1466 space and nuclear applications.

1467 3.4.6 Single-Event Burnout (SEB) and Gate Rupture (SEGR)

1468 Single-event burnout (SEB) and single-event gate rupture (SEGR) are catas-
 1469 trophic destructive effects confined primarily to power MOSFETs and bipolar tran-
 1470 sistors operating at high drain-source voltages. SEB occurs when a heavy-ion strike
 1471 in the epitaxial drain region of an OFF-state power MOSFET triggers a parasitic
 1472 n-p-n transistor, causing sustained high-current conduction that thermally destroys
 1473 the device [86]. SEGR results from the momentary field enhancement in the gate
 1474 oxide caused by the ion-track plasma, which can exceed the oxide breakdown field
 1475 and permanently rupture the gate dielectric. Figs. 3.6, 3.7, and 3.8 illustrate the
 1476 cross-sectional geometry and onset conditions for these effects.

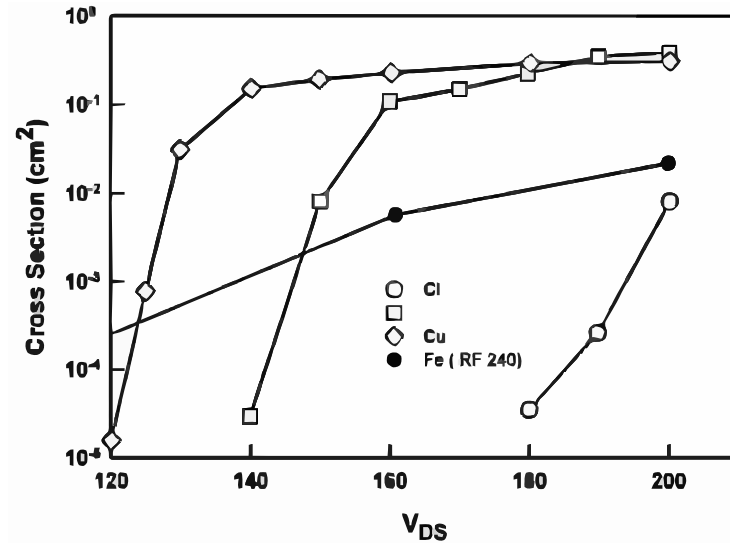


Figure 3.6: Cross-sectional schematic of single-event burnout (SEB) in an n-channel power MOSFET. The heavy-ion track (dashed) deposits charge in the epitaxial drain that forward-biases the parasitic n-p-n transistor base-emitter junction, initiating regenerative feedback and destructive current [86].

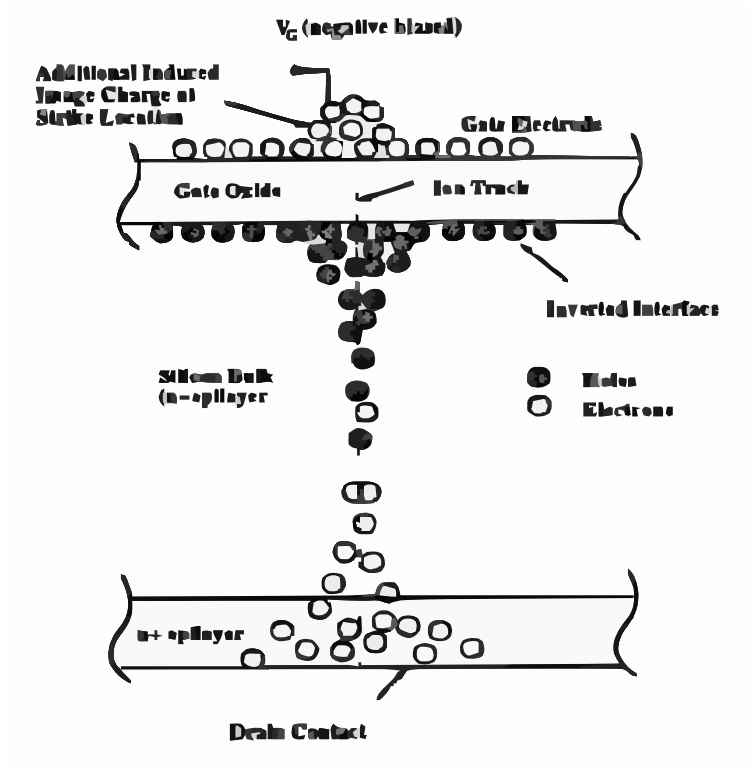


Figure 3.7: Single-event gate rupture (SEGR) mechanism. The transient increase in gate-oxide field caused by the ion-track plasma can locally exceed the intrinsic breakdown field of the SiO_2 , permanently rupturing the gate dielectric and rendering the transistor non-functional [86].

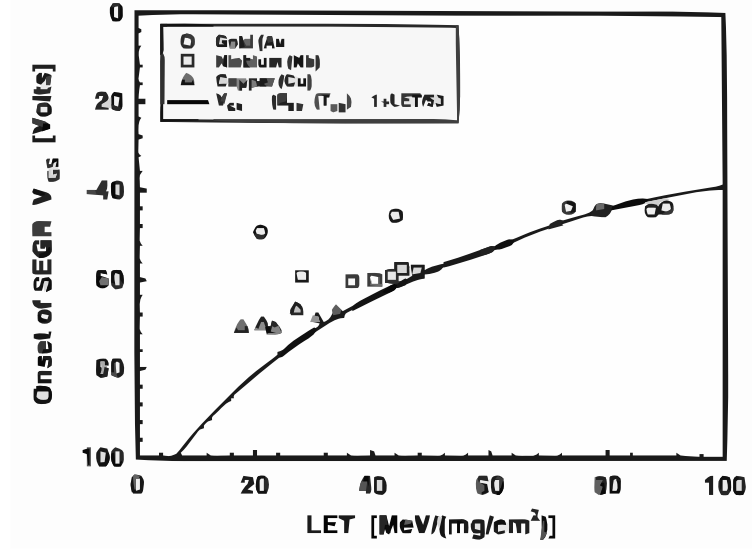


Figure 3.8: Measured SEGR onset V_{GS} as a function of heavy-ion LET for several gate-metal species. The onset voltage decreases with increasing LET, imposing a derating requirement on operating voltage for power devices in heavy-ion environments [86].

SEB and SEGR are not directly applicable to logic-level SRAM FPGAs, which operate at supply voltages far below the breakdown thresholds of their internal power transistors. They are, however, relevant to the power supply circuitry and gate drivers that support an FPGA-based system, and must be considered at the system level in any radiation-hardened design.

3.5 FPGA Architecture and Radiation Vulnerability

3.5.1 SRAM-Based FPGA Architecture

Modern high-density FPGAs are organized as a two-dimensional array of configurable logic blocks (CLBs), interconnected by a programmable routing fabric and surrounded by dedicated I/O, clock management, block RAM, DSP, and hard-IP

peripherals. The entire programmable function of the device is encoded in a configuration memory bitstream stored in an on-chip SRAM array, which for large devices may contain 50–500 million configuration bits [49]. The configuration SRAM is loaded from external non-volatile storage (typically QSPI flash or EEPROM) on power-up and retained in SRAM for the duration of operation.

This architecture has a fundamental radiation vulnerability: every bit of the configuration SRAM is continuously susceptible to SEU throughout device operation. An SEU in the configuration memory can alter the function of a lookup table (LUT), change the routing of a signal, enable or disable a flip-flop reset, or modify the address decode of a block RAM — depending on which configuration bit is struck. Unlike an SEU in a user-data register, which affects only the current computation, a configuration memory SEU silently corrupts the device’s programmed behavior until it is detected and corrected [45].

The sensitivity of different FPGA configuration bit types to functional corruption varies substantially. Bits that control LUT contents are always functionally significant; bits that control unused routing resources may have no effect. Studies of Xilinx Virtex and Kintex FPGAs have found that roughly 15–30% of configuration bits are “essential” — their upset causes a detectable output error — with the remainder corresponding to unused or redundant resources [81, 50]. This essential bit fraction is critical for predicting the functional upset rate from a measured raw upset rate.

3.5.2 Radiation Sensitivity Across FPGA Generations

Quinn et al. surveyed the radiation reliability concerns of Xilinx FPGAs from the Virtex through Virtex-5 generations, finding that each new process node brought

1510 reduced Q_{crit} (worsening raw SEU sensitivity per bit) but also reduced bit count
 1511 per logic function (improving functional upset rate per implemented operation) [81].
 1512 SRAM cell cross-sections measured at particle accelerators for 90-nm-class devices
 1513 (Virtex-4/5) are typically in the range of 10^{-14} – 10^{-13} cm²/bit at high LET, corre-
 1514 sponding to functional upset rates of 10^{-3} – 10^{-2} upsets per device per day in a rep-
 1515 resentative LEO orbit. Rakhman et al. characterized the SRAM configuration cells
 1516 of a Xilinx Kintex-7 FPGA under both heavy-ion and proton irradiation, finding a
 1517 saturation cross-section of approximately 5×10^{-14} cm²/bit and a proton cross-section
 1518 of $\sim 10^{-15}$ cm²/bit at 200 MeV [82].

1519 For nuclear plant environments, the neutron-induced upset rate at control-room
 1520 locations (fast-neutron flux $< 10^3$ n/cm²/s) is orders of magnitude lower than in
 1521 space and may be further reduced by modest concrete shielding. However, design-
 1522 basis accident scenarios that expose I&C systems to elevated fluences over hours to
 1523 days require that the system either be immune to configuration upsets or be capable
 1524 of detecting and correcting them within the plant’s safety response time.

1525 **3.6 Fault-Tolerant Mitigation Techniques**

1526 **3.6.1 Triple Modular Redundancy**

1527 The foundational technique for SEU mitigation in digital systems is triple mod-
 1528 ular redundancy (TMR), originally proposed by Lyons and Vanderkulk in 1962 for
 1529 improving the reliability of vacuum-tube computers [55]. In TMR, the protected
 1530 function is replicated three times, and a majority voter selects the output of the two
 1531 (or three) agreeing copies. A single SEU in any one replica is masked because the two
 1532 healthy copies outvote the corrupted one, and the voter output remains correct. The

1533 technique requires the voter itself to be robust, either because it is implemented in
 1534 radiation-hard technology or because the voter is a simple gate whose critical charge
 1535 greatly exceeds that of the storage elements it protects.

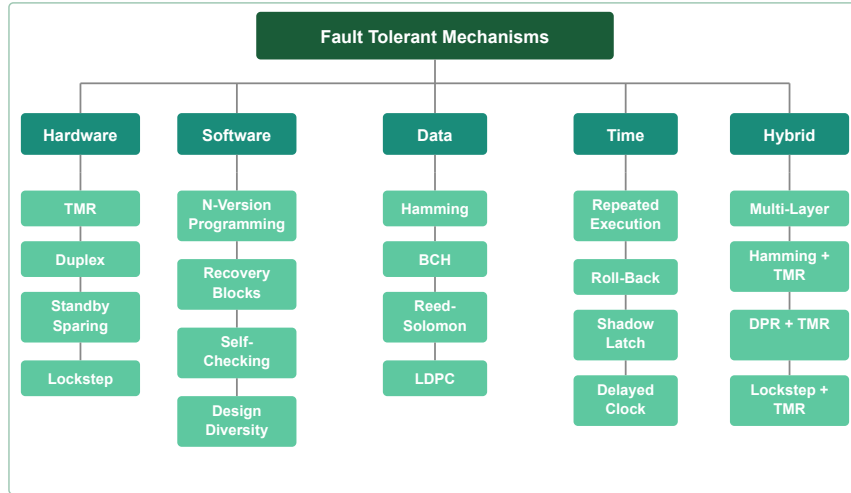


Figure 3.9: Overview of fault-tolerant mechanisms for SRAM-based FPGAs, ranging from single-level flip-flop TMR to full-system architectural redundancy. Each level adds correction capability at the cost of increased resource utilization [45].

1536 In FPGA implementations, TMR is applied at multiple granularities. *Full TMR*
 1537 triplicates every logic element, flip-flop, and routing resource in the design, provid-
 1538 ing maximum fault coverage at the cost of roughly $3\times$ resource usage and power.
 1539 *Partial TMR* applies triplication only to the most safety-critical paths, reducing
 1540 overhead while protecting against the most consequential upsets [80]. *Register-file*
 1541 *TMR* protects only the storage elements (flip-flops and block RAMs) while leaving
 1542 combinational logic unprotected, trading reduced area for acceptance of transient
 1543 SET-induced errors that are not captured [88]. Fig. 3.9 illustrates these trade-offs,
 1544 and Fig. 3.10 shows the voter logic in detail.

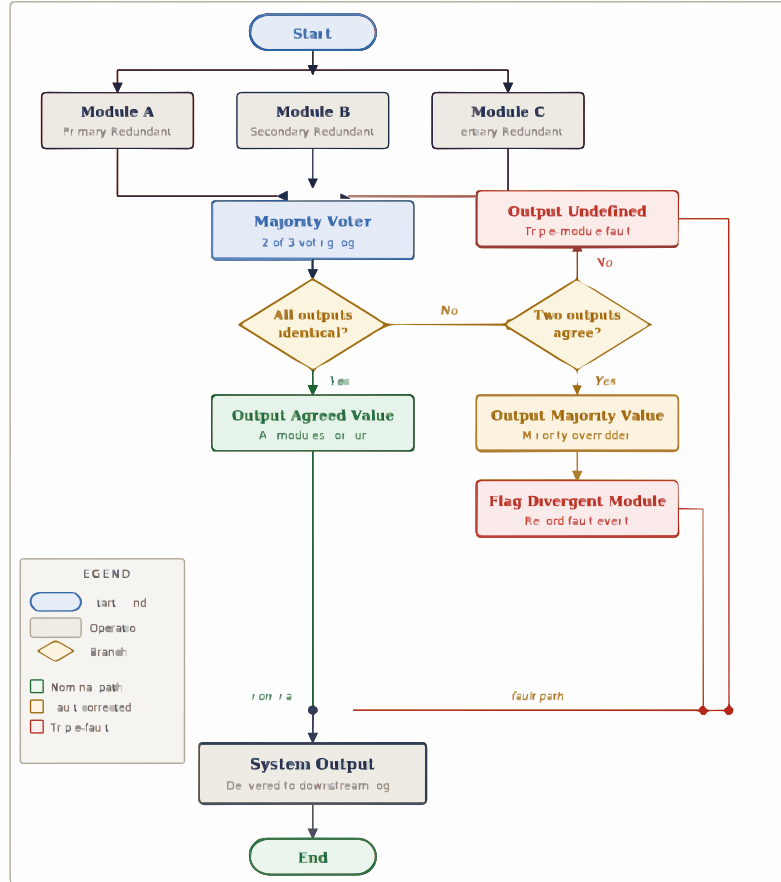


Figure 3.10: Voter logic flowchart for a TMR system. Each output bit from the three replicas is majority-voted independently; disagreement among replicas can be flagged as a fault indicator without interrupting correct output generation [55, 43].

1545 The TMR approach is well-established in space applications and has been demon-
1546 strated to reduce functional upset rates by approximately two orders of magnitude
1547 in Xilinx Virtex-class FPGAs at representative GCR fluences [105]. For nuclear I&C
1548 applications, Sciber et al. demonstrated a TMR-protected FPGA-based safety system
1549 for a research reactor that maintained correct output through simulated configuration
1550 upsets at a rate $500\times$ the expected plant environment level [26].

1551 3.6.2 Configuration Scrubbing

1552 TMR corrects the voter output but does not repair the configuration memory of
1553 the corrupted replica. After a sufficient number of upsets accumulate, two replicas
1554 may be simultaneously corrupted, defeating majority voting. Configuration scrubbing
1555 addresses this by periodically refreshing the FPGA's configuration memory from a
1556 known-good reference stored in non-volatile memory [9].

1557 In *external scrubbing*, an external processor reads the current configuration bit-
1558 stream from the FPGA, compares it bit-by-bit with the golden reference, and writes
1559 back corrected frames where differences are detected. This approach can detect and
1560 correct errors at rates determined by the scrub period, which must be short com-
1561 pared to the mean time between double upsets in the TMR replicas [35]. For a device
1562 with 10^7 configuration bits, a bit-upset rate of 10^{-3} upsets/bit/day, and a required
1563 double-upset probability below 10^{-6} per mission, the required scrub period is on the
1564 order of seconds — achievable with standard scrubbing controllers operating at JTAG
1565 bitstream rates [106].

1566 *Partial reconfiguration scrubbing* uses the FPGA vendor’s partial reconfiguration
 1567 infrastructure to overwrite individual configuration frames without resetting the en-
 1568 tire device, allowing scrubbing to proceed concurrently with normal device opera-
 1569 tion [80]. Xilinx Virtex/UltraScale and Intel Stratix devices both support partial
 1570 reconfiguration, and open-source scrubbing controllers (e.g., the Scrub Engine devel-
 1571 oped at Brigham Young University) implement this approach [106]. Herrera-Alzu
 1572 and López-Vallejo systematically characterized the scrubber performance for Xilinx
 1573 Virtex-5 FPGAs, demonstrating that scrub periods below 50 ms reduced the effective
 1574 functional upset rate below 10^{-6} upsets/day at LEO conditions [36].

1575 **3.6.3 Combined TMR-Scrubbing and Hybrid Approaches**

1576 For the highest reliability requirements, TMR and scrubbing are deployed to-
 1577 gether: TMR provides instant correction of single upsets, while scrubbing prevents
 1578 the accumulation of uncorrected errors that would eventually defeat TMR [45]. The
 1579 combination reduces the effective uncorrectable upset rate to the probability of two
 1580 upsets occurring within a single scrub period before the first is corrected — typically
 1581 several orders of magnitude lower than either technique alone.

1582 Hybrid approaches extend these concepts to the system level. Stoddard et al.
 1583 demonstrated a configuration in which an FPGA running a spacecraft attitude con-
 1584 trol algorithm was protected by TMR at the register level, partial-reconfiguration
 1585 scrubbing at the configuration level, and watchdog-timer reset at the system level,
 1586 achieving an estimated mean time between failures of $> 10^5$ hours in a GEO radia-
 1587 tion environment [90]. The architecture developed in Chapter ?? for nuclear plant
 1588 monitoring adopts a similar layered approach, with TMR protecting the safety logic,

1589 periodic configuration scrubbing maintaining the configuration integrity, and a redun-
1590 dant processor monitoring the voter-disagreement signal to detect any uncorrected
1591 upset before it propagates to a safety output.

1592 **3.6.4 Error Detection and Correction in Block RAMs**

1593 Block RAMs (BRAMs) in FPGAs store application data — not configuration bits
1594 — and are equally susceptible to SEUs. Xilinx and Intel FPGAs provide optional
1595 built-in single-error correction, double-error detection (SECDED) Hamming codes for
1596 BRAM data, which correct single-bit upsets transparently and flag double-bit upsets
1597 as uncorrectable [58]. For nuclear applications where the BRAM stores measurement
1598 data or plant state variables, SECDED with periodic scrubbing provides an acceptable
1599 level of protection, while dual-port BRAM with cross-checking between two separate
1600 memory arrays provides an additional layer of protection against MCU events that
1601 span multiple bits within the same Hamming word.

1602 **3.7 COTS FPGAs in Nuclear and Radiation-Critical Applica-** 1603 **tions**

1604 **3.7.1 The Cost-Performance Trade-off**

1605 Purpose-built radiation-hardened FPGAs — such as the Microchip RTG4 (130-nm
1606 flash-based, inherently SEU-immune configuration) and the AMD/Xilinx XQRKU060
1607 (16-nm FinFET, rad-hard by design) — provide guaranteed TID tolerance exceeding
1608 300 krad(Si) and quantified SEL immunity, but at unit prices of \$50,000–\$500,000 and
1609 with logic densities one to two generations behind commercial equivalents [57, 3]. For
1610 nuclear plant I&C modernization programs that may require hundreds of processing
1611 nodes, this cost premium is prohibitive.

Commercial off-the-shelf (COTS) FPGAs from the Xilinx 7-series, UltraScale, and Intel Cyclone/Arria families offer 50–100× lower cost per gate, higher logic density, and access to modern soft-IP ecosystems, at the cost of no radiation-hardness guarantee from the manufacturer [11]. The engineering question is therefore not whether COTS FPGAs are radiation-tolerant, but whether their radiation response — combined with appropriate mitigation — can be characterized to a level of assurance sufficient for the safety classification of the application.

3.7.2 Demonstrated COTS FPGA Performance in Radiation Environments

Several accelerator and nuclear programs have demonstrated that COTS FPGAs with TMR and scrubbing can meet stringent reliability requirements at radiation levels consistent with nuclear plant I&C. Lee et al. irradiated a Xilinx Kintex-7 FPGA (28-nm) with protons at TRIUMF and with neutrons at Los Alamos National Laboratory’s Weapons Neutron Research facility, measuring a total accumulated dose of 300 krad(Si) with no functional failure, and recording zero configuration upsets attributable to TID at that dose level [50]. Wirthlin et al. demonstrated a Leon3 soft-core processor implemented on a Kintex-7 FPGA that maintained correct program execution through 3×10^{10} n/cm² of 14-MeV neutron irradiation with TMR+scrubbing active, corresponding to a ground-level exposure equivalent of several decades of control-room operation [107].

The CERN high-energy physics community has deployed COTS FPGAs extensively in the LHC detector electronics, where annual doses in some locations reach tens of krad and the SEE environment from secondary shower particles is comparable to LEO. Systematic characterization programs at CERN show that 28-nm Kintex-7 and

UltraScale devices exhibit no destructive effects (SEL, SEB, SEGR) at bias voltages within datasheet ratings at LET up to $62 \text{ MeV}\cdot\text{cm}^2/\text{mg}$, and that TMR+scrubbing reduces functional upset rates below the $10^{-9}/\text{hour}$ target imposed by LHC safety systems [23].

Pignol reviewed the use of COTS components in safety-critical space avionics and concluded that the combination of architectural mitigation (TMR, watchdog), physical mitigation (shielding, latchup current limiters), and comprehensive testing is sufficient to achieve failure-in-time rates competitive with purpose-built rad-hard parts, at substantially lower cost [77]. The same conclusion drives the DOE GEMINA program, which targets AI/ML-enabled COTS digital systems for advanced reactor I&C [96], and the ANS-recommended roadmap for I&C modernization at the existing LWR fleet [4].

3.7.3 Regulatory Pathway for COTS FPGAs in Nuclear Plants

The NRC’s regulatory framework for digital I&C in nuclear plants (10 CFR 50 Appendix A, GDC 21–25; Regulatory Guide 1.152; NUREG/CR-6263) does not prohibit COTS devices but requires that the licensee demonstrate that the device meets the qualification requirements for its safety classification, including radiation qualification per IEEE Std 7-4.3.2 and post-accident monitoring per RG 1.97 [63]. The key qualification documentation for an FPGA-based safety function includes: (1) a radiation characterization data package demonstrating TID tolerance, SEL immunity, and SEU cross-section measurement; (2) a functional upset rate prediction for the plant-specific radiation environment; (3) a mitigation architecture description showing that the functional upset rate with TMR+scrubbing meets the specified single

1659 failure criterion; and (4) a test program demonstrating correct operation under the
 1660 combined effects of irradiation and functional loading [37, 62].

1661 3.8 Testing Methodologies for FPGA Radiation Qualification

1662 The qualification of FPGAs for radiation environments requires experimental mea-
 1663 surement of both TID and SEE parameters, since analytical prediction from process
 1664 parameters alone is not sufficiently accurate for safety-relevant applications. Fig. 3.11
 1665 illustrates the principal testing approaches and their relationship to the environment
 1666 being simulated.

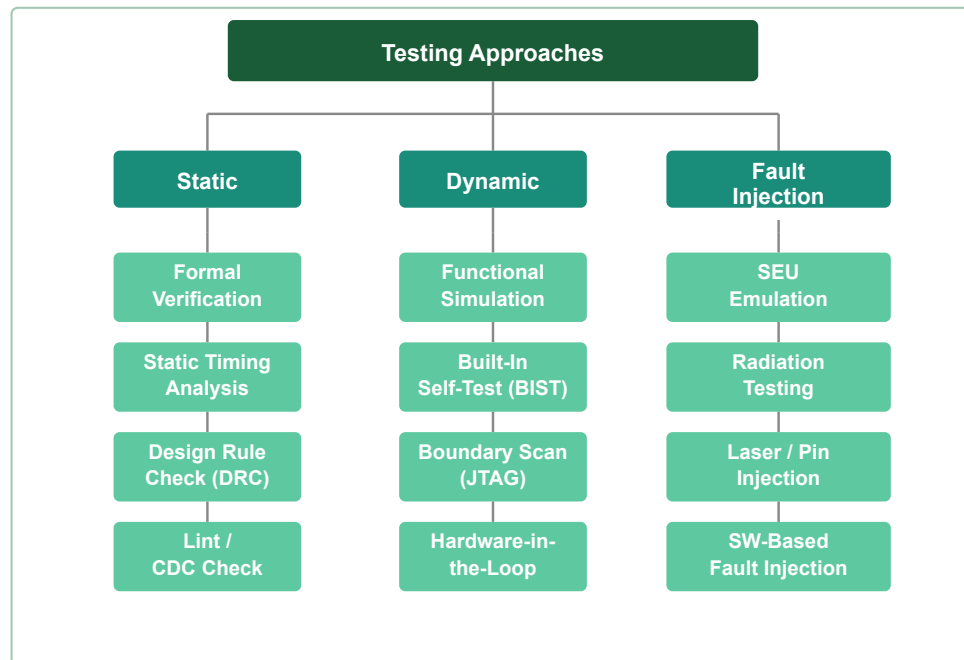


Figure 3.11: Summary of testing approaches for radiation hardness assurance of digital electronics, organized by radiation source, simulated environment, and measured effect. Each approach has characteristic advantages and limitations regarding representativeness, cost, and achievable fluence [85, 17].

1667 **3.8.1 Heavy-Ion Testing**

1668 Heavy-ion irradiation at particle accelerators is the primary technique for mea-
1669 suring SEE cross-sections as a function of LET and for identifying the SEL, SEB,
1670 and SEGR thresholds of a device. Facilities commonly used for FPGA radiation test-
1671 ing include the Brookhaven National Laboratory Tandem Van de Graaff (LET range
1672 1–100 MeV·cm²/mg), the Lawrence Berkeley National Laboratory 88-Inch Cyclotron
1673 (LET up to 85 MeV·cm²/mg), the Texas A&M K500 Cyclotron, and the GANIL
1674 facility in Caen, France [62].

1675 During a heavy-ion test, the device under test (DUT) is mounted in a vacuum
1676 chamber, biased and clocked at normal operating conditions, and exposed to a uniform
1677 beam of a selected heavy ion species at controlled LET and fluence. For FPGA
1678 testing, two complementary readout strategies are used: *static testing*, in which the
1679 device is irradiated with configuration memory pre-loaded and then readback to count
1680 configuration bit flips, and *dynamic testing*, in which the FPGA executes a test design
1681 that monitors its own outputs for errors in real time [81]. Dynamic testing captures
1682 both configuration upsets and user-register upsets but is limited in the depth of
1683 design validation per test run; static testing provides complete configuration memory
1684 coverage but misses upsets in the user logic flip-flops.

1685 **3.8.2 Proton Testing**

1686 Proton irradiation at energies of 50–200 MeV provides a practical and cost-effective
1687 alternative to heavy-ion testing for devices whose SEE cross-section vs. LET curve
1688 is required through the proton-relevant range, and for devices intended for use in
1689 proton-dominated environments (trapped proton belts, low-altitude orbits). Protons

1690 cause SEEs primarily through nuclear elastic scattering (which produces silicon or
1691 oxygen recoil nuclei with typical LET of 5–15 MeV·cm²/mg) and through non-elastic
1692 nuclear reactions (which produce heavy recoil fragments with higher LET) [76]. The
1693 resulting SEU cross-section is lower than the heavy-ion saturation cross-section by
1694 one to two orders of magnitude, but the high proton fluences available at accelerators
1695 like TRIUMF (Vancouver), IUCF (Bloomington), and the Paul Scherrer Institute
1696 (Villigen) allow statistically meaningful measurements [85].

1697 Proton testing also accumulates significant TID during the SEE test (at 100 MeV,
1698 10¹¹ p/cm² corresponds to ~1 krad(Si)), which must be accounted for in test planning
1699 to avoid confusing TID degradation with SEE events.

1700 3.8.3 Neutron Testing

1701 Neutron testing is the most directly relevant modality for nuclear plant appli-
1702 cations and for ground-level qualification [89]. The upset mechanisms for neutrons
1703 above ~10 MeV are similar to protons — nuclear elastic scattering producing silicon
1704 recoil nuclei plus non-elastic spallation reactions producing heavier fragments — but
1705 the neutron’s lack of charge means it produces no ionization along its own track and
1706 the DUT need not be operated in vacuum. This simplifies the experimental setup
1707 considerably.

1708 Several neutron irradiation facilities are available for FPGA testing. The Los
1709 Alamos Neutron Science Center (LANSCE) ICE House provides a quasi-monoenergetic
1710 neutron spectrum peaked at 800 MeV that is often used as a reference for atmospheric
1711 cosmic-ray testing [107]. The Indiana University Cyclotron Facility (IUCF) produced

1712 monoenergetic 200-MeV neutrons, now primarily replaced by the Triangle Universi-
1713 ties Nuclear Laboratory (TUNL) and the nTOF facility at CERN for intermediate
1714 energies. For reactor neutron environments, fission reactor irradiation facilities such
1715 as the MIT reactor’s fission converter beam [32] and the Ohio State University Re-
1716 search Reactor (OSURR) provide thermal-to-fast neutron spectra representative of
1717 ex-core I&C locations.

1718 The standard approach for neutron SEE testing of FPGAs follows the JEDEC
1719 JESD89A protocol: the device is irradiated with a known neutron fluence at the
1720 test facility’s reference spectrum, and the measured upset count is scaled to the
1721 terrestrial sea-level equivalent fluence using the spectral overlap integral [42]. Stirk et
1722 al. compared three neutron testing approaches (monoenergetic, white-spectrum, and
1723 reactor) for a complex system-on-chip and found that the accelerator white-spectrum
1724 test at LANSCE gave the closest agreement with reactor exposure results, while
1725 monoenergetic beam tests required more careful spectral correction [89].

1726 **3.8.4 Total Ionizing Dose Testing**

1727 TID testing is performed by exposing the device to a controlled dose from a ^{60}Co
1728 gamma source or an X-ray irradiator at a specified dose rate, with the device biased
1729 in a representative operating condition. The JEDEC/MIL-STD-883 Method 1019
1730 protocol specifies the irradiation conditions, readout intervals, and post-irradiation
1731 anneal procedure required for military and space qualification [85].

1732 A critical subtlety in TID testing is dose-rate dependence: many MOS devices
1733 exhibit enhanced low-dose-rate sensitivity (ELDRS), in which the damage at low dose
1734 rates ($<0.1 \text{ rad(Si)/s}$) exceeds that at high dose rates ($1\text{--}100 \text{ rad(Si)/s}$) for the same

total dose [29]. ELDRS occurs in bipolar transistors and certain MOS linear circuits but is less pronounced in digital CMOS; it is nonetheless a qualification concern for reactor I&C systems that accumulate dose over decades at rates of millirad per hour rather than the rad-per-second rates used in accelerated laboratory tests. The MIL-STD-883 Method 1019 elevated-temperature anneal procedure is designed to provide a conservative bound on ELDRS without requiring multi-year low-dose-rate exposures [87].

3.8.5 Ground-Level and Accelerated System Testing

For applications where the primary concern is the terrestrial cosmic-ray neutron environment, system-level testing at the actual deployment site or at a representative high-altitude location provides the most directly applicable qualification data [17]. However, at sea-level neutron flux ($\sim 10^{-2}$ n/cm²/s above 10 MeV), accumulating 10^{10} n/cm² — a fluence corresponding to several decades of ground-level exposure — would require $\sim 3 \times 10^{10}$ seconds, making direct ground-level testing impractical for qualification purposes. Accelerated testing at mountain-top observatories (e.g., Mauna Kea, 5 km altitude, flux $\sim 10\times$ sea level) or accelerator facilities bridging this gap.

Adell et al. and Artola et al. developed simulation frameworks using GEANT4 to predict ground-level neutron upset rates from device-level cross-section measurements, validating the approach against measured field failure data from commercial servers [1, 5]. The GEANT4 Monte Carlo toolkit [2], which models hadronic and

1756 electromagnetic interactions from eV to TeV energies, has become the standard sim-
1757 ulation infrastructure for predicting SEE rates in complex radiation environments
1758 including both atmospheric and reactor neutron spectra.

1759 **3.9 Summary**

1760 This chapter has established the physical and engineering framework for the
1761 FPGA-based nuclear instrumentation system developed in Chapter ???. The three
1762 radiation environments of concern — space, nuclear plant, and terrestrial cosmic-ray
1763 — each impose distinct TID and SEE demands on digital electronics. TID damage in
1764 MOS gate oxides accumulates throughout device operation and degrades transistor
1765 performance at doses that, for control-room I&C in nuclear plants, remain well within
1766 the tolerance of modern CMOS processes. The more consequential threat is stochastic
1767 single-event effects, which occur at arbitrarily low dose rates: SEUs corrupt FPGA
1768 configuration memory silently and persistently, SEFIs can interrupt device operation,
1769 and SELs can destroy devices if power cycling is not promptly applied.

1770 Triple modular redundancy, applied at the register and routing level within the
1771 FPGA fabric, corrects single upsets instantaneously without interrupting the safety
1772 function; periodic configuration scrubbing prevents upset accumulation that would
1773 eventually defeat TMR. The combination of these two techniques reduces the un-
1774 correctable upset probability to levels that can be demonstrated to meet the single-
1775 failure criterion of nuclear safety standards. COTS FPGAs from the 28-nm and 16-nm
1776 Kintex-7 and UltraScale families have been qualified for radiation environments sub-
1777 stantially more demanding than nuclear plant I&C through heavy-ion, proton, and
1778 neutron testing at national accelerator facilities, and the regulatory pathway for their

1779 use in safety-related applications is established in IAEA SSG-39 and NRC Regulatory
1780 Guide 1.152. The following chapter implements and experimentally validates these
1781 principles in a digital plant monitoring system built on a COTS Kintex-7 FPGA.

Chapter 4: $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ Perovskite Solar Cells as Ionizing Radiation Monitors

The preceding chapter established that halide perovskites combine the electronic properties requisite for radiation detection — high effective atomic number, adequate charge-carrier mobility-lifetime products, and tunable bandgap — with solution-processable fabrication routes inaccessible to conventional compound semiconductors. The work described in this chapter exploits a distinct but complementary strategy: rather than deploying a thick single crystal as a direct charge-collection detector, it repurposes a photovoltaic-architecture thin-film device as a *current-mode radiation monitor*. The rationale is twofold. First, modern perovskite solar cells have reached certified power-conversion efficiencies exceeding 27% [51], implying that the absorber layers have been optimized over a decade of intense materials research for the very properties — long carrier diffusion lengths, low trap-state densities, and efficient carrier extraction — that also govern sensitivity to ionizing radiation. Second, photovoltaic modules are inherently scalable in area, enabling the construction of sensor arrays whose geometric aperture is far larger than what is practical with single-crystal bulk detectors, and whose short-circuit current provides a direct analog output suitable for interfacing with standard nuclear instrumentation.

1800 The $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ composition is specifically chosen because partial substitution
 1801 of cesium for formamidinium suppresses the spontaneous $\alpha \rightarrow \delta$ phase transition of
 1802 pure FAPbI_3 while preserving its 1.48 eV bandgap and superior charge transport rela-
 1803 tive to MAPbI_3 [111]. The *pin* diode device structure (Section 4.1) further suppresses
 1804 dark current relative to a simple ohmic-contact geometry, and its inverted polarity
 1805 places the transparent electrode on the illuminated face — an arrangement that si-
 1806 multaneously enables photovoltaic operation under sunlight and direct exposure of
 1807 the perovskite absorber to externally incident radiation.

1808 This chapter reports visible-light temporal response measurements (Section 4.2),
 1809 X-ray response under pulsed and continuous illumination (Section 4.3), gamma-ray
 1810 response during continuous ^{137}Cs and ^{60}Co irradiation including a six-day and sixteen-
 1811 day stability study (Section 4.4), and steady-state neutron response measurements
 1812 performed at the Ohio State University Nuclear Reactor Laboratory (Section 4.5).

1813 4.1 Device Fabrication and Photovoltaic Performance

1814 Devices were fabricated by Prof. Jinsong Huang’s group at the University of North
 1815 Carolina at Chapel Hill using a standard inverted (*pin*) architecture developed for
 1816 high-efficiency perovskite photovoltaics. The composition of the absorber layer is
 1817 $\text{Cs}_{0.1}\text{FA}(\text{CH}_3\text{NH}_2)_{0.9}\text{PbI}_3$, where the 10 mol% Cs^+ substitution entropy-stabilizes the
 1818 cubic α -phase at ambient conditions as described in Section 2.1. Device fabrica-
 1819 tion proceeded by sequential solution deposition of the hole transport layer, the per-
 1820 ovskite precursor solution, the electron transport layer, and the top metal contact
 1821 onto glass/ITO substrates. The layer stack, illustrated in Figure 4.1, is ITO / hole
 1822 transport layer / $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ / electron transport layer / Ag.

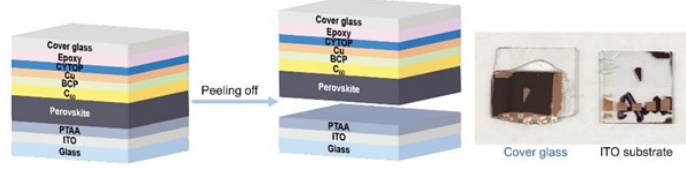


Figure 4.1: Cross-sectional schematic of the *pin* perovskite device structure used throughout this chapter. The ITO/hole-transport-layer (HTL) contact serves as the illuminated face for both photovoltaic and radiation-monitoring operation; the Ag back contact completes the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ absorber sandwich.

Two device geometries were characterized: a small-area cell ($\sim 8 \text{ mm}^2$ active area) for high-throughput screening and an encapsulated module (63.7 cm^2 aperture area, four cells wired in series) intended for radiation monitoring applications. Representative current density–voltage (J–V) curves measured under 1-sun AM1.5G illumination are shown in Figure 4.2.

The small-cell PCE of 21.3% is consistent with champion efficiencies reported for the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ composition in the contemporary literature [52], confirming that the devices received for radiation testing represent the current state of the art in perovskite photovoltaic fabrication. The module efficiency of 16.9% reflects the geometric fill-factor penalty inherent in a series-connected configuration and is comparable to certified module efficiencies from the same composition reported by independent groups [51].

4.2 Temporal Photoresponse Characterization

Before proceeding to ionizing radiation measurements, the time-domain photoresponse of the devices was characterized under visible light to establish the inherent

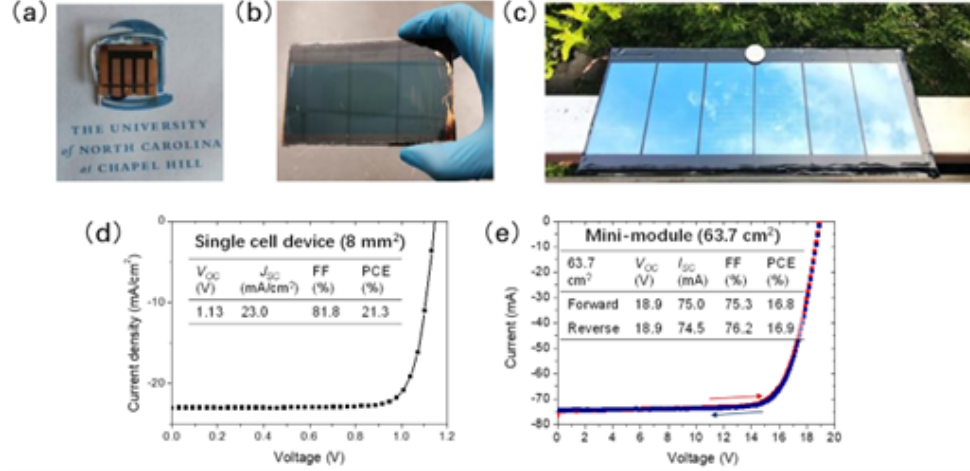


Figure 4.2: Photovoltaic performance of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ devices used in this study. (a) J-V curve for the 8 mm² small cell (PCE = 21.3%, V_{OC} = 1.13 V, J_{SC} = 23.0 mA cm⁻², FF = 81.8%). (b) J-V curve for the 63.7 cm² module (PCE = 16.9%, V_{OC} = 18.9 V, J_{SC} = 74.5 mA cm⁻², FF = 76.2%). All measurements performed at UNC Chapel Hill under AM1.5G illumination prior to shipment.

bandwidth of the detector and to identify any device-physics limitations on pulse-following ability. Three complementary measurement modalities were employed: a DC chopped-beam experiment to measure the turn-on and turn-off time constants, a pulsed-laser experiment to probe the high-frequency limit, and an AC-coupled grid-simulation experiment to assess the feasibility of power-grid synchronization using the solar module.

4.2.1 DC Transient Response

The DC transient measurement used a 532 nm continuous-wave laser attenuated to approximately 0.45 sun equivalent irradiance and modulated by a mechanical optical chopper. Short-circuit current transients were acquired on a digital oscilloscope and analyzed to extract the 10–90% rise time and the 90–10% fall time.

1849 The 8 mm² cell exhibited a rise time of approximately 33 ms, while the 28.6 cm²
1850 module required approximately 60 ms to reach 80% of its peak short-circuit current
1851 after the chopper opened. Both values are substantially longer than the RC time
1852 constants computed from the device capacitance and load resistance, confirming that
1853 the observed response time is limited by the chopper blade transit speed rather than
1854 by any intrinsic device physics. This conclusion is consistent with the known sub-
1855 microsecond free-carrier lifetimes in high-quality Cs-FA perovskite films [31].

1856 4.2.2 Pulsed Laser Response

1857 To access the true high-frequency bandwidth of the devices, a pulsed diode laser
1858 (Thorlabs NPL41B, 405 nm central wavelength) was used in place of the chopped CW
1859 source. The laser was operated at a repetition rate of 1 kHz with a pulse width of
1860 20 ns (variable 6–38 ns range, up to 10 MHz maximum repetition rate). Photoresponse
1861 pulses were amplified by a transimpedance amplifier with 10⁵ V A⁻¹ gain and 1 MHz
1862 bandwidth before acquisition on a Keysight MSO-X-3104T oscilloscope.

1863 Two series-parallel module assemblies, denoted Cell A and Cell B, each consist-
1864 ing of four Cs_{0.1}FA_{0.9}PbI₃ cells mounted on a printed breadboard, were character-
1865 ized. Cell A exhibited an output pulse rise time of approximately 84 ns under these
1866 conditions; Cell B produced a comparable waveform. The measured rise time is
1867 dominated by the 1 MHz bandwidth of the transimpedance amplifier rather than
1868 the device itself: the -3 dB amplifier bandwidth corresponds to a rise-time limit of
1869 0.35/10⁶ Hz $\approx$ 350 ns, and the shorter observed value reflects the partial bandwidth
1870 contribution of the oscilloscope front-end and cable parasitics. The intrinsic device

1871 response is therefore faster than the approximately 84 ns system-limited value, consis-
1872 tent with the sub-microsecond carrier lifetimes expected from the $\mu\tau$ products typical
1873 of $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ thin films reported in Section 2.4.2.

1874 4.2.3 AC Grid Simulation

1875 To evaluate whether the solar module could track the periodic voltage waveform
1876 produced by a single-phase power inverter — a potential application for verifying
1877 grid integrity or synchronizing a radiation monitor to a power-cycle reference —
1878 the four-cell module was connected to the input of a commercial single-phase DC-
1879 to-AC microinverter under indoor fluorescent illumination. The inverter consumed
1880 approximately 4 W at its rated load, but the module could deliver only ~ 1 W under
1881 the available illumination, leaving the inverter operating in a deeply underloaded
1882 regime. No distinct 50/60 Hz modulation of the module output current was discernible
1883 above the noise floor in this configuration. The negative result is attributed entirely
1884 to the power deficit rather than to any bandwidth limitation of the perovskite device:
1885 achieving an observable AC response would require either a larger-aperture module,
1886 a light concentrator, or a calibrated illumination source at close to 1-sun intensity.

1887 4.3 X-Ray Response

1888 X-ray characterization was performed using an Amptek Mini-X miniature X-ray
1889 tube with a gold anode, operated at tube voltages up to 50 kV. For continuous-
1890 wave measurements, the module was placed in the beam path and J-V characteristics
1891 were recorded as a function of X-ray tube voltage and current. Figure 4.3 shows the
1892 resulting I-V curves; the short-circuit current I_{SC} increases monotonically with tube
1893 voltage, consistent with the increasing photon fluence and mean photon energy.

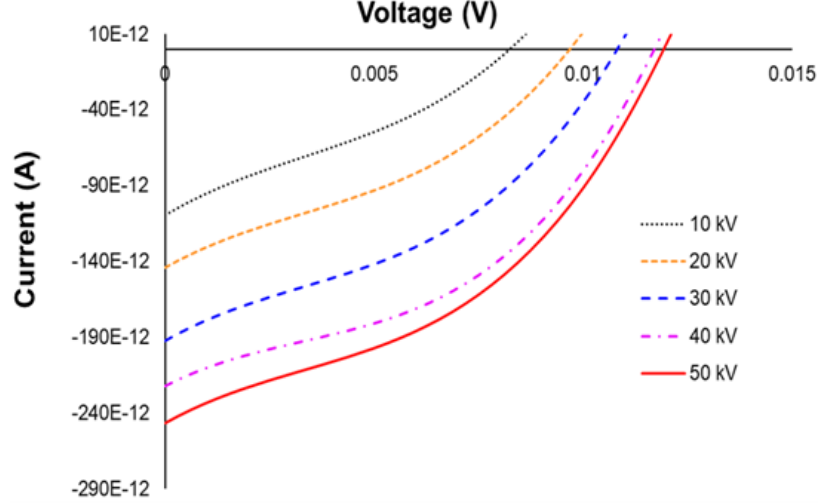


Figure 4.3: Current-voltage characteristics of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module under Amptek Mini-X X-ray tube illumination at tube voltages from 10 kV to 50 kV. Inset: I_{SC} at 50 kV as a function of irradiation time, showing the characteristic slow decline associated with ion migration under the built-in field [93].

1894 The time-resolved I_{SC} at 50 kV, shown in the inset of Figure 4.3, exhibits the
1895 characteristic slow-decline behavior previously reported for perovskite photovoltaic
1896 devices under sustained illumination in the presence of a built-in electric field. The
1897 decline is attributed to ion migration: mobile halide vacancies drift under the de-
1898 vice's built-in field, accumulating at the charge-transport layer interfaces and partially
1899 screening the collection field [93, 21].

1900 The absolute I_{SC} induced by the X-ray source is substantially smaller than that
1901 observed in the silicon reference device operated under the same conditions, owing to
1902 the much smaller active area of the perovskite module relative to the silicon photo-
1903 diode used for comparison ($\sim 64 \text{ cm}^2$ vs. several hundred cm^2). On a per-unit-area
1904 basis, the perovskite compares favorably due to its higher photoelectric absorption
1905 cross section at the relevant photon energies.

For pulsed X-ray measurements, the Amptek Mini-X beam was modulated by a Thorlabs MC2000 optical chopper equipped with a 10-slot blade (760 μm stainless steel, slot width matched to achieve pulse frequencies from 10 Hz to 1 kHz). The pulsed response confirmed that the device faithfully follows the beam modulation at frequencies up to the chopper bandwidth, consistent with the pulsed laser measurements in Section 4.2.2.

4.4 Gamma-Ray Response

4.4.1 Measurement Setup and Dose Rate Calibration

Gamma-ray irradiation experiments were performed at the Ohio State University Nuclear Reactor Laboratory (OSU NRL) using two independent sources: a benchtop ^{137}Cs irradiator (six source rods, $E_\gamma = 662$ keV) and a ^{60}Co irradiator submerged in the reactor pool ($E_\gamma = 1.17$ and 1.33 MeV). Because the dose rate calibration values provided with each irradiator are given in units of roentgen per hour for air, conversion to the absorbed dose rate in the perovskite film was performed using the relation

$$\dot{D}_{\text{perovskite}} = 0.877 \times \frac{(\mu_{\text{en}}/\rho)_{\text{perovskite}}}{(\mu_{\text{en}}/\rho)_{\text{Si}}} \times \dot{X}, \quad (4.1)$$

where $\dot{X}$ is the exposure rate in roentgen per hour, 0.877 rad/R is the roentgen-to-rad conversion factor for soft tissue [48], and the ratio of mass energy-absorption coefficients (μ_{en}/ρ) was evaluated at the mean photon energy of each source using NIST XCOM tabulations [10]. This procedure yielded dose rates of 44.16 rad hr $^{-1}$ in the perovskite layer for the ^{137}Cs irradiator and 355.61 rad hr $^{-1}$ for the ^{60}Co source.

1926 4.4.2 Cesium-137 Irradiation

1927 The $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module was inserted into and withdrawn from the ^{137}Cs irra-
 1928 diator while the short-circuit current was monitored in real time. Figure 4.4 shows a
 1929 time-series of I_{SC} recorded during a representative insert/withdraw cycle. A clear and
 1930 reproducible step change in I_{SC} is observed upon insertion, with the current increasing
 1931 by an amount proportional to the gamma-induced electron-hole pair generation rate
 1932 within the absorber. The induced current density at $44.16 \text{ rad hr}^{-1}$ was measured as
 1933 2.19 nA cm^{-2} , which represents the net gamma-to-current conversion sensitivity of
 1934 the device under the prevailing irradiation geometry.

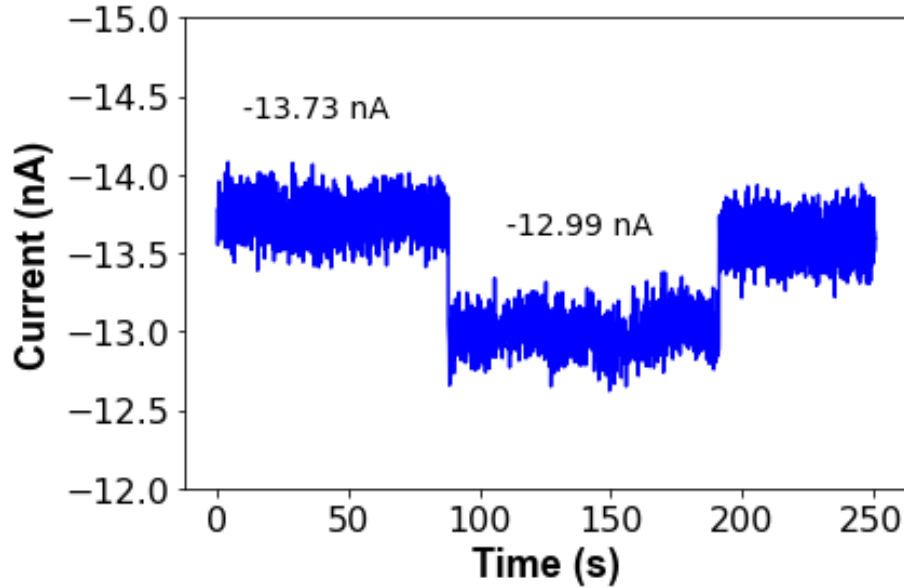


Figure 4.4: Short-circuit current I_{SC} of the $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ module as a function of time during insertion into and withdrawal from the ^{137}Cs irradiator (dose rate = $44.16 \text{ rad hr}^{-1}$ in perovskite). The step changes in I_{SC} provide an unambiguous current-mode detection signal.

1935 To assess the long-term stability of the perovskite module as a radiation monitor,
 1936 the device was subjected to continuous ^{137}Cs irradiation for six days without removal
 1937 from the source. The accumulated absorbed dose over the six-day period was com-
 1938 puted by integrating the dose rate over the irradiation time. Figure 4.5 shows the
 1939 evolution of I_{SC} as a function of cumulative absorbed dose. A linear decrease in I_{SC}
 1940 with dose is observed, with a slope of $-0.74 \text{ nA rad}^{-1}$, corresponding to a relative
 1941 degradation rate of approximately $[I_{\text{SC}}(0) - I_{\text{SC}}(D)] / I_{\text{SC}}(0)$ per unit absorbed dose.

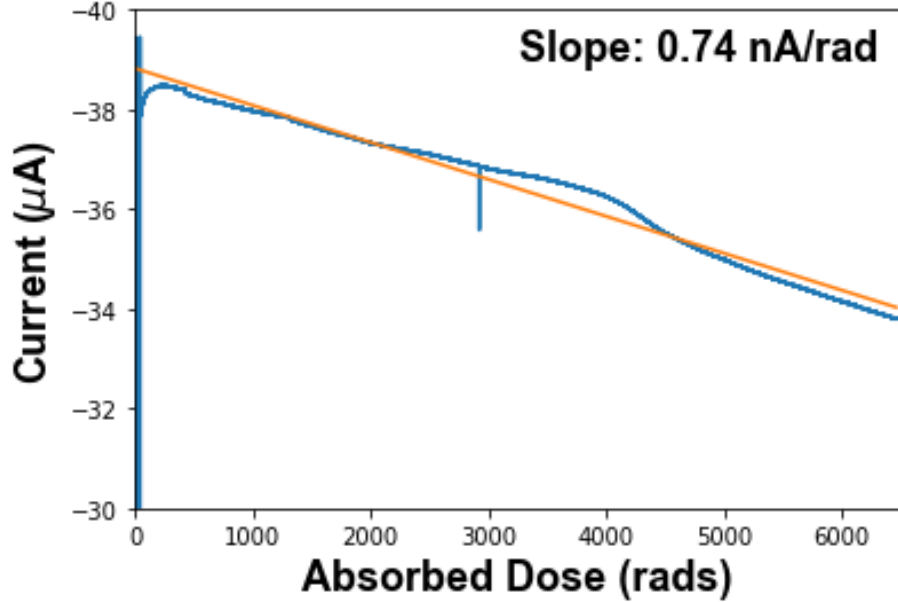


Figure 4.5: Long-term evolution of I_{SC} under continuous ^{137}Cs irradiation ($44.16 \text{ rad hr}^{-1}$) over six days. The linear decline in I_{SC} with accumulated dose is attributed to a combination of radiation-induced defect creation in the perovskite absorber and progressive reduction in the optical transmittance of the glass superstrate [13].

1942 The observed linear decline is consistent with the creation of electrically active
 1943 defects — primarily iodide Frenkel pairs — proportional to the absorbed dose, as

has been documented for ^{60}Co irradiation of MAPbI_3 single crystals at doses up to 1000 krad [13]. An additional contribution to the I_{SC} decline comes from cumulative darkening of the borosilicate glass superstrate, which reduces the visible-light transmittance and hence the photovoltaic current even in the absence of absorber degradation; this contribution has been documented separately [14].

4.4.3 Cobalt-60 Irradiation

The module was subsequently irradiated in the OSU NRL reactor pool using the underwater ^{60}Co irradiator at a dose rate of $355.61 \text{ rad hr}^{-1}$ in the perovskite layer — approximately eight times higher than the ^{137}Cs rate. The measurement geometry was otherwise identical: I_{SC} was recorded continuously throughout a 16-day irradiation. The induced current density at $355.61 \text{ rad hr}^{-1}$ was 3.56 nA cm^{-2} .

Figure 4.6 shows the I_{SC} time series for the ^{60}Co irradiation. As with ^{137}Cs , a downward linear trend in the baseline I_{SC} is observed over the 16-day period, consistent with progressive defect accumulation proportional to absorbed dose.

These results demonstrate that arrays of calibrated perovskite modules could, in principle, be deployed in a spatial configuration around a radiation source to triangulate the detonation location or measure the spatial dose distribution by comparing the I_{SC} responses of individual array elements. The linear dose–current relationship established here provides the calibration function necessary for such an application.

4.5 Neutron Response

Neutron response measurements were performed at the OSU Nuclear Reactor Laboratory using the facility’s Fast Beam Facility (FBF) beamport. The experimental geometry placed the perovskite module alternately outside the beam (background),

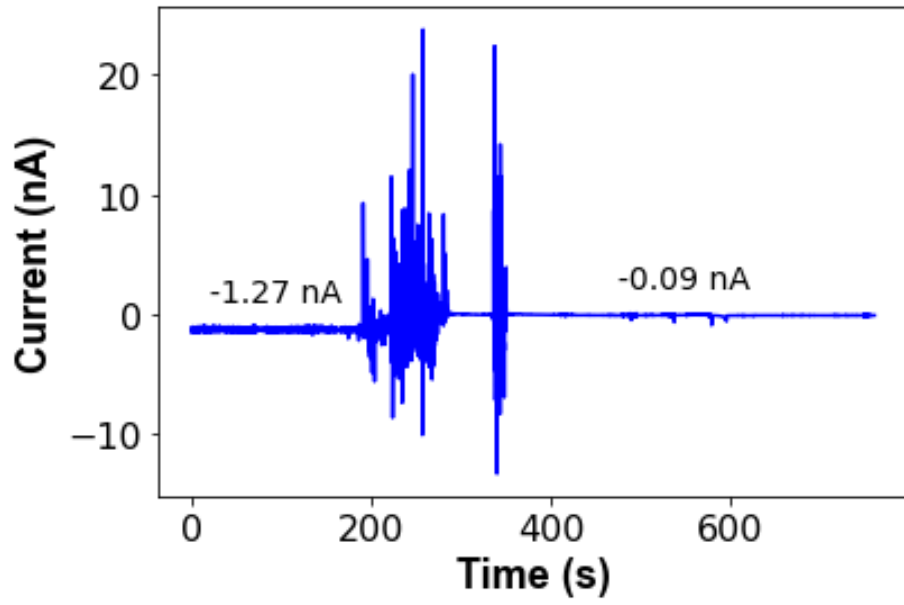


Figure 4.6: Short-circuit current I_{SC} of the module during 16-day continuous ^{60}Co irradiation (dose rate = $355.61 \text{ rad hr}^{-1}$ in perovskite). The downward trend mirrors that observed for ^{137}Cs (Figure 4.5) and scales approximately with the higher absorbed dose rate.

1967 in the beam with the gamma-suppression filter closed, and in the beam with the
 1968 filter open, enabling separation of the neutron-induced and gamma-induced current
 1969 components.

1970 4.5.1 Steady-State Reactor Measurements

1971 Table 4.1 summarizes the short-circuit current I_{SC} measured under each condi-
 1972 tion, referenced to reactor power levels of 0 kW_{th} (reactor shutdown) and 450 kW_{th}
 1973 (operating power).

Table 4.1: Steady-state short-circuit current of the Cs_{0.1}FA_{0.9}PbI₃ module at OSU
 NRL FBF. Uncertainties are $\pm 1\sigma$ over the measurement interval. “Gamma filter
 closed” and “gamma filter open” refer to the polyethylene/lead shielding assembly that
 preferentially attenuates the reactor gamma flux while transmitting fast neutrons.

Condition	Reactor Power	I_{SC} (pA)
Background (reactor off)	0 kW _{th}	51.29 ± 6.60
Background (reactor on)	450 kW _{th}	57.50 ± 7.39
Gamma filter closed	450 kW _{th}	57.62 ± 7.93
Gamma filter open	450 kW _{th}	59.77 ± 8.62

1974 The data in Table 4.1 reveal a slight but statistically marginal increase in I_{SC} upon
 1975 opening the gamma filter (from 57.62 ± 7.93 pA to 59.77 ± 8.62 pA). The difference of
 1976 ~ 2.2 pA represents a $\sim 0.26\sigma$ effect relative to the combined quadrature uncertainty,
 1977 which is not statistically significant at the 95% confidence level. The result is therefore
 1978 classified as inconclusive: a neutron-induced contribution cannot be ruled out, but
 1979 the signal-to-noise ratio of a single 8 mm²–63.7 cm² module is insufficient to resolve
 1980 it above the fluctuating gamma and ambient backgrounds present at the FBF port.

1981 These results motivate the use of a larger-area array for future neutron sensing
1982 experiments. Scaling the active area by a factor of ten to twenty, combined with the
1983 differential signal extraction method described by Panaccione et al. [73], is expected
1984 to raise the neutron-to-background ratio above the detection threshold.

1985 4.5.2 Chopped-Beam Transient Measurements

1986 In an attempt to resolve a time-modulated neutron signal against the steady-state
1987 gamma background, the reactor beam was mechanically chopped at frequencies be-
1988 tween 1 Hz and 100 Hz using a borated polyethylene blade. Synchronous detection
1989 (lock-in analysis) of the I_{SC} transient was implemented in software. No neutron-
1990 induced current component was detected above the noise floor under any chopping
1991 frequency in this configuration, consistent with the steady-state analysis: the signal
1992 magnitude is simply too small relative to the electronic noise of the current mea-
1993 surement system at the flux levels available from the 450 kW_{th} reactor at the FBF
1994 port.

1995 4.6 Summary

1996 This chapter has demonstrated that photovoltaic-architecture Cs_{0.1}FA_{0.9}PbI₃ per-
1997 ovskite solar cells can serve as sensitive current-mode monitors for a broad spectrum
1998 of ionizing radiation. The key results are as follows.

1999 Temporally, the device bandwidth is limited by the measurement electronics rather
2000 than the intrinsic perovskite photoresponse: rise times of 33–84 ns were recorded
2001 depending on the amplifier configuration, and the actual carrier lifetime in the film
2002 is substantially shorter than the measured system rise time.

2003 Under X-ray irradiation (Amptek Mini-X, up to 50 kV), the module produces a
2004 measurable short-circuit current proportional to the beam intensity, with the char-
2005 acteristic slow I_{SC} decline associated with ion-migration field screening under the
2006 device's built-in field [93].

2007 Under ^{137}Cs gamma irradiation at $44.16 \text{ rad hr}^{-1}$, an induced current density of
2008 2.19 nA cm^{-2} was measured, and a distinct step response was observed upon inser-
2009 tion and withdrawal. Long-term (six-day) continuous irradiation revealed a linear
2010 decline of $\sim 0.74 \text{ nA rad}^{-1}$ attributable to cumulative defect creation and glass dark-
2011 ening [13, 14]. The ^{60}Co measurement at $355.61 \text{ rad hr}^{-1}$ yielded 3.56 nA cm^{-2} and
2012 an equivalent linear degradation trend over sixteen days, confirming the scalability
2013 of the current-mode detection approach across a dose-rate range spanning nearly an
2014 order of magnitude.

2015 Neutron response at the OSU NRL FBF was statistically inconclusive for a single
2016 module at 450 kW_{th} ; the signal-to-noise analysis indicates that scaling the detector
2017 array area by a factor of ten to twenty is necessary to resolve the neutron-induced
2018 current above the combined gamma and electronic noise backgrounds.

Chapter 5: Digital Electronics Verification Methods for Radiation Environments

5.1 Introduction

Radiation environments impose two fundamentally distinct threats on digital electronics: displacement damage and total ionizing dose (TID) acting as cumulative material-degradation mechanisms, and single-event effects (SEEs) acting as stochastic, event-driven disturbances. Of these, single-event upsets (SEUs) are of particular concern for SRAM-based field-programmable gate arrays (FPGAs) because such devices store both user data and their own circuit definition in volatile static-RAM cells that are as vulnerable to ionizing particles as any other SRAM. Assessing and mitigating this vulnerability requires purpose-built verification instruments that can observe radiation-induced state changes in real time, with sufficient state-volume coverage to yield statistically meaningful upset rates, yet compact enough to deploy inside a beam facility with limited infrastructure.

This chapter describes the design, implementation, and experimental results of the Flip-Flop Shift Register (FFSR) test structure used to evaluate Xilinx Spartan-7 FPGAs under fast neutron irradiation at the Ohio State University Research Reactor (OSURR) Fast Neutron Beam Facility (FNBF). Section 5.2 reviews relevant radiation

2037 effects in FPGAs. Section 5.3 discusses verification challenges specific to FPGA-based
2038 systems in nuclear environments. Section 5.4 presents the FFSR RTL architecture
2039 and governing algorithms. Section 5.5 describes the test configuration and radiation
2040 environment. Section 5.6 presents the experimental results from two reactor runs
2041 and the intervening overnight monitoring period. Section 5.7 discusses the observed
2042 upset mechanisms and their implications, and Section 5.8 summarises the principal
2043 findings.

2044 5.2 Radiation Effects in FPGAs

2045 SRAM-based FPGAs are widely used in nuclear instrumentation and space-borne
2046 electronics because of their reconfigurability, high logic density, and short design
2047 cycles. However, their reliance on SRAM storage cells for both configuration memory
2048 and user state makes them susceptible to a range of radiation-induced failure modes.

2049 **Single-Event Upsets.** An SEU occurs when a charged particle deposits suffi-
2050 cient energy in an SRAM cell to flip its stored bit. In the user data fabric, an SEU
2051 manifests as a transient or persistent logical error in the circuit under test. In the
2052 configuration SRAM, the consequence is more severe: a bit flip can alter LUT con-
2053 tents, routing multiplexers, embedded constant strings, or timing-control registers,
2054 effectively changing the implemented circuit itself. Because the configuration SRAM
2055 is not typically protected by on-chip error correction in commercial devices, such
2056 upsets persist until the device is power-cycled or a scrubbing operation rewrites the
2057 affected cell.

2058 **Single-Event Latch-up.** High-energy particle strikes can trigger parasitic bipo-
2059 lar structures in bulk CMOS, creating a low-impedance path between supply and

2060 ground. The resulting latch-up condition draws large currents, halts device opera-
2061 tion, and can cause permanent damage if not cleared by a prompt power cycle. The
2062 probability of latch-up increases with particle flux and LET.

2063 **Total Ionizing Dose Effects.** Prolonged irradiation builds up trapped charge
2064 in gate oxides and field oxides, increasing leakage current, shifting threshold voltages,
2065 and ultimately degrading transistor drive strength. TID effects accumulate monoton-
2066 ically with fluence and manifest at the system level as a gradual increase in supply
2067 current drawn by the device.

2068 **Displacement Damage.** Fast neutrons displace lattice atoms through nuclear
2069 recoil, creating interstitial-vacancy pairs that act as recombination centres and carrier
2070 traps. In CMOS logic this mechanism is generally less significant than in minority-
2071 carrier devices (e.g., solar cells or bipolar transistors), but at high fast-neutron fluences
2072 it contributes to parametric degradation.

2073 5.3 FPGA Verification Challenges in Nuclear Systems

2074 Several practical constraints complicate the experimental characterisation of FPGA
2075 upset susceptibility in nuclear beam facilities.

2076 **State-volume coverage.** An SEU cross-section measurement requires observing
2077 a statistically significant number of events. The cross-section for a single memory
2078 cell is of order 10^{-14} to 10^{-12} cm², so at a flux of 10^5 – 10^7 n/cm²/s a single cell
2079 would require impractically long exposures to accumulate even one event on average.
2080 Maximising the monitored state population is therefore essential.

2081 **Observability without intrusion.** Reading every flip-flop off-chip individually
2082 during irradiation is infeasible for large state populations and introduces additional

2083 wiring susceptible to radiation and noise. A compact on-chip checker that compresses
2084 the health of the state array into a small digital flag is preferable.

2085 **Configuration vs. data upsets.** Any experimental design must be capable of
2086 distinguishing a user-data upset from a configuration memory upset. The two mech-
2087 anisms require different mitigation strategies: data-path redundancy (DMR/TMR)
2088 addresses the former, while periodic configuration scrubbing addresses the latter.

2089 **Telemetry reliability.** The UART or other communication channel used to
2090 export results is itself implemented in FPGA resources and therefore subject to the
2091 same configuration upsets. A robust transmitter design reduces the risk that the
2092 observability channel itself is corrupted during an experiment.

2093 5.4 FFSR RTL Architecture

2094 5.4.1 System-Level Overview

2095 The FFSR test structure is an FPGA-resident self-test instrument designed to
2096 address the observability and state-volume challenges described in Section 5.3. Its
2097 purpose is to instantiate a large, controllable population of sequential elements, ex-
2098 ercise those elements with known scan/capture behaviour, and report any deviation
2099 from the expected internal state in real time. The design contains four cooperating
2100 subsystems:

- 2101 1. A **top-level experiment controller** (`top_selftest_100mhz_dbg`) that gen-
2102 erates experiment timing, reset sequences, capture windows, and telemetry ca-
2103 dence.
- 2104 2. A **scan-farm fabric** (`ff_scanfarm / ff_scanfarm_sig`) that instantiates many
2105 shift-register chains in parallel, scaling the monitored state volume.

2106 3. A **signature checker** that computes an expected compact signature from the
2107 chain contents and compares it against the observed signature to produce live
2108 and latched error flags.

2109 4. A **UART telemetry path** that serialises concise status words for logging on
2110 a host computer, enabling offline correlation with beam current, fluence, and
2111 shutter state.

2112 Rather than streaming every monitored bit off chip, the design compresses the
2113 health of a large state space into a smaller set of mismatch indicators and coun-
2114 ters. This separation of upset *occurrence* from complete state readout enables larger
2115 monitored populations and simpler experimental integration while still supporting
2116 controlled injection tests and persistent error latching.

2117 5.4.2 Module Descriptions

2118 Table 5.1 summarises the primary RTL blocks.

2119 **top_selftest_100mhz_dbg**

2120 The `top_selftest_100mhz_dbg` module is the supervisory block of the FFSR
2121 experiment. It establishes the operating clock domain, applies reset, sequences the
2122 scan and capture phases, clears latched errors when requested, and routes status
2123 information into the UART telemetry path. In a typical implementation, operation
2124 is divided into repeated frames or epochs: the design allows the farm to evolve or
2125 hold state, then enables capture, then enables scan shifting, then examines mismatch
2126 outputs, and finally serialises the results.

Table 5.1: Summary of primary RTL blocks in the FFSR design.

Module	Primary Role	Key Parameters / Signals	Main Idea	Algorithmic
<code>top_selftest_100mhz_dbg</code>	Top-level experiment controller and telemetry wrapper	Clock, reset, scan enable, capture enable, <code>clear_errors</code> , UART handshake, status/counter outputs	Coordinates test phases; periodically packages status for serial transmission	
<code>ff_scanfarm</code>	Wrapper layer for the scan farm	<code>CHAINS</code> , <code>BITS_PER_CHAIN</code> , <code>INJECT_EN</code> , <code>INJ_TAP</code> , <code>error_live</code> , <code>error_any</code>	Preserves a stable top-level interface while instantiating the robust signature-based implementation	
<code>ff_scanfarm_sig</code> / scan-chain fabric	Core monitored state array	<code>clk_scan</code> , <code>scan_en</code> , <code>capture_en</code> , <code>scan_in/out</code> , signature bits, mismatch flags	Updates each chain per shift/capture rules; compares compact state signature against expected value	
<code>scan_chain</code> primitive	One lane of monitored sequential storage	Serial input/output, per-stage registers, optional injection tap	Implements shift-register recurrences; optional controlled bit-flip insertion for self-verification	
<code>prbs31</code>	Pseudo-random pattern source	31-bit state, enable/reset, PRBS output bit/word	Maximal-length LFSR using $x^{31} + x^{28} + 1$; $2^{31} - 1$ states before repetition	
<code>uart_tx</code> / <code>uart_tx_robust</code>	UART serialiser; robust variant adds framing protection	<code>tx_start</code> , <code>tx_data</code> , baud divider, <code>tx_busy</code> , <code>txd</code>	Converts parallel byte to start bit + 8 data bits + stop bit at programmable baud rate; robust variant prevents malformed frames	

2127 **ff_scanfarm and ff_scanfarm_sig**

2128 The **ff_scanfarm** module acts as a named interface to the monitored sequential
2129 fabric. In the wrapper variant used in this project, it preserves a stable external inter-
2130 face while instantiating the more robust internal implementation **ff_scanfarm_sig**.
2131 This decouples the project hierarchy from changes in the checker implementation.
2132 The key outputs are a real-time mismatch indicator, **error_live**, and a sticky fault
2133 memory, **error_any**.

2134 The scan-farm core is parameterised by the number of parallel chains C and the
2135 number of bits per chain B , giving a total monitored state volume of:

$$N_{\text{state}} = C \times B. \quad (5.1)$$

2136 **scan_chain Primitive**

2137 At the lowest level, the **scan_chain** logic implements the per-stage shift-register
2138 recurrence. A chain receives a serial input bit, shifts stored data through successive
2139 registers when **scan_en** is asserted, and presents the final stage at **scan_out**. An
2140 optional injection feature can flip a selected stage, allowing the checker path to be
2141 validated without radiation. Because the chain state is deterministic under com-
2142 manded scan and capture sequences, any uncommanded state change is detectable
2143 through signature mismatch.

2144 **5.4.3 Core Algorithms**

2145 **Shift-Register State Update**

2146 Let $q_{c,i}[k]$ denote the state of stage i in chain c at clock step k , where $i =$
2147 $0, 1, \dots, B-1$ and $c = 0, 1, \dots, C-1$. During scan mode the chain obeys the standard

2148 shift recurrence:

$$q_{c,0}[k+1] = u_c[k], \quad (5.2)$$

$$q_{c,i}[k+1] = q_{c,i-1}[k], \quad i > 0, \quad (5.3)$$

2149 where $u_c[k]$ is the applied serial scan input for chain c . In hold mode, $q_{c,i}[k+1] =$
 2150 $q_{c,i}[k]$.

2151 **Optional Upset Injection for Verification**

2152 To validate the detector path without radiation exposure, a controlled injection
 2153 can be applied at tap t_{inj} :

$$q_{c,t_{\text{inj}}}[k+1] = q_{c,t_{\text{inj}}}^{\text{nom}}[k+1] \oplus \text{inj}[k], \quad (5.4)$$

2154 where $\text{inj}[k]$ is asserted only during a commanded test window and q^{nom} denotes the
 2155 nominal update in the absence of injection.

2156 **Signature Computation and Mismatch Detection**

2157 A compact checker is formed by reducing chain contents into parity-like signatures.
 2158 The signature for chain c is:

$$s_c[k] = \bigoplus_{i=0}^{B-1} q_{c,i}[k]. \quad (5.5)$$

2159 The fabric-level signature vector is $\mathbf{S}[k] = [s_0[k], s_1[k], \dots, s_{C-1}[k]]$. The instantane-
 2160 ous mismatch flag is:

$$e_{\text{live}}[k] = \begin{cases} 1 & \text{if } \mathbf{S}_{\text{obs}}[k] \neq \mathbf{S}_{\text{exp}}[k], \\ 0 & \text{otherwise.} \end{cases} \quad (5.6)$$

2161 **Sticky Error Memory**

2162 A sticky fault flag preserves the occurrence of any prior mismatch even if the chain
2163 later returns to agreement:

$$e_{\text{any}}[k+1] = (e_{\text{any}}[k] \vee e_{\text{live}}[k]) \wedge \overline{\text{clear_errors}[k]}. \quad (5.7)$$

2164 This form is essential in beam testing because transient upsets are not lost between
2165 telemetry packets.

2166 **PRBS31 Recurrence**

2167 The `prbs31` module generates a pseudo-random binary sequence using a 31-bit
2168 linear feedback shift register with the primitive polynomial $p(z) = z^{31} + z^{28} + 1$. Using
2169 zero-based indexing with x_{30} as the most-significant stage, the feedback bit is:

$$f[k] = x_{30}[k] \oplus x_{27}[k], \quad (5.8)$$

2170 and the next state is:

$$\mathbf{x}[k+1] = \{x_{29}[k], x_{28}[k], \dots, x_0[k], f[k]\}. \quad (5.9)$$

2171 Provided the all-zero state is avoided, this recurrence cycles through $2^{31} - 1$ non-
2172 zero states before repeating, yielding strong transition density and good statistical
2173 coverage for stressing the sequential fabric.

2174 **UART Frame Formation**

2175 For an 8-bit telemetry byte $D = [d_0, d_1, \dots, d_7]$, the UART transmitter outputs:

$$\text{TX frame} = [0, d_0, d_1, \dots, d_7, 1], \quad (5.10)$$

2176 where 0 is the start bit and 1 is the stop bit. The nominal bit-period divider is:

$$N_{\text{baud}} = \left\lceil \frac{f_{\text{clk}}}{f_{\text{baud}}} \right\rceil, \quad (5.11)$$

2177 where $\lceil \cdot \rceil$ denotes rounding to the nearest integer. The `uart_tx_robust` variant
 2178 extends the baseline transmitter with stronger start-request handling, safer state
 2179 transitions, and additional protection against malformed back-to-back frames under
 2180 transient control behaviour.

2181 5.5 Test Configuration

2182 5.5.1 Devices Under Test

2183 Five Digilent CMOD S7 boards, each loaded with the FFSR dual modular redun-
 2184 dancy (DMR) firmware on a Xilinx Spartan-7 FPGA, were monitored simultaneously.
 2185 Each device was connected to the host data acquisition PC via a dedicated USB-to-
 2186 serial COM port at 115 200 baud. The DMR comparator within the FFSR design
 2187 monitored two redundant data paths and incremented a 32-bit upset counter when-
 2188 ever a mismatch was detected. The counter value was reported over UART every
 2189 5 seconds. A non-zero counter output indicated detection of one or more SEUs in the
 2190 user data path. Table 5.2 summarises the device configuration and test timeline.

Table 5.2: CMOD S7 device configuration and test timeline.

Device	COM Port	Baseline Start	Reactor Start	Reactor End
FFSR1	COM8	21:21, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR2	COM7	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR3	COM6	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR4	COM9	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27
FFSR5	COM4	22:41, Jan 26	11:42, Jan 27	16:34, Jan 27

2191 5.5.2 Radiation Environment

2192 The test was conducted at the OSURR Fast Neutron Beam Facility over two days
2193 using distinct beam configurations that produced substantially different neutron flux
2194 and fluence levels.

2195 **Day 1 (gamma shutter closed, January 27).** The reactor operated at approx-
2196 imately 300 kW (steady-state), reached within ~ 10 minutes of start-up and sustained
2197 for approximately 4.7 hours. The fast neutron flux (steady-state) was approximately
2198 2×10^5 n/cm²/s; the thermal component was approximately one order of magnitude
2199 lower. Total accumulated fast neutron fluence was approximately $6\text{--}7 \times 10^9$ n/cm².
2200 The fast and total fluence curves were nearly coincident throughout, confirming fast-
2201 neutron dominance.

2202 **Day 2 (gamma shutter open, January 28).** Reactor power remained at
2203 approximately 300 kW, but the open-shutter configuration produced a fast neutron
2204 flux of approximately 1.5×10^7 n/cm²/s—roughly two orders of magnitude higher
2205 than Day 1. The thermal flux was approximately 6×10^6 n/cm²/s (within a factor
2206 of 2–3 of the fast flux). Total accumulated fast neutron fluence was approximately
2207 $8\text{--}9 \times 10^{11}$ n/cm², also approximately two orders of magnitude greater than Day 1.

2208 Figures 5.1–5.3 show the neutron fluence, flux, and power profiles for the Day 1
2209 run; Figures 5.4–5.6 show the corresponding plots for the Day 2 run. Figures 5.7–5.9
2210 provide overlay comparisons of total fluence, flux, and reactor power between the two
2211 runs.

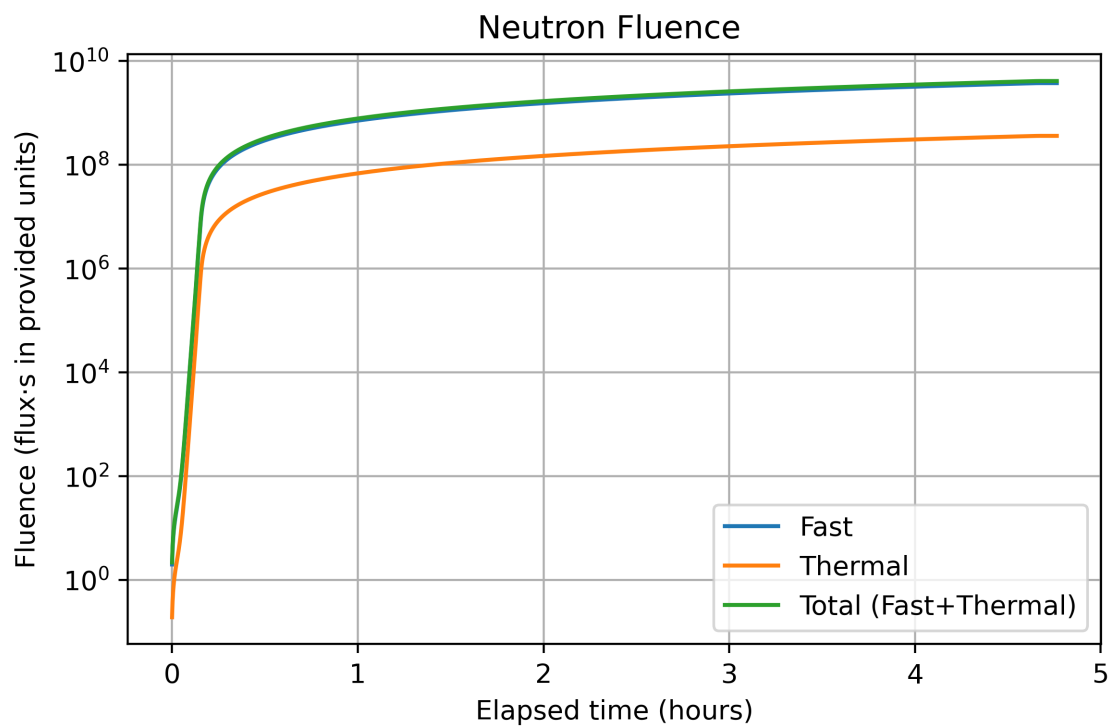


Figure 5.1: Neutron fluence (fast, thermal, and total) accumulated during the Day 1 (gamma shutter closed) reactor run at the OSURR FNBF. The fast and total curves are nearly coincident, confirming fast-neutron dominance at this beam configuration.

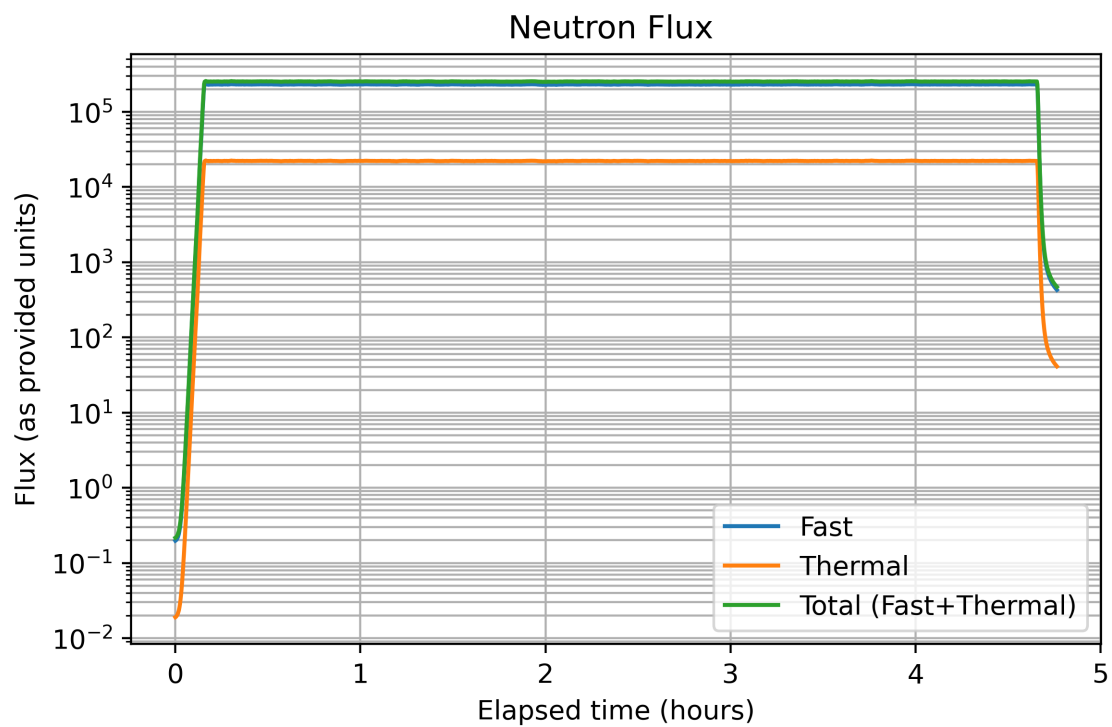


Figure 5.2: Neutron flux (fast, thermal, and total) during the Day 1 (gamma shutter closed) run. Fast flux reached a steady-state value of approximately 2×10^5 n/cm²/s; the thermal component was approximately one order of magnitude lower.

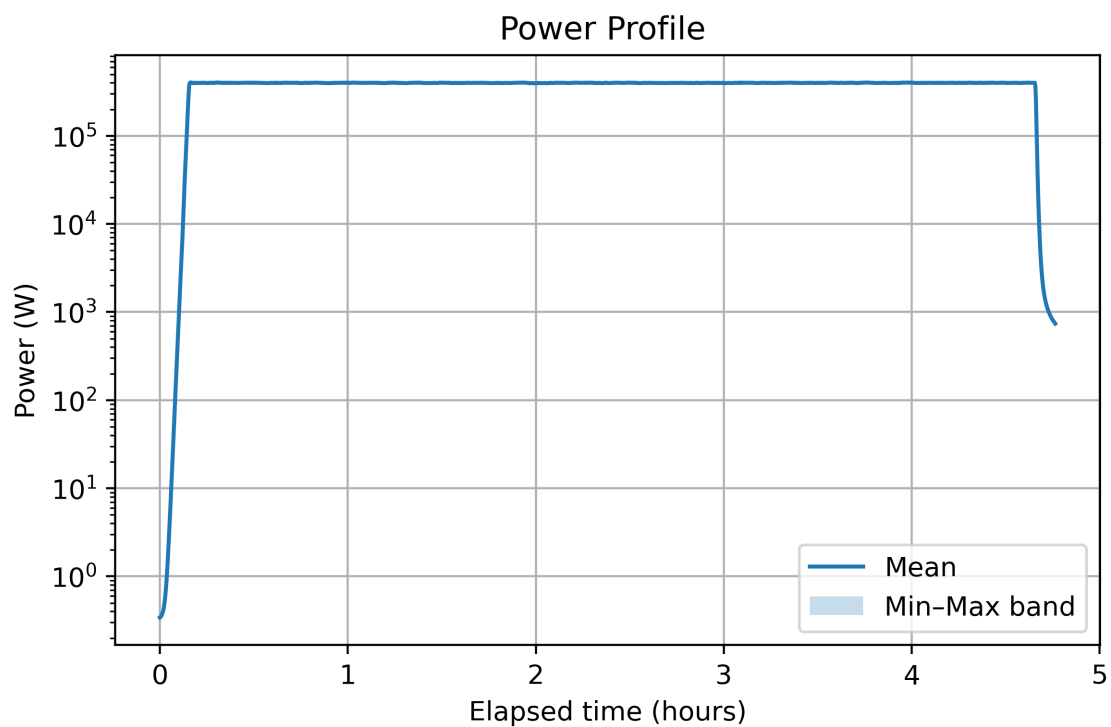


Figure 5.3: Reactor power profile during the Day 1 run. Full power (~ 300 kW) was reached within approximately 10 minutes of start-up and sustained for approximately 4.7 hours before controlled shutdown.

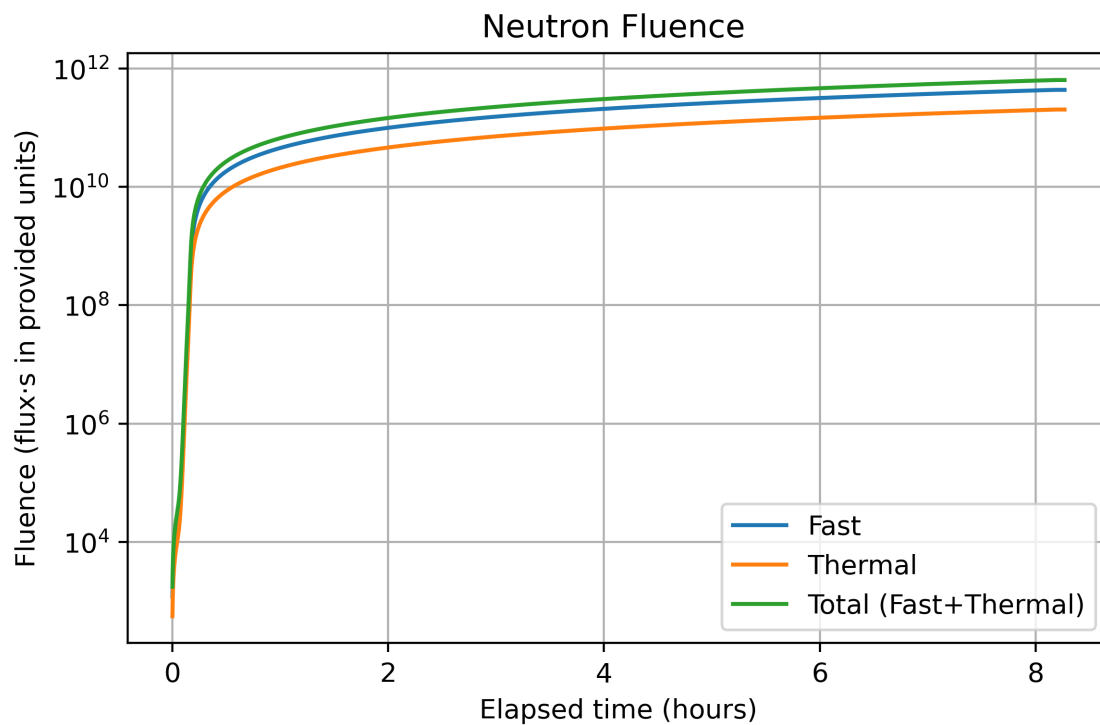


Figure 5.4: Neutron fluence (fast, thermal, and total) accumulated during the Day 2 (gamma shutter open) reactor run. Total fast fluence reached approximately $8\text{--}9 \times 10^{11} \text{ n/cm}^2$, roughly two orders of magnitude greater than Day 1.

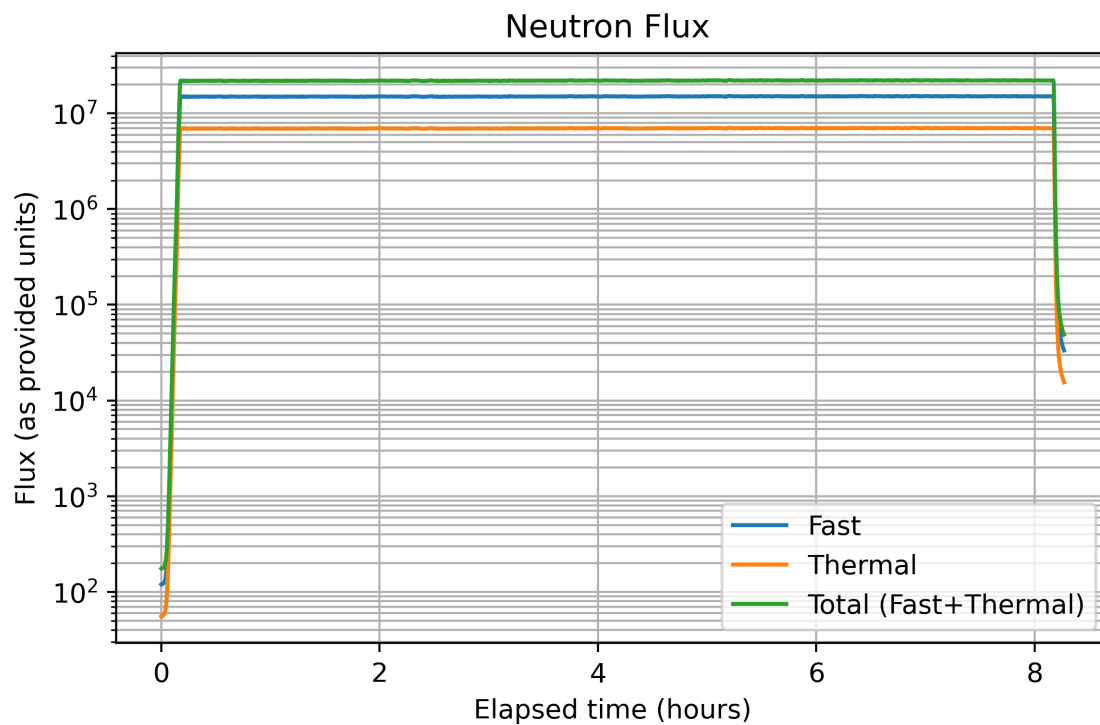


Figure 5.5: Neutron flux during the Day 2 (gamma shutter open) run. Fast flux reached approximately 1.5×10^7 n/cm²/s and thermal flux approximately 6×10^6 n/cm²/s, narrowing the fast-to-thermal ratio compared with Day 1.

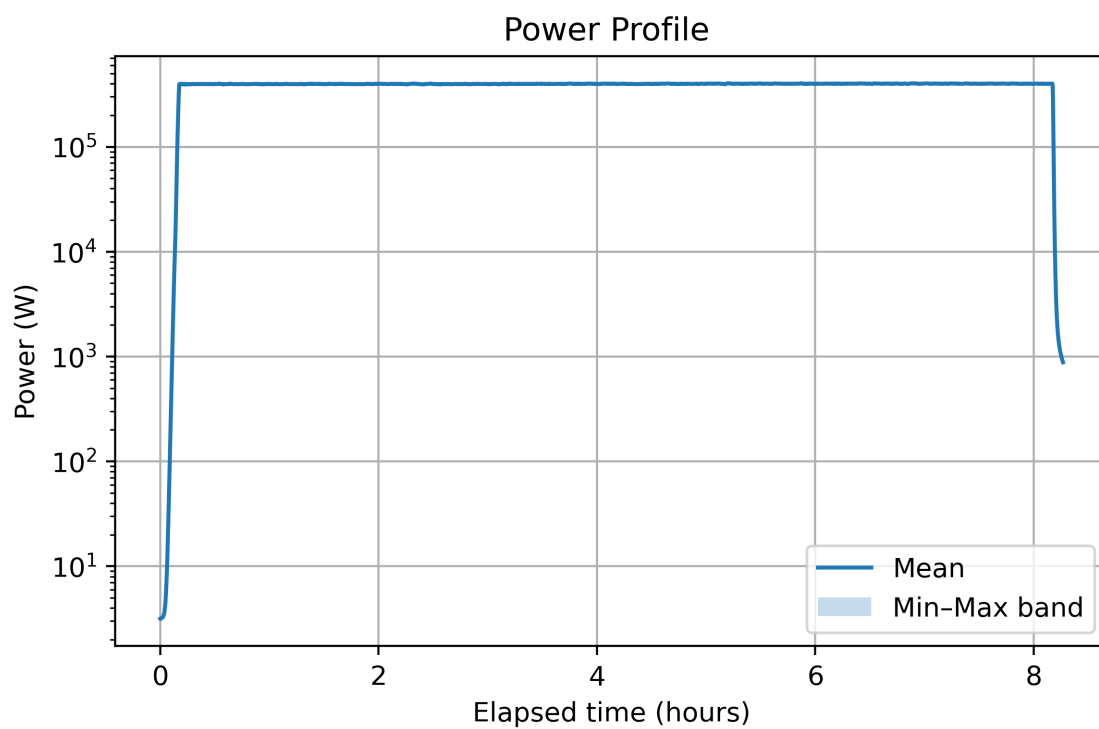


Figure 5.6: Reactor power profile during the Day 2 run. Full power was sustained for approximately 8.5 hours before shutdown.

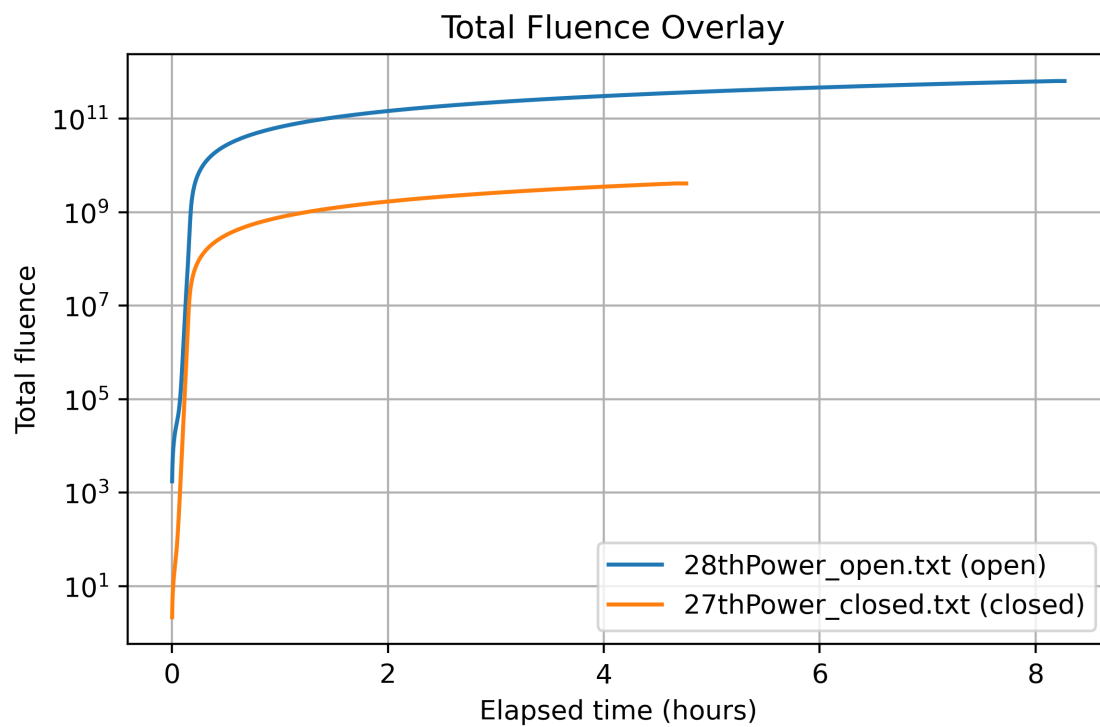


Figure 5.7: Overlay of total neutron fluence (fast + thermal) for the Day 1 (shutter closed) and Day 2 (shutter open) runs, illustrating the approximately two-orders-of-magnitude difference in accumulated fluence between the two configurations.

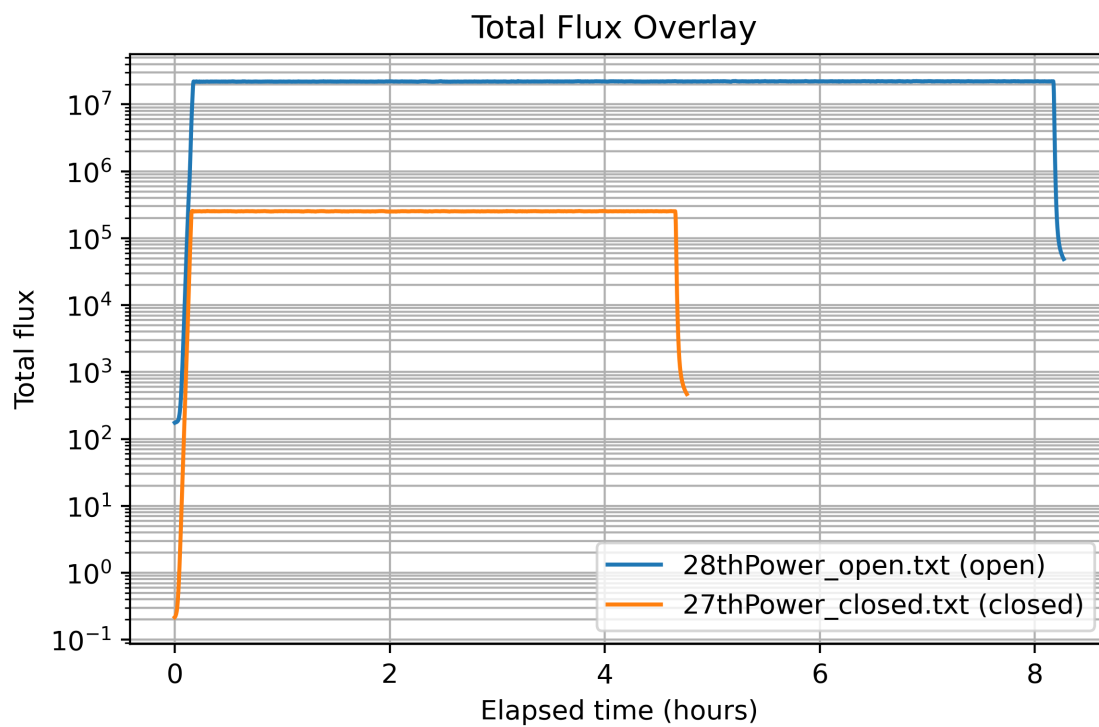


Figure 5.8: Overlay of total neutron flux for both runs. The Day 2 (open shutter) flux exceeded the Day 1 (closed shutter) flux by approximately two orders of magnitude throughout the steady-state period.

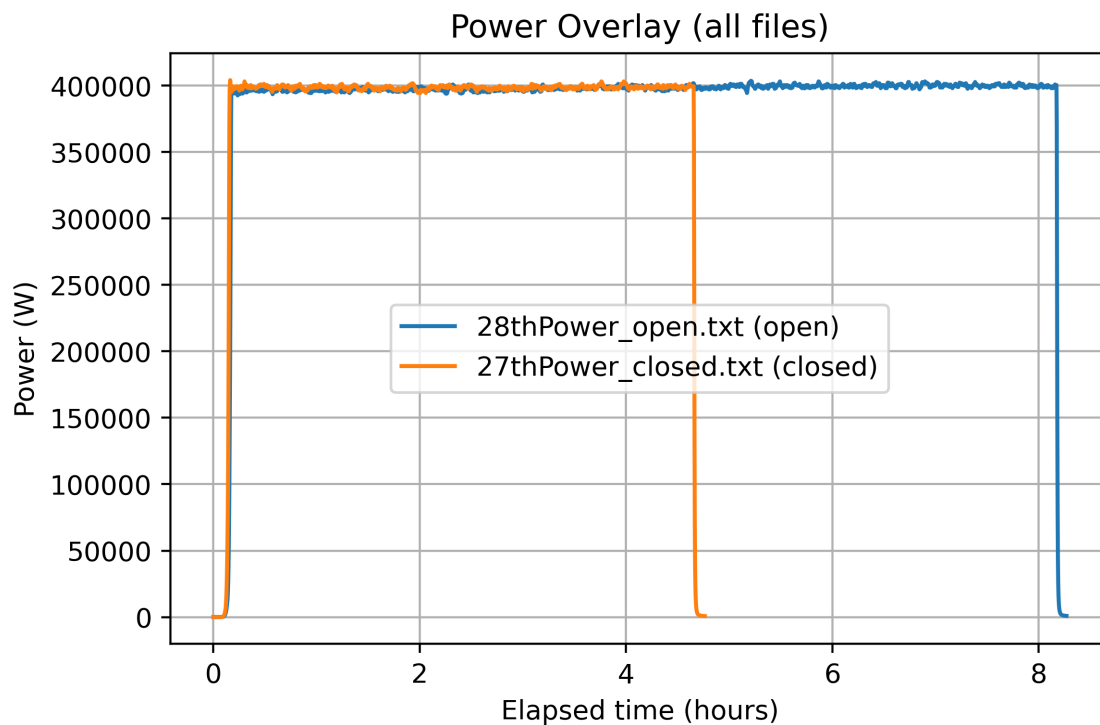


Figure 5.9: Reactor power overlay for both runs on a linear scale. Both days operated at the same nominal reactor power ($\sim 300\text{--}400$ kW); the large difference in neutron flux between the two days arose entirely from the gamma shutter configuration, not from a change in reactor power.

2212 **5.6 Experimental Results**

2213 **5.6.1 Baseline Test (January 26, Out of Beam)**

2214 Prior to transport to the reactor facility, all five devices were operated at ambi-
2215 ent background radiation levels to establish a baseline and to confirm correct data
2216 acquisition system operation. FFSR1 began logging at 21:21 on January 26; FFSR2–
2217 5 began at 22:41. All devices reported CNT:00000000 continuously throughout the
2218 baseline period, confirming zero false positives from the test setup and demonstrating
2219 that the DMR detection logic was functioning correctly.

2220 Logging on all five devices terminated simultaneously at approximately 22:59 due
2221 to a Windows Update reboot on the acquisition PC. This was not a device failure.
2222 After the reboot, logging was not restarted until the devices were at the reactor
2223 facility the following morning. Future tests should disable automatic OS updates on
2224 the acquisition PC for the duration of the experiment.

2225 **5.6.2 Day 1 Reactor Run (January 27)**

2226 Logging resumed at the reactor facility at approximately 11:42 on January 27 for
2227 all five devices. The reactor reached full power within the first 10–15 minutes. All
2228 devices monitored continuously through reactor shutdown at approximately 16:34,
2229 providing approximately five hours of reactor run data.

2230 **Upset Count Results**

2231 Zero SEUs were detected on any of the five devices throughout the entire Day 1
2232 reactor run at a total fast neutron fluence of approximately $6\text{--}7 \times 10^9$ n/cm². All
2233 devices reported a consistent CNT:00000000 output for the full irradiation duration.
2234 Table 5.3 summarises the Day 1 results.

Table 5.3: Day 1 reactor run upset detection results. *FFSR1 count includes 250 CNT-format and $\approx 3\,252$ CNZ-format readings (see Section 5.6.2).

Device	COM	Start	End	Readings	SEUs
FFSR1	COM8	11:42	16:34	$\approx 3\,502^*$	0
FFSR2	COM7	11:42	16:34	3 491	0
FFSR3	COM6	11:42	16:34	3 496	0
FFSR4	COM9	11:42	16:34	3 497	0
FFSR5	COM4	11:42	16:34	3 496	0

2235 PSU Current Monitoring—Day 1

2236 Power supply current was monitored for the CMOD board array throughout the
 2237 Day 1 run. As shown in Figure 5.10, the current profile exhibits a power-on inrush
 2238 transient from approximately 0.175 A, rising rapidly to a steady-state value of approx-
 2239 imately 0.283–0.286 A within the first ~ 200 seconds as the boards initialised. The
 2240 current then remained essentially flat for the entire $\sim 17\,500$ -second run (~ 4.9 hours).

2241 One small transient spike to approximately 0.289 A is visible at $\sim 12\,500$ s, consis-
 2242 tent with a brief radiation-induced transient event; the current returned immediately
 2243 to the baseline level with no permanent change. Critically, no monotonic TID-induced
 2244 current increase is evident across the Day 1 run. The flat current profile is consistent
 2245 with the Day 1 fluence level being insufficient to produce measurable TID-related
 2246 leakage current degradation in the CMOS circuitry, providing a useful baseline for
 2247 comparison with the Day 2 profile (Section 5.6.4).

2248 FFSR1 UART Format Change—Candidate Configuration Memory SEU

2249 At approximately 12:03 on January 27, the UART output format on FFSR1
 2250 changed from CNT:00000000 to CNZ0000. The label changed from CNT to CNZ, the

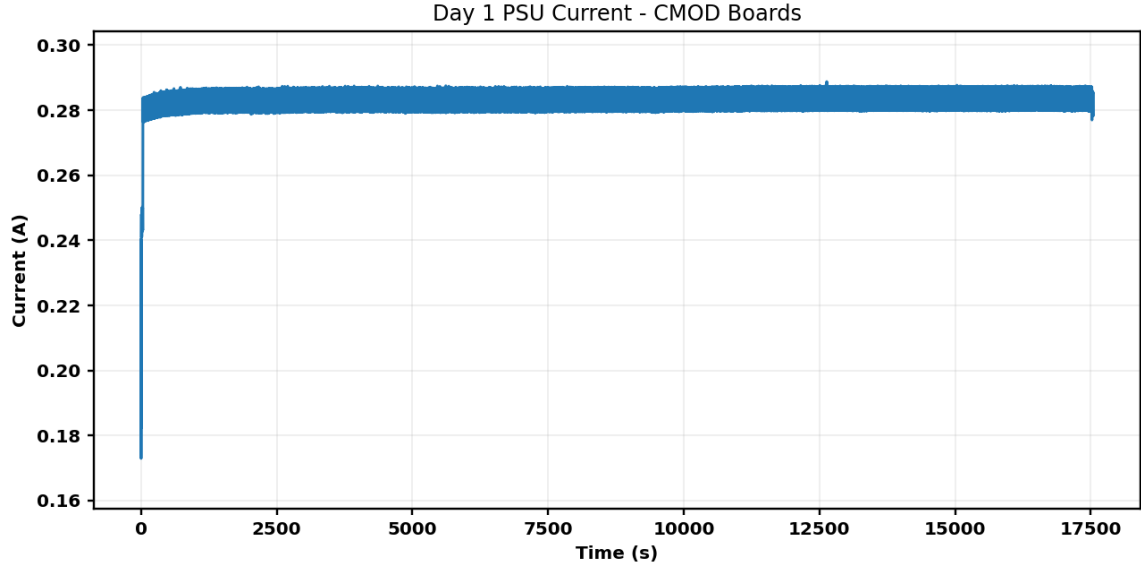


Figure 5.10: PSU current for the CMOD board array during the Day 1 (gamma shutter closed) reactor run. Current stabilises near 0.284 A after initial power-up and remains flat throughout, indicating no measurable TID-induced leakage accumulation at the Day 1 fluence level ($\sim 6\text{--}7 \times 10^9 \text{ n/cm}^2$).

2251 colon delimiter disappeared, and the counter field width reduced from 8 hex digits
 2252 to 4. This format change persisted for the remainder of the run; the upset count
 2253 remained zero throughout in both formats; and FFSR2–5 showed no similar change.

2254 This behaviour is consistent with a single-event upset in the FPGA configuration
 2255 memory (SRAM) rather than in the user data path. The Spartan-7 is an SRAM-
 2256 based FPGA, meaning the design implementation—LUT contents, routing resources,
 2257 and embedded string constants—is stored in volatile SRAM configuration bits as
 2258 susceptible to neutron-induced upsets as any other SRAM. A bit flip in one or more
 2259 configuration cells controlling the UART formatting logic could plausibly alter the
 2260 output label and field width in exactly the structured manner observed.

2261 This event is significant for two reasons. First, the DMR logic correctly protected
 2262 the user flip-flop data path and reported zero upsets throughout, but it does not pro-
 2263 tect the configuration memory; the observation highlights that data-path redundancy
 2264 alone is insufficient for SRAM-based FPGAs in radiation environments. Second, the
 2265 single affected device (out of five) is consistent with the stochastic nature of SEUs at
 2266 the Day 1 fluence level.

2267 5.6.3 Overnight Monitoring (January 27–28)

2268 Following the Day 1 reactor run, all five devices were left running overnight out of
 2269 beam at the reactor facility. Logging ran from approximately 16:42 on January 27 to
 2270 approximately 07:13 on January 28, providing roughly 14.5 hours of out-of-beam mon-
 2271 itoring at ambient background levels. All five devices reported zero upsets throughout
 2272 the overnight period, as expected. Table 5.4 summarises the results.

Table 5.4: Overnight (out-of-beam) monitoring results.

Device	COM	Log Start	Log End	Readings	SEUs
FFSR1	COM8	16:48, Jan 27	07:13, Jan 28	13 981	0
FFSR2	COM7	16:48, Jan 27	07:13, Jan 28	10 389	0
FFSR3	COM6	16:48, Jan 27	07:13, Jan 28	10 354	0
FFSR4	COM9	16:48, Jan 27	07:13, Jan 28	10 353	0
FFSR5	COM4	16:48, Jan 27	07:13, Jan 28	10 353	0

2273 Notably, FFSR1 continued to output in the CNZ0000 format throughout the en-
 2274 tire overnight period, including after three reconnect attempts (16:42, 16:45, 16:48).
 2275 For an SRAM-based FPGA, configuration memory is volatile—loaded from flash at
 2276 power-up and held in SRAM during operation. A bit flip in the configuration SRAM

will persist indefinitely until either the device is power-cycled or a scrubbing operation overwrites the affected cell. The persistence of the CNZ format through logging restarts, which do not cause a power cycle, but the expectation that it would be cleared by a full power cycle, is consistent with a configuration SRAM upset. This observation reinforces the importance of distinguishing between logging-session restarts and full device power cycles when interpreting FPGA behaviour in radiation test environments.

Figure 5.11 shows the PSU current during the overnight period. Current was stable near 0.2835 A with occasional small transient spikes, confirming normal device operation at ambient background levels and providing a clean pre-Day-2 baseline.

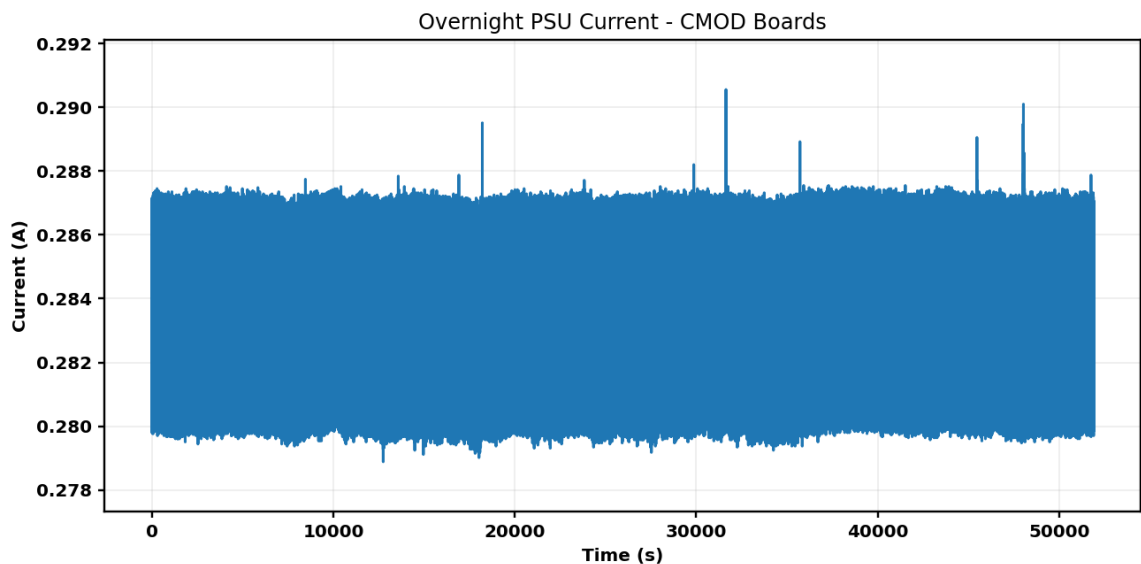


Figure 5.11: PSU current for the CMOD board array during the overnight (out-of-beam) monitoring period (January 27–28). Current is stable near 0.284 A with occasional small transient spikes, confirming normal device operation at ambient background radiation levels.

2287 5.6.4 Day 2 Reactor Run (January 28)

2288 The Day 2 reactor run was conducted with the gamma shutter open, producing
 2289 a fast neutron flux approximately two orders of magnitude higher than Day 1. All
 2290 five devices began logging at 07:21 and the reactor reached full power within approx-
 2291 imately 20 minutes. The reactor ran for approximately 8.5 hours before controlled
 2292 shutdown. Table 5.5 summarises the Day 2 results.

Table 5.5: Day 2 (gamma shutter open) upset detection results.

Device	COM	Log Start	Log End	Readings	SEUs / Notes
FFSR1	COM8	07:21, Jan 28	15:54, Jan 28	715	CNZ0800 detected (Sec. 5.6.4)
FFSR2	COM7	07:21, Jan 28	14:44, Jan 28	5 298	0
FFSR3	COM6	07:21, Jan 28	08:10, Jan 28	580	0—early dropout (~49 min)
FFSR4	COM9	07:21, Jan 28	07:57, Jan 28	431	0—early dropout (~36 min)
FFSR5	COM4	07:21, Jan 28	15:55, Jan 28	6 144	0

2293 FFSR3 and FFSR4 Early Dropouts

2294 FFSR4 ceased transmitting at approximately 07:57 (~36 minutes into the run)
 2295 and FFSR3 at approximately 08:10 (~49 minutes in). Both devices stopped cleanly
 2296 with no error messages; data simply ended mid-run and neither device recovered.
 2297 Two plausible mechanisms are consistent with these observations:

- 2298 • **Radiation-induced latch-up.** A high-energy neutron interaction could trig-
 2299 ger a latch-up condition in the FPGA I/O or core circuitry, causing the device
 2300 to lock up and halt UART output. Latch-up typically requires a power cycle to
 2301 recover and can cause permanent damage if not cleared promptly.

2302 • **Critical configuration memory upset.** A severe or multi-bit upset in a
2303 critical portion of the configuration memory (e.g., clock management or I/O
2304 control logic) could halt device operation entirely, even without a true latch-up.

2305 The PSU current data (Figure 5.12) shows step increases at approximately 2800 s
2306 and 5000 s into the run. FFSR4 stopped at approximately 2160 s and FFSR3 at
2307 approximately 2940 s, placing both dropouts near the step increase at ~ 2800 s. An
2308 abrupt change in aggregate board current at this time is consistent with a latch-up
2309 event altering one or both devices' current draw. These early dropouts represent
2310 the most severe functional failures observed during the two-day campaign and are
2311 consistent with the two-orders-of-magnitude increase in fast neutron flux on Day 2.

2312 **PSU Current Monitoring—Day 2**

2313 Figures 5.12 and 5.13 show the CMOD board aggregate PSU current during the
2314 Day 2 run. Key observations are:

2315 • **Part 1 ($\sim 07:21$ – $14:15$):** Current increased steadily from approximately 0.28 A
2316 at run start to approximately 0.315 A by the end of the logging window—an
2317 increase of approximately 12.5%. This is in stark contrast to the flat Day 1 pro-
2318 file and is consistent with TID effects accumulating in the FPGA and support
2319 circuitry, increasing leakage current monotonically under the much higher Day 2
2320 fluence. Step increases near 2800 s and 5000 s correspond to the approximate
2321 dropout times of FFSR4 and FFSR3 respectively.

2322 • **Part 2 ($\sim 14:28$ – $16:00$):** Current was approximately 0.315 A at the start of
2323 this window with a continued slow increase to ~ 0.319 A. Brief sharp transients

2324 at approximately 4 300 s and 5 200 s into this window are consistent with tran-
 2325 sient radiation events affecting the boards.

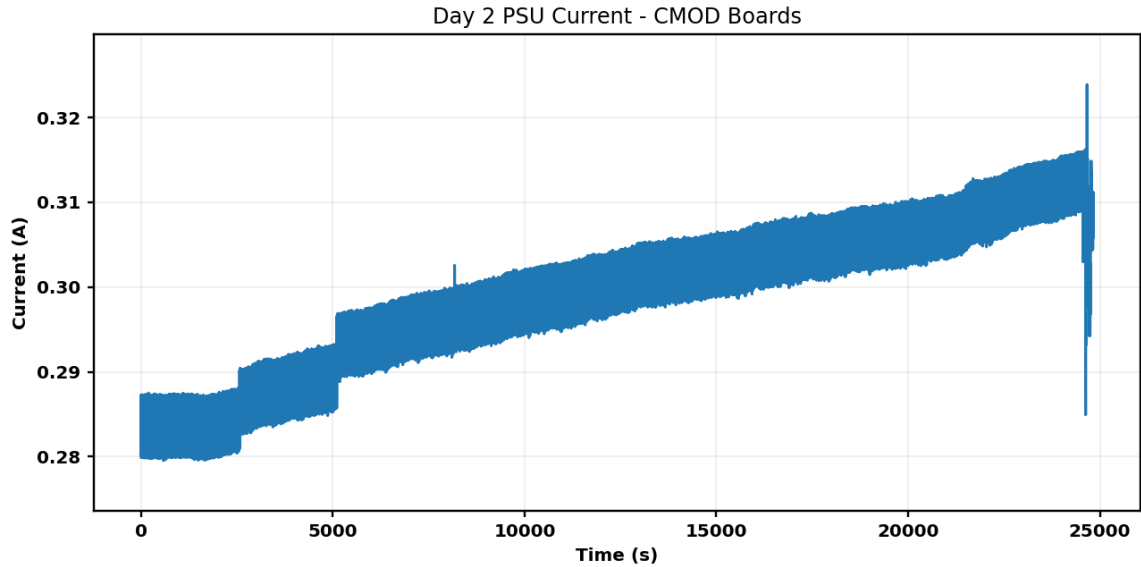


Figure 5.12: PSU current for the CMOD board array during the first part of the Day 2 (gamma shutter open) reactor run. A steady monotonic increase from approximately 0.28 A to 0.315 A ($\sim 12.5\%$) is visible over the ~ 7 -hour logging window, consistent with TID-induced leakage current accumulation. Step increases near 2 800 s and 5 000 s correspond to the approximate times of the FFSR4 and FFSR3 dropouts.

2326 **FFSR1 End-of-Run Burst Event and Confirmed DMR Upset**

2327 The most significant finding of the entire test campaign occurred on FFSR1 at
 2328 approximately 15:54:44 on January 28. Under normal operation, FFSR1 transmitted
 2329 one CNZ output line every ~ 5 seconds. At 15:54:44, the device abruptly began trans-
 2330 mitting at an extremely high rate—approximately 400 readings within a 2-second win-
 2331 dow, roughly $500\times$ the normal output rate. Within this burst the following anomalies
 2332 were observed:

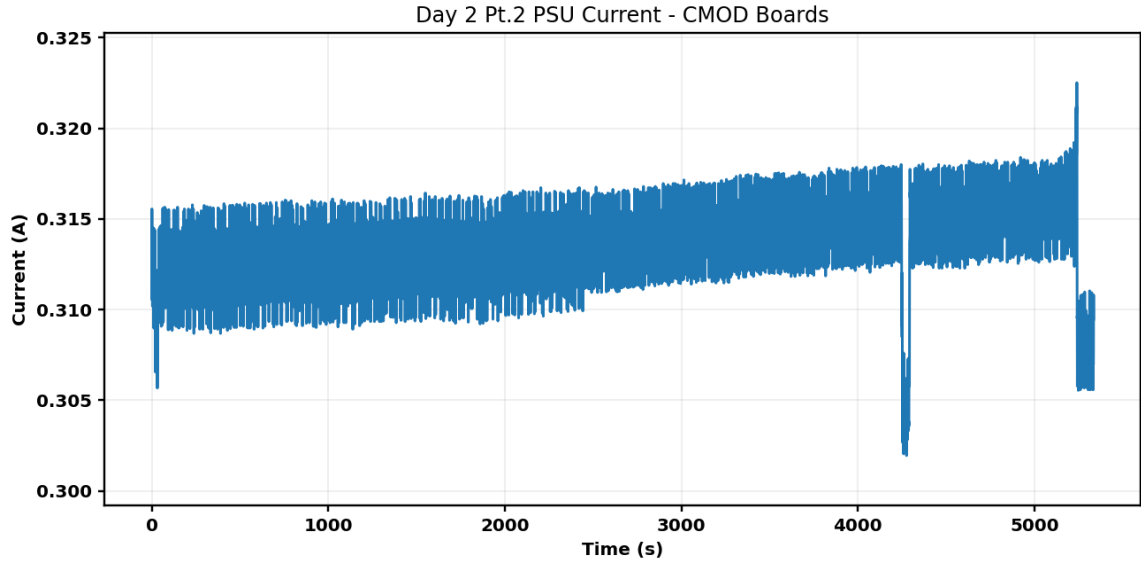


Figure 5.13: PSU current for the CMOD board array during the second part of the Day 2 run. Current continues near 0.315 A with a slow upward trend; brief sharp transients are consistent with transient radiation-induced events.

- 2333 • Multiple corrupted character substitutions: `cNZ0000` (lowercase ‘c’), `CLZ0000`
- 2334 (‘L’ for ‘N’), `KNZ0000` (‘K’ for ‘C’), `CN[0000` (‘[’ for ‘Z’).
- 2335 • A truncated output: `CNZ0` (partial transmission).
- 2336 • A non-zero upset count: `CNZ0800` (hexadecimal `0x0800` = 2048 decimal).

2337 The value `CNZ0800` is the *first and only non-zero DMR upset count recorded across*
 2338 *the entire two-day test campaign*, representing a confirmed detection of one or more
 2339 single-event upsets in the FFSR user data path by the DMR comparator logic.

2340 The burst event itself—the $\sim 500\times$ increase in UART transmission rate—is strongly
 2341 indicative of a second configuration memory SEU on FFSR1, this time affecting the
 2342 UART baud rate generation logic. The baud rate divider is implemented in FPGA
 2343 configuration resources; a bit flip in the timing register can reduce the inter-character

2344 delay to near zero, producing exactly the rapid-fire transmission pattern observed.
2345 The coincidence of the baud rate upset, the character corruption events, and the
2346 non-zero DMR count within a 2-second window is consistent with a high-rate SEU
2347 cluster—multiple configuration and data memory upsets occurring in rapid succession
2348 near the end of the reactor run when cumulative fluence was at its maximum.

2349 **5.7 Discussion**

2350 **5.7.1 SEU Sensitivity of the DMR Design**

2351 The absence of any DMR-detected upsets across all five devices at the Day 1 fast
2352 neutron fluence of approximately $6\text{--}7 \times 10^9$ n/cm² indicates either that the DMR
2353 design successfully masked data path upsets that would otherwise be detectable,
2354 or that the Spartan-7 shift-register fabric did not experience a sufficient number of
2355 upset events to register a DMR mismatch at this fluence level. To distinguish between
2356 these interpretations, a companion test without DMR protection (e.g., a bare shift-
2357 register comparison) would be needed to establish the intrinsic SEU cross-section of
2358 the Spartan-7 fabric under this beam condition.

2359 On Day 2, the first confirmed DMR data path SEU (counter value 2048) was
2360 observed on FFSR1 at the highest fluence level of the campaign ($\sim 8\text{--}9 \times 10^{11}$ n/cm²
2361 total fast fluence). The two-orders-of-magnitude increase in flux and fluence from
2362 Day 1 to Day 2 is qualitatively consistent with the transition from zero detectable
2363 data-path upsets on Day 1 to device failures, latch-up events, and confirmed DMR
2364 detection on Day 2.

2365 **5.7.2 Configuration Memory Vulnerability**

2366 The FFSR1 observations across both days provide clear evidence of configuration
2367 memory SEUs manifesting as persistent, structured changes in device behaviour. The
2368 Day 1 UART label change (Section 5.6.2) and the Day 2 burst event (Section 5.6.4)
2369 both represent configuration memory upsets whose effects persisted until the de-
2370 vice was power-cycled, consistent with the volatile SRAM nature of the Spartan-7
2371 configuration store. These observations highlight that SRAM-based FPGAs have
2372 two distinct radiation vulnerability mechanisms: upsets in the user data registers
2373 (addressed by DMR/TMR) and upsets in the configuration SRAM (addressed by
2374 periodic scrubbing). Radiation-hardened designs intended for extended exposure en-
2375 vironments must address both.

2376 **5.7.3 TID Effects in PSU Current Data**

2377 The PSU current monitoring data provides a consistent and clear picture of TID
2378 accumulation across the two days. The Day 1 current profile was flat throughout
2379 (~ 0.284 A), indicating negligible TID-induced leakage at the Day 1 fluence level. In
2380 contrast, the Day 2 current profile exhibited a steady monotonic increase of approxi-
2381 mately 12.5% over the approximately 7-hour run. This is consistent with TID effects
2382 in the FPGA and support circuitry at the much higher Day 2 fluence. The step
2383 increases in the Day 2 current profile near 2800 s and 5000 s provide independent
2384 corroborating evidence for latch-up events at the times of the FFSR4 and FFSR3
2385 dropouts.

2386 5.7.4 Implications for FPGA Radiation Hardening

2387 Taken together, the results of this two-day test campaign demonstrate several
2388 conclusions relevant to the use of SRAM-based FPGAs in fast neutron environments.

2389 First, at moderate fluence levels ($\sim 10^9$ n/cm²), configuration memory SEUs can
2390 occur before data-path SEUs are detectable at the DMR counter level, suggesting
2391 that configuration scrubbing should be considered even in designs that rely on DMR
2392 or TMR for data-path protection.

2393 Second, at higher fluence levels ($\sim 10^{11}$ n/cm² and above), latch-up, multiple si-
2394 multaneous SEUs, and TID-induced leakage all become significant concerns that re-
2395 quire active mitigation strategies including latch-up detection and recovery circuitry,
2396 TID-hardened component selection, and TID-aware power budget margins.

2397 Third, the FFSR architecture proved effective as a radiation test instrument.
2398 The DMR comparator correctly identified the single confirmed data-path upset event
2399 while maintaining consistent zero-count output throughout normal operation, and the
2400 UART telemetry path provided a reliable real-time window into device state for the
2401 duration of both reactor runs.

2402 5.8 Summary

2403 This chapter presented the design and experimental results of the FFSR DMR
2404 irradiation test conducted at the OSURR Fast Neutron Beam Facility on January 26–
2405 28, 2026. The principal findings are as follows.

2406 **Day 1 (gamma shutter closed, $\sim 6\text{--}7 \times 10^9$ n/cm² fast fluence).** Zero DMR-
2407 detected SEUs across all five devices during approximately five hours of irradiation.

FFSR1 exhibited a UART output format change consistent with a configuration memory SEU while the DMR upset counter remained zero. No measurable TID effects were observed in the PSU current data.

Overnight (out of beam, January 27–28). Zero upsets across all five devices during approximately 14.5 hours of ambient monitoring. The FFSR1 CNZ format persisted through all logging restarts, confirming a persistent SRAM hardware state change rather than a software artefact.

Day 2 (gamma shutter open, $\sim 8\text{--}9 \times 10^{11}$ n/cm² fast fluence). FFSR3 and FFSR4 experienced early functional dropouts at approximately 36 and 49 minutes respectively, consistent with radiation-induced latch-up supported by the PSU current data. FFSR2 and FFSR5 completed the full run with zero DMR upsets. FFSR1 experienced a severe multi-SEU cluster event near the end of the run, producing the first and only confirmed non-zero DMR upset count of the campaign (CNZ0800, $0x0800 = 2048$) along with a second configuration memory SEU affecting the UART baud rate. PSU current monitoring showed a sustained $\sim 12.5\%$ increase over the Day 2 run, consistent with TID-induced leakage current accumulation.

Overall, these results demonstrate that SRAM-based FPGAs operating in this fast neutron environment are susceptible to configuration memory SEUs at moderate fluence levels, and to data-path SEUs, latch-up, and TID degradation at higher fluence levels. The FFSR DMR architecture successfully fulfilled its design objective of detecting data-path upsets while operating reliably throughout the test campaign, and highlights the need for configuration memory scrubbing and latch-up recovery circuitry for SRAM-based FPGAs intended for extended operation in radiation environments.

2432

Appendix A: The Data on Cows

2433 This is the data that was used to produce the table in Table ?? . In 1990, 57
2434 people had cows. In 1991, 80 people had cows. In 1992, 199 people had cows.

2435 **Appendix B: Important commands defined in osudissert96**

2436 The following is a list of all commands available in osudissert96.

```
2437 \author{First Middle Last}
2438 \title{The title of the thesis}
2439 \authordegrees{degree1, degree2}
2440 \unit{Department of Whatever The Name Is}
2441 \degree{Doctor of Philosophy}
2442 \committee{Dissertation}
2443 \advisorname{Name of advisor}    % Possible usage "Prof. Big Dude"
2444 \member{Name of committee member}
2445
2446 \thesis    % this makes it a thesis rather than a dissertation,
2447            % similar to the documentclass option [masters] or [ms].
2448
2449 \maketitle
2450
2451 \disscopyright            % or \blankpage
2452
2453 \begin{abstract}
2454 \end{abstract}
2455
2456 \begin{externalabstract}
2457 \end{externalabstract}
2458
2459
2460 \dedication{This is dedicated to \ldots}
2461
2462 \begin{acknowledgements}
2463 \end{acknowledgements}
2464
2465 \begin{vita}
```

```

2466 \dateitem{Important Date}{ Why its important}
2467
2468 \begin{publist}
2469 \researchpubs
2470 \pubitem{Bibliography item (from BibTeX?)}
2471
2472 \instructpubs
2473 \pubitem{Bibliography item (from BibTeX?)}
2474 \end{publist}
2475
2476 \begin{fieldsstudy}
2477 \majorfield*          % which uses \unit above
2478 % \majorfield{Your Major Field}
2479 \begin{studieslist}
2480 \studyitem{Topic 1}{Prof.\ 1}
2481 \studyitem{Topic 2}{Prof.\ 2}
2482 \studyitem{Topic 3}{Prof.\ 3}
2483 \end{studieslist}
2484 %% Note:  If there were only one field of study, the following list
2485 %%          would best be done using the following command:
2486 %% \onestudy{Only Topic}{Only Professor}
2487 \end{fieldsstudy}
2488
2489 \end{vita}

```

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